Farm Size Distribution, Weather Shocks, and Agricultural Productivity*

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First Version: June 2023
This Version: June, 2026

Abstract

We study how weather shocks affect the farm-size distribution and agricultural productivity. We document new patterns about farm-size dynamics and the effects of weather shocks across developing countries. Drawing on administrative data from Colombia with land-transaction records, census-based farm sizes, and household surveys, we show that weather shocks intensify land-market activity and reduce farm size within regions. To understand mechanisms, we quantify a heterogeneous-agent model with endogenous farm-size distribution. Results reveal that temporary shocks have long-lasting effects on farm size, that agricultural risk substantially distorts the farm-size distribution, and that small changes in weather regimes significantly reduce output per farmer.

*We thank Chris Barret, Lauren Falcao Bergquist, Jeff Campbell, Juan Esteban Carranza, Andrew Foster, Xavier Giné, Douglas Gollin, Steven Helfand, Clément Imbert, Joe Kaboski, Rachid Laajaj, Karen Macours, Jeremy Majerovitz, Mushfiq Mobarak, Frederik Noack, Tommaso Porzio, Mica Sviatschi, and Jack Willis for conversations. We are grateful to Paola Poveda, Juliana Quigua, Salvador Traettino, and Xinyi Li for excellent research assistance. We thank also participants in presentations at AAEA, Barcelona Summer Forum, BREAD/Princeton, CEPR Development, Columbia, Cornell, DIL–University of Chicago, EPGE, EESP, IEA World Congress, INSPER, LACEA, NEUDC, Northeastern, PacDev, PUC-Rio, RIDGE, Universidad de los Andes Chile, Universidad de Los Andes Colombia, Universidad de Rosario, University of British Columbia, University of Chile, UC–Davis, the World Bank, and Harvard/MIT Environment Series. We thank Arturo García, Nury Bejarano, and Jonathan Beltrán, and the Econometría team for their support with data collection. The findings, interpretations, and conclusions expressed in this paper are solely those of the authors and do not necessarily represent the views of the IADB or Banco de la Republica.

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1 Introduction

In low- and middle-income countries, two empirical regularities stand out. First, agriculture is dominated by small, low-productivity farms, a fact that has motivated an extensive literature on the determinants and consequences of the farm-size distribution—see reviews by Gollin (2019); Helfand and Taylor (2021). Second, rural households are highly exposed to uninsured income shocks, which has inspired a large body of work that studies how these shocks shape production, investment, and asset-holding decisions, including farmland sales, through consumption-smoothing motives—see, for example, Rosenzweig and Wolpin (1993); Udry (1995); Krishna (2010); Donovan (2021). These two literatures are central to development economics, yet they have evolved in parallel. This paper brings them together.

We ask how uninsured weather shocks—by inducing distress land sales and exit from farming—shape the farm-size distribution and what this implies for agricultural productivity. We proceed in three main steps. First, using longitudinal household surveys from eleven developing countries, we document two novel patterns: (i) there is significant exit from farming—especially among small farmers—and (ii) weather shocks increase farm exit while simultaneously increasing entry of landless households. Second, we turn our focus to Colombia, which provides uniquely rich administrative panel data with information on land transactions and the universe of agricultural properties. We show that weather shocks substantially increase land sales, expand the number of small farms, and reduce average and median farm size within municipalities, with no evidence of large farms absorbing smaller ones. In addition, Colombian household-level data reveal that shocks reduce income and consumption, and increase asset sales, including sales of agricultural land.

Third, to move beyond these reduced-form patterns and study mechanisms and productivity implications, we quantify a two-sector heterogeneous-agent model in which the farm-size distribution emerges endogenously from occupational and landholding decisions. We discipline the model to match our reduced-form estimates of the response to shocks and key features of the Colombian agricultural setting. In our model, income shocks lead smaller farmers to sell their land to smooth consumption, lowering land prices and increasing the incentives for agents who are not affected by the shock to enter agriculture. Our results show that the effects of temporary shocks take a long time to fade out, generating persistent farm fragmentation; that uninsured income shocks significantly distort the farm-size distribution, reducing the proportion of large farms; and that modest increases in the frequency of weather shocks can translate into sizable declines in output per farmer.

Taken together, our findings reveal an important and underexplored channel linking uninsured shocks to the organization of agriculture in developing countries. These results carry implications for understanding the determinants of the farm-size distribution, the consequences of climate change, and the broader costs of missing insurance and credit markets. We now turn to describing our three steps in more detail.

Our study starts by documenting changes in farm size over time using household-level data for several developing countries. In particular, there is significant exit from farming, especially among smaller farms. However, smaller farmers who stay experience faster farm size growth than surviving large farms. Moreover, there is substantial entry into farming, with most new farmers coming from non-agricultural backgrounds. We find that weather shocks increase the exit probability of incumbent farmers, for both smaller and larger farms. In stark contrast, landless households are more likely to enter farming and operate a positive amount of land. This asymmetry indicates that the incentives underlying the decision of being a farmer after weather shocks differ sharply between landless households and incumbent farmers. Notably, these patterns emerge consistently within each regional group in our sample (e.g., Latin America versus West Africa), despite the economic and institutional differences between these regions.

While this evidence reveals consistent patterns in farm dynamics across countries, there are limits to how much we can say about specific mechanisms underlying these changes. In these survey data, we do not observe land transactions, ownership changes, or the evolution of the farm-size distribution at the regional level. To better understand these farm dynamics, we turn our focus to Colombia, where uniquely rich administrative records permit us to track all of these aspects of land markets over time at the municipality level. Specifically, we present three reduced-form results. First, administrative data on land transactions show that weather shocks substantially increase land market activity in a municipality. Both the number of farm sales—full and partial property sales—and the number of mortgage contracts rise following such shocks. Much of this increase in sales is driven by purchases made by buyers who did not previously own any land in the region, which is consistent with our previous finding that weather shocks induce entry among households that operate no land.

Second, using our administrative census data on the universe of rural properties, which allow us to measure the farm-size distribution, we show that weather shocks increase the number of property owners in a municipality and reduce the mean and median farm size.[1] This result indicates that the entry of new farmers following adverse shocks more than offsets the exit of existing ones. We find that this increase in the number of farmers is driven primarily by a rise in the number of small farmers, with no evidence of larger farms absorbing smaller ones. These patterns imply not only that there are more farmers overall—thereby reducing the average farm size—but also that the new entrants tend to operate small farms.

[1] In our data, while rare, a property owner may have more than one farm registered in the system.

Third, using a longitudinal household survey from rural Colombia, we show that weather shocks lead to declines in both income and consumption, with a larger drop in income. Households are also more likely to migrate to another municipality and to sell land, livestock, and durable assets, such as televisions and refrigerators. These results are consistent with the increases in land sales observed in the administrative data. Moreover, a larger fall in income than in consumption, together with a higher likelihood of migration and the sale of a broad range of assets, suggests that land sales are part of a consumption-smoothing strategy.

To further characterize the agents who enter farming after a shock, we report results from an original survey of rural community leaders across the country. These results confirm that land sales are a common consumption smoothing mechanism. Moreover, the majority of respondents report that land is not purchased by farmers in their community. This pattern again indicates that buyers are households who were not exposed to weather shocks.

Our reduced-form results are robust to an extensive battery of checks. These include different measures of weather shocks (e.g., using different sources of weather data, splitting results between hot and cold days, or exploiting different lag structures), alternative farm-size definitions, varying sampling restrictions, controls for state-specific trends, more conservative standard errors that account for spatial correlation, and alternative sets of controls, including controls for expected weather shocks induced by pre-trends.[2] Beyond these specification checks, we show that the lack of land consolidation is not driven by institutional or geographic barriers, including land regulation, ease of property conversion into recreational use, contiguity between small and large farms within municipalities, and regional conflicts.

While the empirical patterns documented thus far are consistent with consumption-smoothing motives fragmenting the farm-size distribution, they leave several relevant questions open. How much does the consumption-smoothing behavior implied by our data distort the farm-size distribution? Under what economic conditions do weather shocks generate land fragmentation instead of land consolidation? What are the aggregate productivity effects of changing the frequency of weather shocks, as measured by output per worker, average farm productivity, and output per unit of land? To address these questions, we develop and quantify a dynamic, heterogeneous-agent model tailored to our agricultural setting.

In the model, agents can be either farmers or non-agricultural workers and must choose their occupation each period. Those who choose to farm decide, before the harvest season, how much land to buy for production. Those who choose to work sell all their land—if they have any—and earn wages. Agents are heterogeneous in their initial landholdings as well

[2]We follow here the approach recently proposed by Jones et al. (2026). The idea is to generate counterfactual distributions of temperatures for every region and year based on the temperature trends implied by climate change, and use these counterfactual distributions to formulate the probability of being under a given bin of the temperature distribution.

as in their productivity in agricultural and non-agricultural activities. These productivities are subject to idiosyncratic shocks, which introduce uncertainty. Reflecting the institutional context of rural markets in developing countries, households lack access to agricultural insurance, credit, and land rental markets. Within this structure land serves a dual role: it is a productive asset and it can be sold to smooth consumption. The farm-size distribution emerges endogenously from agents' joint decisions over land accumulation and occupational choice. We calibrate the model to match several moments of the farm-size distribution in Colombia and also to match our reduced-form estimates of the impact of weather shocks on consumption.[3] Our calibrated model performs well when we compare its predictions to both targeted and non-targeted moments.

Our first structural result concerns the implications of uninsured risk for land allocation. Because land serves as a precautionary saving device, less productive farmers hold more land than is optimal, while more productive farmers—who are better self-insured—hold less. This mechanism has important aggregate implications: keeping the number of farmers and the average farm size fixed, an efficient economy in which the marginal product of land is equalized across farmers would feature a larger fraction of large farms and 42 percent higher agricultural output. To put this gain in perspective, using World Bank WDI/ILOSTAT data on agricultural value added per worker in 2015, the ratio of US to Colombian agricultural productivity was about 13. A 42 percent increase in Colombian output would lower this ratio to about 9.

We then simulate the economy's response to a single weather shock—modeled as a disturbance to aggregate agricultural TFP—starting from the stationary equilibrium. The effects are large and persistent. Average farm size falls by 0.12 percent on impact and remains 0.06 percent below its pre-shock level even twenty years later, long after aggregate TFP has returned to its steady-state value. Crucially, this persistence extends to agricultural productivity: because the number of farms remains elevated for an extended period, average farm size stays lower, and the farmers who remain in agriculture have, on average, lower agricultural productivity than before the shock. As a result, output per farmer falls and stays low well beyond the duration of the shock itself.

The model also offers a natural explanation for the striking asymmetry between incumbent farmers and landless households in their responses to adverse shocks—and for the seemingly counterintuitive result that the total number of farmers increases following a negative agricultural productivity shock. The mechanism operates through supply and demand in

[3]Specifically, in the stationary equilibrium, we target the mean and variance of farm size, the share of households engaged in agriculture, and the exit rates of farmers across the farm-size distribution. In addition, we also experiment with a calibration that allows for multiple locations and find no significant differences in our results.

the land market. On the supply side, adverse shocks compress farm incomes, induce farmers to sell their land to smooth consumption, and reduce the price of land. On the demand side, because the shock is short-lived, it leaves the present value of becoming a farmer largely unchanged for landless households, who are not directly exposed to the income loss. Landless households therefore face an attractive entry opportunity: the expected returns to farming are intact, but land has become cheaper. They enter, crowding out the potential expansion of larger incumbent farmers whose incomes are under pressure because of the shock. When we simulate scenarios with more persistent shocks than those observed in our data, the opposite occurs: the total number of farmers falls and average farm size increases. Our results highlight a broader point that, to the best of our knowledge, has not been emphasized before: the persistence of a productivity shock—not just its magnitude— can fundamentally alter equilibrium outcomes.

Finally, we move beyond the response of the economy to a single shock to study the evolution of a stochastic economy subject to a sequence of shocks.[4] We ask how aggregate outcomes change when the frequency of atypical temperature days rises permanently and find that even modest increases have large consequences: a 5 percent increase in that frequency reduces aggregate output by 3 percent. Because average farm size also falls as shocks become more frequent, the effect on output per farmer is roughly twice as large. Even if farmers fully anticipate the change in average TFP, the reduction in average farm size induced by weather shocks still amplify the reduction in output per farmer by 40 percent. These results point to a new mechanism through which a changing climate can depress agricultural productivity in developing economies—one that operates not through the direct damage of any single event, but through the cumulative effect of more frequent shocks on the farm-size distribution itself.

Related Literature. This paper contributes to a rich literature on the causes and consequences of the farm-size distribution in developing countries. Discussions linking farm size to productivity date back at least to the seminal work of Sen (1962), and have since become a staple of development economics (Gollin, 2019; Helfand and Taylor, 2021; Barrett et al., 2010). With a micro-oriented approach, studies have explored a range of driving factors, including frictions in land and labor markets (Foster and Rosenzweig, 2022; Acampora et al., 2025), price risk (Barrett, 1996), and access to urban markets (Gáfaro and Pellegrina, 2022; Pellegrina, 2022; Rao et al., 2022; Madhok et al., 2022). Some recent papers show that an important consequence of small farm size is that it prevents mechanization and potential agricultural productivity gains (Foster and Rosenzweig, 2022; Caunedo and Kala,

[4] We follow the approach proposed by Auclert et al. (2021) and Boppart et al. (2018), which provides a local approximation to the stochastic equilibrium. Because this approach is valid in a neighborhood of the steady state, we focus on small changes in the frequency of shocks.

2021). Taking a more macro perspective, others have examined how land institutions shape economy-wide productivity (Adamopoulos and Restuccia, 2020; Chen et al., 2022). Our main contribution here is to document a new mechanism shaping the farm size distribution: uninsured income shocks lead to land sales and the fragmentation of farms. By doing so, we bring a dynamic perspective to our understanding of the farm-size distribution, an object that has mainly been treated as static in the literature. Moreover, our results indicate that low agricultural productivity can be exacerbated by the aggregate consequences of individual farm size choices made under uninsured risk, pointing to a novel link between income shocks, land fragmentation, and the notoriously low agricultural productivity of developing countries (Gollin et al., 2014; Donovan, 2021).

We also contribute to growing research that links well-identified microeconomic mechanisms to their macroeconomic consequences—for a review, see Buera et al., 2023. Our paper is particularly close to recent papers that employ heterogeneous-agent macro tools, building on the seminal contributions of Hopenhayn (1992) and Aiyagari (1994), to questions related to agriculture. For example, Brooks and Donovan (2020) show how improved market access and reductions in income risk affect labor supply and farm investment; Manysheva (2022) studies how land and credit frictions shape resource allocation and the effects of land reform; Gottlieb and Grobovšek (2019) analyze how communal land tenure impacts occupational choice and land allocation; Mazur and Tetenyi (2024) study how agricultural input subsidies affect aggregate welfare when markets are incomplete; and Trouvain (2025) studies the role of land in shaping consumption dynamics. The key distinguishing feature of our paper is that we focus on the farm-size distribution as an equilibrium object, one that is endogenous to households' occupational and landholding decisions. This allows us to study how weather shocks reshape the farm-size distribution and the further implications for transition dynamics and aggregate outcomes like average farm-size and agricultural productivity.

Finally, we speak to a rich literature in development economics examining the effects of income shocks on agricultural households. A substantial body of work has documented the various strategies households employ to smooth consumption when faced with negative income shocks (Paxson, 1992; Townsend, 1994; Udry, 1994; Morduch, 1995; Dercon, 2002).[5] Closer to our work, some studies show that negative productivity shocks often force poor landowners to liquidate assets to smooth consumption (Deaton, 1989; Rosenzweig and Wolpin, 1993; Carter and Zimmerman, 2003; Kazianga and Udry, 2006). Beyond asset depletion, other work has highlighted a broader set of behavioral responses—including ad-

[5] Taking a more macro-perspective, recent papers show that governments also react to weather shocks, for example by intervening in food trade policy or by adjusting subsidies (see Hsiao et al. (2026) and Crews et al. (2025)).

justments in labor and input use, loan repayment, crop choice, migration, and investment in human capital (Jayachandran, 2006; Jessoe et al., 2018; Colmer, 2021; de Roux, 2021; Jagnani et al., 2021; Aragón et al., 2021). Consistent with our findings, Kaboski et al. (2022) show that farmers in Uganda invest in land when they win a large lottery. We add to this literature by documenting that land sales constitute an important margin of adjustment for farmers facing negative productivity shocks. Because land is a key asset for most farmers in developing economies, land sales can have strong, long-lasting effects on their future income. In addition, because climate change increases the severity and frequency of adverse weather shocks (IPCC, 2021), our results highlight an additional mechanism through which these changes can deepen the agricultural performance gap between poor and rich economies.

2 Farm-Size Dynamics, Weather Shocks, and Rural Land Markets in Developing Countries

Drawing on household-level longitudinal data covering eleven countries—three in Latin America and eight in Africa—this section provides new evidence on farm-size dynamics and how weather shocks influence them. For each country, we construct a panel using two survey rounds conducted approximately three years apart, which we merge with subnational region- and year-specific measures of weather shocks. Appendix OA describes the surveys used for each country. Throughout the analysis, we define farm-size as the land operated by the household.[6] Given our focus on Colombia in subsequent analysis, when the data allows, we present the empirical patterns separately for that country. We conclude this section by discussing key characteristics of land markets in developing countries, which motivate the structure of the model we develop in Section 5.

2.1 Farm-size Dynamics and the Impact of Weather Shocks

Farm-size Dynamics. We document two novel patterns in farm-size dynamics: (i) exit from farming is substantial between survey waves, with higher exit rates concentrated among small farms; and (ii) on average, small farms expand while large farms shrink. Figure 1 shows results from local polynomial regressions with the sample of households that operated land in the first survey round. In Panel (a), the outcome is a dummy equal to one if the

[6] By focusing on operated land, we avoid issues related to land ownership. At this point, we are agnostic about how farmers change their operating status—for example, through land transactions, communal land redistribution, within-family reallocation, or land rentals. Our conclusions here do not hinge on these institutional characteristics. We provide a detailed discussion of the institutional features of land markets in developing countries later in the section.

household does not operate any land in the second survey round, which we interpret as exiting farming.[7] In Panel (b), the sample is restricted to households that continue operating land in the second period, and the outcome is the log growth rate of farm size between periods. In both panels, the horizontal axis is the farm size in the first round relative to the average farm size in the respective country (in logs). Panel (a) shows that the exit probability is positive across all farm sizes and that smaller farmers are more likely to exit than larger farmers. The magnitudes are sizable: in Colombia, within the three-year period between surveys, farmers operating below the average farm size are on average 75 percent more likely to exit (a probability of 0.07 versus one of 0.04). In West African countries, farmers below the average farm size are twice as likely to exit as those above (0.16 versus 0.08). The patterns for the other Latin American countries are similar.

Farm exit is matched by significant entry. Entrants account for between 13 to 43 percent of all households who increase the area of land operated between survey waves (Appendix Figure O.1). These new farmers differ markedly from incumbents: they operate at considerably smaller scale—their farm size is on average 40 percent smaller in Africa and 70 percent smaller in Latin America (Appendix Figure O.2)—and most of them are not employed in agriculture in the baseline period. Only 20 percent of entrant household heads are employed in agriculture at baseline, compared to 80 percent of incumbents (Appendix Table O.1). These patterns indicate a dynamic process in which exiting farmers are continuously replaced by new ones.[8]

Panel (b) of Figure 1 shows that small farms tend to grow, whereas large farms tend to shrink.[9] In Colombia, farms below the average farm size grow by a factor of 12, whereas those above the average shrink by about 5 percent; patterns for the other Latin American countries are similar. In West Africa, farms below the average grow by a factor of about 13, while those above the average shrink by roughly 30 percent. Much of the large growth among small farms comes from the extremely low base: in the West African countries in our sample the average farm size is 3.5 hectares, whereas in Colombia it is 2.6.[10]

[7] In the Colombian data, migration is closely tied to exiting farming. About 30 percent of farmers who stop operating land between 2010 and 2013 migrate to an urban area or another rural municipality; while only 1 percent of migrants continue farming.

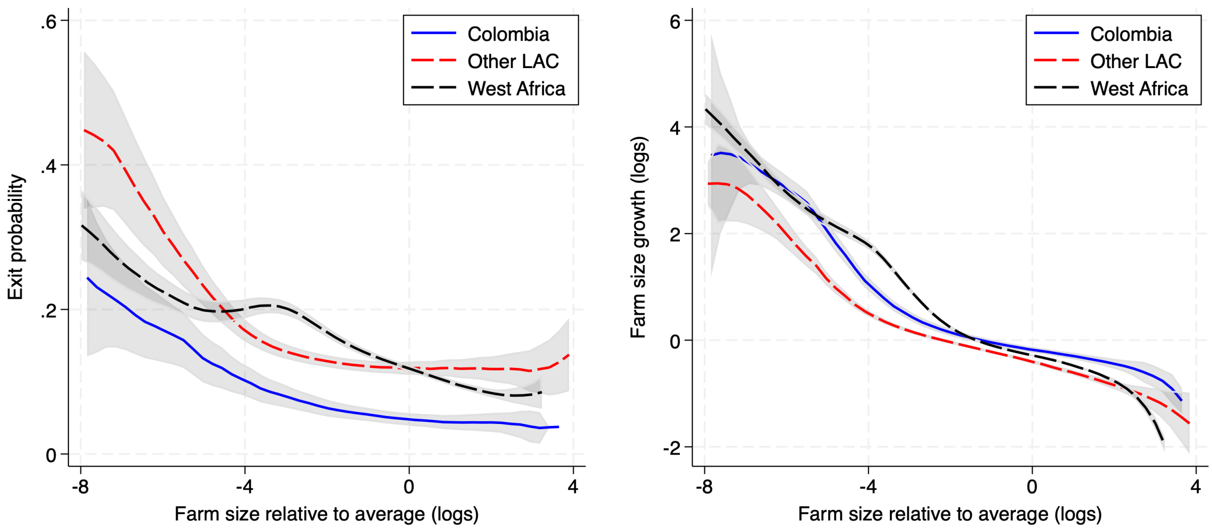
[8] Appendix Table O.1 also shows that entrants tend to be younger and have smaller household sizes than incumbents—characteristics that are typical of recent migrants and first-time entrepreneurs. Furthermore, roughly 30% of entrants were located in urban areas at baseline.

[9] A small proportion of surviving farmers experience no change in operated farm size between survey rounds: about 6% show exactly no change, and 11% percent show a change smaller than 10 percent. While some changes in farm size between rounds may reflect measurement error in survey responses, it is unlikely that such error explains the negative relationship between growth and initial size. Carletto et al. (2013) show that smaller farms tend to over report and larger farms to under report land, which would go against finding a negative relationship if farm size is measured more accurately in later rounds.

[10] As discussed in Section 3, the Colombian household survey data used in these exercises are only repre-

Figure 1: Farm Size Dynamics across Countries

(a) Exit Probability (b) Farm Size Growth



Notes: This figure shows results from local polynomial regressions (bandwidth = 1). Panel (a) uses households that operated land in the first survey round; the outcome is an exit indicator equal to 1 if the household operates no land in the second period. Panel (b) restricts the sample to households that also operate land in the second period; the outcome is the log of the ratio of farm size between periods. In both panels, the right-hand-side variable is farm size in the first period relative to the country's average (in logs), which is also shown on the x-axis. The data cover Colombia, Mexico, and Peru in Latin America, and Benin, Burkina Faso, Côte d'Ivoire, Guinea-Bissau, Mali, Niger, Senegal, and Togo in Africa. Survey rounds are 2010–2013 for Colombia, 2002–2005 for Mexico, and 2018–2021 for West Africa. For Peru, the data come from an annual rotating panel (2007–2023); we use households observed three years apart.

Weather Shocks. To explore the impact of weather shocks on households' entry and exit decisions, we construct a weather shock variable based on temperature information from ERA5 data. This dataset contains global reanalysis information on temperature with a horizontal resolution of $0.25° \times 0.25°$ ($\approx$ 28 km by 28 km) at an hourly frequency since 1979.[11] We aggregate hourly data to daily averages at the municipality level and construct a municipality-level measure of exposure to atypical temperature days.[12] To account for geographical and seasonal variations in temperature within countries, we proceed as follows. First, for each municipality and each quarter of the year, we compute the distribution of daily temperatures using data for 1979–2016. This yields four distributions per municipality, one

sentative of small farms in a few regions of the country. For comparison, we note that in the 2014 agricultural census, the average farm size in the country is 5.8 hectares.

[11]ERA5 reanalysis information combines climate models and observational data from satellites and ground sensors. Data are provided by the Copernicus Climate Change Service (C3S) of the European Center for Medium-Range Weather Forecasts (ECMWF).

[12]Across surveys, we use "municipality" to refer to units identified by *Admin 2* codes. In Colombia, these units are municipalities, but in other countries the corresponding administrative units have different names, such as provinces in Burkina Faso or cercles in Mali. The only exception is Peru, where we use *Admin 3* units.

for each quarter. Next, we classify a day as atypical in a municipality if its temperature falls below the 20^{th} percentile or above the 80^{th} percentile of its respective municipality-quarter distribution of daily temperature. Formally, indexing days by d, quarters by q, years by t, and municipalities by i, and denoting the 20^{th} and 80^{th} percentiles of municipality-quarter daily temperature distributions as $\mu^{20}_{i,q(d)}$, and $\mu^{80}_{i,q(d)}$, we define an atypical temperature day in a municipality as

$$AtipicalDay_{i,t,d} = \mathbb{1}(\text{Temperature}_{i,t,d} \leq \mu^{20}_{i,q(d)}) + \mathbb{1}(\mu^{80}_{i,q(d)} \leq \text{Temperature}_{i,t,d}). \tag{1}$$

Defining a single variable that captures the exposure of a municipality i in year t to atypical days requires addressing two challenges. First, there may be a lag between farmers' responses to atypical weather and the registration of changes in land ownership, land transactions, or land operation in the data. Second, the impact of the shock may influence farmers' decisions and outcomes with a lag—for example, atypical days experienced by a farmer at the end of calendar year $t-1$ can have an effect on crop revenues only in t, during the harvest. To address this potential mismatch between the timing of atypical days and economic outcomes we define the shock as the sum of atypical days lagged over the past two years:

$$TempShocks_{i,t} = \sum_{d=1}^{365} AtypicalDay_{i,t-1,d} + \sum_{d=1}^{365} AtypicalDay_{i,t-2,d}. \tag{2}$$

For ease of exposition, we divide $TempShocks_{i,t}$ by 100, so coefficients can be interpreted as the effect of 100 additional atypical days over the previous two years. More details on the construction of weather shocks are provided in Appendix OA.5.

Weather Shocks Effects. We estimate the impact of weather shocks on farm-size dynamics using the following specification:

$$\begin{aligned} y_{h,t} = \beta_1 \, TempShocks_{i(h),t} &+ \beta_2 \, TempShocks_{i(h),t} \times \mathbb{1}\{\text{Landless}\}_h \\ &+ \beta_3 \, TempShocks_{i(h),t} \times \mathbb{1}\{\text{Small Farm}\}_h + \alpha_h + \gamma_{r(h),t} + \varepsilon_{h,t}, \end{aligned} \tag{3}$$

where h indexes households, $i(h)$ the municipality of household h, and t the survey year. The dependent variable $y_{h,t}$ equals one if household h operates any land in period t. The terms α_h and $\gamma_{r(h),t}$ denote household and region–year fixed effects, respectively, where region represent one level of aggregation above municipalities.[13] Our measure of weather

[13] Regions correspond to *Admin 1* units in the surveys. In Colombia, these units are *departamentos*.

shock, $TempShocks_{i(h),t}$, is the total number of atypical temperature days (in hundreds) in municipality i over $t-1$ and $t-2$, which we calculate as described above. The indicator $\mathbb{1}\{\text{Landless}\}_h$ identifies households that did not operate any land in the first survey wave, while $\mathbb{1}\{\text{Small Farm}\}_h$ identifies those whose operated area was below the median farm size in their respective region. $\varepsilon_{h,t}$ denotes the error term, which we cluster at the municipality (i) level. We exclude Colombian households from this estimation because, as described in Appendix Section OA, the survey for Colombia only samples households operating land in the first wave, preventing us from observing entry into farming.

Figure 2 plots the estimated coefficients from equation (3), and Appendix Table O.2 reports the regression results alongside estimates from alternative specifications. We highlight two findings. First, both in Latin America and West Africa, weather shocks increase the likelihood of exiting farming regardless of the farm size, with slightly larger point estimates for smaller farms. Second, weather shocks increase the likelihood that initially landless households will operate land in the subsequent period.

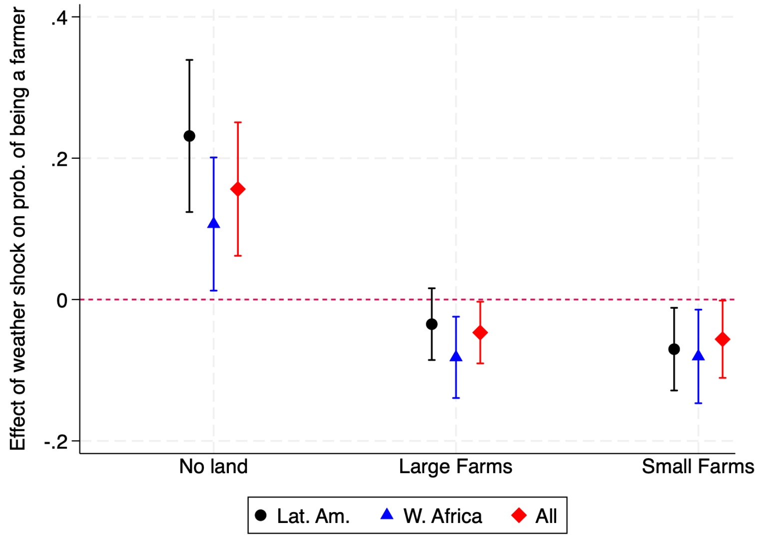
This figure illustrates that the incentives to engage in farming after weather shocks differ sharply between landless households and incumbent farmers. Later in Section 5, we provide an economic explanation for this pattern, relating it to the short-run nature of weather shocks and the fact that both small and large farmers are directly affected, while potential entrants may be less exposed. Before turning to the empirical results using the Colombian administrative data, we next discuss key characteristics of land markets and agriculture in developing countries that will guide the interpretation of our results.

2.2 Rural Land Markets in Developing Countries

Three institutional features of land markets and agriculture in developing countries guide the interpretation of our reduced-form findings and the design of our model. First, the adoption of agricultural insurance is extremely low. Unsubsidized agricultural insurance coverage rates in high-income countries average 41.7%, compared to 8% and 0.5% in lower-middle-income and low-income countries (Mahul and Stutley, 2008). In Colombia, according to DANE (2019), the share of farmers with any form of agricultural insurance is below one percent. In the absence of insurance, farmers facing adverse shocks may be forced to sell their assets to smooth consumption.

Second, survey data indicates that distress land sales are not uncommon in developing countries and that farmers sell land when experiencing financial distress to smooth consumption. In Colombia, for example, between 2013 and 2016, nearly 65% of households who reported selling land did so to pay for household expenses or cover outstanding debts, pay for

Figure 2: Effects of Weather Shocks on Farmers' Exit and Entry



Notes: This figure shows the impact of weather shocks on households' probability of operating land. We plot the estimates $\hat{\beta}_1$ (large farmers), $\hat{\beta}_1 + \hat{\beta}_2$ (landless households), and $\hat{\beta}_1 + \hat{\beta}_3$ (small farmers) from equation (3) and their 95% confidence intervals. Standard errors are clustered at the geographic administrative level at which the weather shock is measured. The sample contains 24,074 household-year observations for LAC and 85,584 for West Africa.

medical treatment, or pay for education fees. Section 4.5 revisits this issue using our original survey of rural community leaders. The use of land sales as a consumption smoothing device is also documented by Krishna (2010) in Africa.

Third, land rental markets in developing countries are generally thin, which means that land ownership tends to be strongly associated with land operation, and that land rental is not a common adjustment mechanism. In Colombia, according the National Agricultural Survey (DANE, 2019), about 85% of plots are operated by owners, and only 8% of plots are operated by renters.[14] These figures are similar to those reported by Acampora et al. (2025), who find that the proportion of households renting land in Kenya is 4% and ranges between 0% and 5% in six other countries across sub-Saharan Africa. This stands in contrast to the U.S and the European Union, where the share of farmland operated under a rental agreement is about 40% and 46%, respectively.[15] These statistics suggest that the patterns documented in Figure 1 likely reflect changes in land ownership.[16]

These institutional characteristics motivate the basic structure of our model in Section 5. To capture the relationship between farm growth and farm size shown in Figure 1, we

[14] Other types of tenancy, such as sharecropping, usufruct, loan-for-use, or de-facto occupation are even less common and jointly account for 5% of plots.

[15] USDA Economic Research Service (2024) and Eurostat (2020).

[16] The prevalence of communal land tenure systems varies across countries. In our sample, the average share of communal land across African countries is 30%, ranging from 70% in Burkina Faso to 10% in Niger. Below, we study changes in land ownership originating from land market transactions in Colombia. However, similar patterns may arise from non-market communal land allocation.

introduce idiosyncratic productivity shocks that induce farmers to expand or shrink their operations. Additionally, agents make occupational choices, in which they can either buy land and become farmers or sell their entire property and exit agriculture. These choices are made in an economic environment with uninsured risk and no land rental markets.

Next, we turn our attention to Colombia, where we assemble a comprehensive longitudinal dataset on land transactions and farm sizes. We use it to uncover a set of empirical facts that unpack the mechanisms behind the effects of weather shocks on the farm-size distribution.

3 Data from Colombia

Our Colombian dataset is structured at two levels of disaggregation: municipality and farm. At the municipality level, it contains annual information on land transactions, moments of the farm-size distribution, and weather shocks for 864 municipalities.[17] At the farm-level, it provides detailed information on household consumption, asset ownership, and investment decisions. Here, we provide a brief description of our datasets, relegating details to Appendix OA. Appendix Table O.3 present descriptive statistics of the different estimation samples. After presenting our data in this section, we turn to the reduced-form effects of weather shocks in Section 4.

Land Transactions. We use administrative, property-level transaction information from the *Superintendencia de Notariado y Registro* (SNR, National Superintendence of Notaries)—the government agency that regulates and oversees Colombia's notarial and public-registry system. Beyond recording real-estate transactions, the SNR supervises notaries and registry offices and guarantees the legal validity of property titles. For each property-level transaction, we observe the buyer, the seller, and whether the transaction was a partial property sale, a full sale, or a mortgage—we do not observe, however, the price or the size of the properties. These data cover all the properties that were once part of a *baldío*—publicly owned vacant land—and were subsequently allocated to private owners. By 2014, they summed to 23 million hectares—more than half of the currently privately-held land in the country (Sánchez and Villaveces, 2016; Arteaga et al., 2017). These data contain all transactions dating back to the early 1900s, but we focus on the period 2000-2011 when our other datasets are available. We construct a balanced panel at the municipality-year level with the number of full and partial sales, and the number of mortgage transactions. For complementary analyses, we

[17] We exclude from our final sample of municipalities large metropolitan areas and municipalities with very few rural properties (i.e. below the 1st percentile). Our final sample covers 85.3% of the rural population in the country.

split the number of sales into those involving buyers who previously owned land and those involving first-time owners in the *departamento*.[18]

Farm Size Distribution. Our farm-size distribution data come from the national cadastre managed by the *Instituto Geográfico Agustín Codazzi* (IGAC, National Geographical Institute of Colombia). This land registry contains information on the universe of farm properties in Colombia.[19] We obtained municipality-level aggregate information from all properties in IGAC's cadastre that are i) privately owned, and ii) categorized as having an agricultural economic purpose. This amounts to roughly 40 million hectares of land, roughly comparable in size to Germany or California. The data form a yearly panel for 2000–2011 at the municipality level and include the number of farms, the number of owners, and the mean and median farm size for specific size ranges.[20]

Household-Level Data. We use household panel data from the rural sample of the *Encuesta Longitudinal Colombiana* (ELCA), a survey conducted by Universidad de los Andes. The data include a sample of 4,800 rural households interviewed over three survey rounds (2010, 2013, and 2016). The baseline sample covers households in 222 rural communities of 17 municipalities. We have detailed information on land ownership, consumption, income, migration, and assets.

Agricultural Census We also use information from the 2014 National Agricultural Census, provided by DANE, Colombia's National Statistical Agency. The census covers agricultural production units (*Unidades de Producción Agropecuaria*, UPAs) across Colombian municipalities and records information on agricultural activity during the 2013 production year.[21] Importantly for our purposes, the census provides georeferenced information for each UPA. In our analysis, we use these data only for complementary exercises, including the construction

[18]The *departamento* is the administrative unit above the municipality, of which there are 28 in our sample. As detailed in Appendix OB.1, we proxy ownership by determining whether the name of the buyer appears in the past in the SNR at the receiving end of a transaction (sale, inheritance, or government allocation) within the same *departamento*. We restrict the search for names within *departamentos* to avoid false matches arising from common names across regions.

[19]The cadastral data-collection process is designed to capture even informal properties that do not have all required formal documentation.

[20]The land registry records rural properties used for agriculture, which we treat as equivalent to farms. In settings with active land-rental markets, property and farm can differ because farmers may operate land they do not own or combine plots from multiple owners. As discussed in Section 2.2, in Colombia land rental is uncommon, so ownership and operation (farm) typically coincide. Consistent with this, land registry data show that owners in a municipality almost always report only one property.

[21]A UPA is a production unit under a single decision-making authority and may include one or more plots, dwellings, or households.

of an alternative measure of plot contiguity and the calibration of moments of the farm-size distribution.

4 Reduced-form Impact of Weather Shocks in Colombia

This section studies the reduced-form effects of weather shocks in Colombia. We begin with municipality-level regressions in Section 4.1. We show that weather shocks increase land sales and mortgages, reduce average farm size, and are associated with the entry of new landowners in affected municipalities. Section 4.2 then shows that these main results are robust to alternative shock definitions, inference procedures, data sources, and specifications that account for location-specific temperature trends. Section 4.3 provides household-level evidence showing that consumption smoothing operates as a supply-side mechanism underlying land transactions. Section 4.4 discusses and discards alternative mechanisms that could explain the decline in average farm size. Finally, Section 4.5 presents results from our survey of community leaders, which we use to validate our interpretation of the results and to characterize new entrants into farming.

4.1 Weather Shocks, Land Transactions, and Farm Size

We estimate the impact of weather shocks on land transactions and farm size using:

$$s_{i,t} = \beta TempShocks_{i,t} + X'_{i,t}\delta + \eta_i + \kappa_t + \varepsilon_{i,t}. \tag{4}$$

Here, $s_{i,t}$ represents the number of land transactions (sales or mortgages), the log number of farm owners, or the log average and median farm size in municipality-year pair (i, t). $TempShocks_{i,t}$ is our measure of weather shocks, capturing the number of atypical temperature days (in hundreds) in the municipality over the previous two years, as defined in Section 2.1. $X_{i,t}$ is a vector that includes time-varying municipality characteristics; specifically, rainfall in year t and five lags, the cumulative number of land allocations in the municipality up to t, a dummy for cadastral updates, and the total municipal land area recorded in the registry, which accounts for changes in registry coverage. The specification also includes municipality fixed effects η_i to control for time-invariant heterogeneity, and year fixed effects κ_t to capture economy-wide shocks that may affect land markets and farm size. Standard errors are clustered at the municipality level.

Equation (4), as well as all subsequent specifications, relies on the identifying assumption that, conditional on the fixed effects and control variables, there are no municipality-specific, time-varying unobservables correlated with temperature shocks. This assumption would be

Table 1: Weather Shocks and Land Transactions

	Land Sales					
	Transaction Type			Buyer Type		
	All	Full	Partial	Landless	Owner	Mortgages
	(1)	(2)	(3)	(4)	(5)	(6)
$TempShocks$	2.537***	2.013***	0.523**	1.866***	0.671***	1.046***
	(0.536)	(0.504)	(0.229)	(0.408)	(0.192)	(0.237)
Observations	10,021	10,021	10,021	10,021	10,021	10,021
Adjusted R^2	0.896	0.893	0.600	0.878	0.823	0.733
Mean Dep. Var.	12.62	10.83	1.79	9.37	3.25	2.62

Notes: This table presents coefficient estimates from equation (4) in a balanced panel at the municipality-year level. The dependent variable in column 1 is the number of land sales (full + partial), in column 2 the number of full sales, in column 3 the number of partial sales. Columns 4 and 5 separate total sales into those involving first-time buyers and those involving buyers who previously owned land in the *departamento*. Buyer type is identified by checking whether the buyer's name appears in earlier SNR records (sales, inheritances, or government allocations) within the same *departamento*. The dependent variable in column 6 is the number of land mortgages. The independent variable is the total number of atypical temperature days in the previous two years divided by 100. All regressions include municipality and year fixed effects, as well as cumulative rainfall and its five lags, the accumulated number of land allocations, the area covered by the land registry, and an indicator for cadastral registry updates. See the text for details. Standard errors clustered at the municipality level reported in parenthesis. Data from the National Superintendency of Notaries (SNR). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

violated if differential temperature trends across municipalities are correlated with trends in land transactions and farm size, generating spurious relationships between weather shocks and our outcomes. Section 4.2 addresses this concern with a series of robustness checks. First, we include state-specific time trends to absorb differential long-run trajectories across regions. Second, we reconstruct the daily temperature distribution using a narrower time window, thereby reducing potential idiosyncratic measurement error in atypical temperature days resulting from municipality-level temperature trends. Third, we apply the approach developed by Jones et al. (2026), which controls for expected weather shocks based on observed municipality-level temperature trends to mitigate spurious correlations. Reassuringly, our results are largely robust to all these checks.

Effects on Land Transactions. Table 1 presents the OLS estimates from equation (4) for different measures of land transactions. Column 1 shows results for the total number of land sales. Columns 2 and 3 separate this into *full* sales (entire-property transfers) and *partial* sales (fractional-property transfers). Column 1 shows that more days with atypical temperature lead to more land sales. In particular, an additional 100 atypical days (roughly two standard deviations) over the previous two years increases the number of land sales by 20%. Columns 2 and 3 show that this effect is driven by a 19% increase in full sales and a 29% increase in partial sales. Column 6 shows that an additional 100 days of atypical temperature in the municipality increases the number of mortgages by 40%. In sum, weather shocks substantially increase land-transaction activity within a municipality. [22]

To explore whether the observed increase in land sales is driven by purchases made by landless households or by farmers who already own land in the municipality or nearby areas, we use our transaction-level data to identify all individuals involved in land transactions. We use this information to construct, for every year and *departamento*, a list of individuals who own land based on prior transactions that confer ownership. We then match this list to land buyers in each municipality and year. This allows us to determine whether buyers already owned land in any of the municipalities within the *departamento*.[23]

We decompose the estimated impact of weather shocks on total land sales by buyer type: landless buyers and farmers who were already landowners at the time of the sale. Columns 4 and 5 of Table 1 show that extreme temperatures increase land purchases made by both types of buyers, but effects are larger among landless buyers, who on average account for 74% of the weather-driven land purchases. These results indicate that, although a substantial share of transactions occurs among existing farmers, weather shocks also trigger sales to households that previously owned no land in the municipality or nearby areas.

Effects on the Farm-Size Distribution. Table 2 presents the results of estimating equation (4) on the number of owners, the mean, and the median of the farm-size distribution.[24] A higher number of atypical-temperature days in a municipality during the previous two years leads to an increase in the number of landowners (column 1). Importantly, this indicates a *net* increase in landowners—some landowners might leave, but overall there must be more entry than exit. Columns 2 and 3 shows that both the average and the median farm size fall. An additional 100 days of atypical temperature over a two-year period increases the number of landowners by 1.5% and reduces average and median farm size by 1.5% and 2.1%, respectively. Weather shocks therefore lead to land fragmentation instead of land consolidation.

Table 2 does not provide information on which types of farms drive the decline in average farm size. This decline could arise from the division of large estates into medium-sized prop-

[22] We also study the impact of shocks on inheritances. First, Appendix Table O.3 shows that inheritances are substantially less frequent than land sales, averaging about 2 transactions per municipality-year compared to 13 land sales. Additional estimations (not reported here) indicate that an increase of 100 atypical-temperature days raises inheritance registrations by 0.35 (s.e. 0.102). This implies that for each additional inheritance associated with atypical temperature days, there are roughly seven additional land sales. These two results indicate that changes in farm size induced by weather shocks are driven mostly by land sales.

[23] We match the lists of owners and buyers using first and last names with a bigram fuzzy matching algorithm. Appendix Section OA.4 provides details on the matching procedure, and Appendix Section OB.1 discusses the robustness of our results to alternative matching techniques and to potential measurement error resulting from the coverage of our transaction data.

[24] Farm size is defined based on the registered size of the plots. Since some farmers might own more than one plot, the average farm size can differ from the ratio of total land to the number of owners.

Table 2: Weather Shocks and Average Farm Size

	Number of Owners (logs) (1)	Mean Farm Size (logs) (2)	Median Farm Size (logs) (3)
TempShocks	0.015***	-0.015***	-0.021*
	(0.005)	(0.005)	(0.012)
Observations	10,021	10,021	10,021
R^2	0.992	0.993	0.974

Notes: This table reports the estimated effect of weather shocks on the farm-size distribution. Dependent variables are the log number of land owners in the municipality (column 1), log average farm size (column 2), log median farm size (column 3). The independent variable is the total number of atypical temperature days in the past two years divided by 100. All regressions include municipality and year fixed effects, as well as cumulative rainfall and its five lags, the accumulated number of land allocations, the area covered by the land registry, and an indicator for cadastral registry updates. Standard errors clustered at the municipality level reported in parenthesis. Data from the National Land Registry (*Catastro Nacional*), maintained by the National Geographical Institute (IGAC). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

erties without any change in the number of small farms, or from small farms fragmenting into even smaller plots, without affecting larger properties. To investigate this, we estimate effects on the number of owners within fixed farm-size bins over time. We begin by splitting each municipality's farm-size distribution in 2000—the first year of the sample—into quantiles.[25] We then compute, for each subsequent year, the number of owners in each fixed-size bin. Consider J quantiles in municipality i, denoted by $\{q_i^1, ..., q_i^J\}$. Let $AreaOwned_{o,i,t}$ denote the total landholdings of owner o in municipality i and year t. For each year, we compute the number of owners whose total landholdings fall within each of these fixed-size bins as:

$$NumOwners_{i,t}^j \equiv \sum_{o \in i} \mathbb{1}[AreaOwned_{o,i,t} \in (q_i^{j-1}, q_i^j)], \tag{5}$$

where $j = 1, ..., J$, and $q_i^0 = 0$ for all i. We use this variable to estimate independent regressions (one per quantile j) of the form:

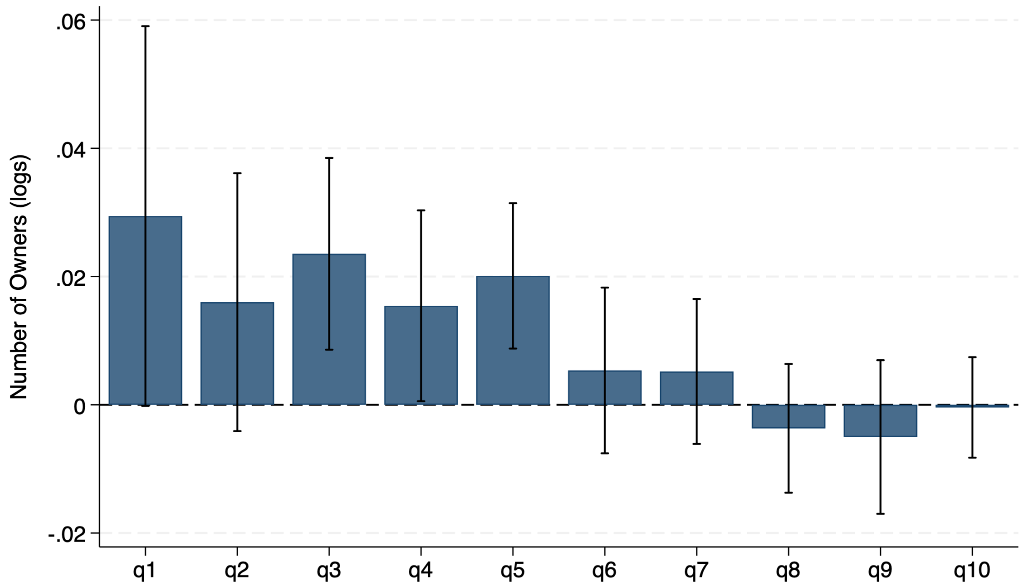
$$\log(NumOwners_{i,t}^j) = \gamma^j \; TempShocks_{i,t} + X_{i,t}'\xi^j + \mu_i^j + \kappa_t^j + \omega_{i,t}^j, \tag{6}$$

where the right-hand variables are the same as in equation (4).

Figure 3 reports estimates of γ^j for $j = 1, \ldots, 10$. Extreme weather shocks cause a sizable increase in the number of owners with farms in the lower five deciles of the initial distribution, but have close to zero and statistically insignificant effects on the number of owners in the

[25] 97% of municipalities have registry information in that year. For the remaining municipalities, we use the first year in which they appear in the land-registry dataset.

Figure 3: Weather Shocks and Number of Owners by Deciles of Initial Distribution



Notes: This figures reports estimates of γ^j for $j = 1, \ldots, 10$ from equation (6). A separate regression is run for each of the ten quantiles of the initial municipality-level distribution of farm sizes. The outcome is the log number of owners whose farm sizes fall within the corresponding decile. The independent variable is the number of atypical temperature days in the past two years, divided by 100. All regressions include municipality and year fixed effects, as well as cumulative rainfall and its five lags, the accumulated number of land allocations, the area covered by the land registry, and an indicator for cadastral registry updates. Error bars indicate 95% confidence intervals. Standard errors are clustered at the municipality level.

top five deciles.[26] The fact that the upper tail of the distribution shows no increase in farm counts—if anything, the point estimate is negative—rules out a land-consolidation story and suggests instead that larger farms are themselves hurt by the shock and unable to absorb smaller ones: if large farms were absorbing small ones, we would observe a decline, not an increase, in small-farm counts.

4.2 Robustness

We now test our main results on land transactions and farm size with an extensive battery of robustness checks. These include alternative specifications, alternative measures of weather shocks, different inference procedures, and accounting for potential bias arising from location-specific trends. Specifically, we present results varying the lag structure of the shocks, modifying the threshold for defining atypical temperature days, allowing for differential effects of hot and cold days, using a drought index as an alternative measure of weather shock, and allowing for broader geographic correlation in the error term. We also assess the robustness of our results to alternative specifications that account for a potential spurious

[26]Regression results are presented in Appendix Table O.5. Appendix Figure O.10 shows analogous estimates for alternative partitions of the initial farm-size distribution ($J = 5$ and $J = 20$).

correlation arising from location-specific trends in temperature. We present the results of these robustness checks for our main outcome variables, which include land transactions, the number of owners, and mean and median farm size. Our conclusions remain unchanged throughout.

Weather Shock Lags. The measure of weather shock in our main specification counts the number of atypical temperature days over years $t-1$ and $t-2$. We explore whether our findings are driven by this choice of timing. In particular, we propose a flexible-lag specification, described in Appendix Section OB.2, that estimates separate coefficients for atypical days that occur in each year from $t-7$ through t. Appendix Figures O.8 and O.9 show that our main conclusions are not affected by the particular aggregation of lags. Extreme temperature days have reduced-form effects that begin the year of the shock and persist for 5 to 6 years. Consistent with this, Panels A and B of Appendix Tables O.6 and O.7 show that when we aggregate atypical days over t to $t-2$ or $t-1$ to $t-3$ our main conclusions are similar to those in our baseline specification.

Atypical Days Thresholds and Temperature Distribution. The definition of atypical days in our main specification focuses on the 20^{th} and 80^{th} percentiles of the municipality-quarter distribution of daily temperatures. We assess the robustness of our results to alternative cutoffs and to using the municipality-level distribution of daily temperature without conditioning on the quarter. First, we use the 5^{th} and 95^{th} percentiles, and the mean $\pm$ 1.5 standard deviations of the municipality-quarter distribution of daily temperature as alternative thresholds. Panels C and D of Appendix Tables O.6 and O.7 show that our main results are robust to these definitions. If anything, we find stronger effects with the more stringent definition of atypical days of the top and bottom 5 percent. Second, we construct a municipality-level distribution of daily temperatures over the entire period, without conditioning on the quarter, and define atypical days as those in the bottom and top 20 percent of this distribution. Panel E of Appendix Tables O.6 and O.7 shows that while coefficient estimates are slightly lower in magnitude and less precise, our overall conclusions are largely unchanged.[27]

Hot and Cold Days. In our main specification, we aggregate atypically hot and cold days into a single measure, capturing the average effect of extreme temperature realizations. We

[27] In our main specification, we condition on the quarter of the year within each municipality to account for seasonal patterns in temperature. Ignoring this seasonal variation may introduce measurement error in our shock variable because some days may be classified as atypical due to regular seasonal variation rather than unusually high or low temperatures. This could explain the relatively smaller magnitude of the coefficient estimates obtained using this alternative measure.

next estimate a more flexible specification that allows for heterogeneous effects across hot and cold days. Specifically, we construct separate measures for the number of days in the bottom and top 20 percent of the municipality–quarter distribution of daily temperatures, and include them jointly in the estimation of equation (4). Appendix Table O.8 shows that both cold and hot days increase the number of land sales and farm owners, while reducing average farm size. These results are consistent with agronomic evidence that both types of temperature extremes can adversely affect agricultural production.[28]

Drought Index. Our main specification uses the number of atypical temperature days as a measure of weather shocks while controlling for total precipitation. Both variables are constructed using ERA5 reanalysis data. We next examine whether our results are sensitive to the choice of a temperature-based measure of shock and to the dataset used to construct it. In particular, we consider the Standardized Precipitation–Evapotranspiration Index (SPEI) as an alternative measure of weather shock. The SPEI is a drought index that captures the balance between precipitation and evapotranspiration—a measure of water demand that depends on temperature, vapor pressure, cloud cover, and wind conditions. The index is standardized, with negative values indicating dry conditions and values below -2 classified as extreme drought. We obtain the SPEI data from SPEIbase v.2.9 (Beguería et al., 2023) and estimate equation (4) using two alternative measures of weather shocks. First, a continuous measure of drought intensity over years $t-1$ and $t-2$ which multiplies the negative values of the index by -1 and sets positive values to zero, so that only drought conditions are considered and larger values correspond to more severe droughts. Second, the number of months in $t-1$ and $t-2$ with a SPEI indicating extreme droughts. Appendix Table O.9 presents these results, which are qualitatively the same as the ones in our main specification.

Location-Specific Trends in Weather Shocks. Appendix Tables O.10 and O.11 present additional robustness exercises in which we consider the possibility that weather shocks are mechanically correlated with location-specific trends in temperature, for example, due to differential exposure to climate change—see discussion in Jones et al. (2026). We implement three alternative approaches to address this concern.. First, we vary the time period used to construct the distribution of daily temperatures. Our baseline relies on temperature realizations from 1979–2016, which extend more than two decades before our sample period (2000–2011). Due to climate change, the distribution of temperatures may have shifted over time, raising the question of whether earlier data remain appropriate for defining atypical

[28]Cold shocks can damage crops through frost events and reduced solar radiation, which slows plant growth. Extreme heat, in turn, increases evapotranspiration and can disrupt flowering and crop development.

conditions during our sample years. Moreover, if climate change affects municipalities differentially, the resulting measurement error may vary systematically across locations. Panel F of Appendix Tables O.6 and O.7 shows that our results are similar to the baseline when we instead compute the temperature distribution using data from 1990–2011. Second, Panel A of Appendix Tables O.10 and O.11 adds to specification (4) *departamento*-specific time trends to account for the possibility that regional trends in temperature shocks may be spuriously correlated with our outcomes. The results are similar to our baseline estimates. Third, Panel B of Appendix Tables O.10 and O.11 follows Jones et al. (2026) by augmenting specification (4) with a control for the expected number of atypical temperature days in each municipality, calculated based on municipality-specific temperature trends. This control mitigates potential bias from spurious correlations between location-specific temperature trends and the outcome variables, for example due to climate change. Appendix Section OB.3 details the construction of this variable.

Inference. In our main results, we cluster the standard errors at the municipality level. This accounts for correlation of the error term within municipalities over time, but assumes independence across municipalities. We assess the sensitivity of our results to allowing more flexible forms of spatial autocorrelation. First, we implement a two-way clustering at the municipality and *departamento* $\times$ year levels. This allows errors to be correlated within municipalities over time and across municipalities within the same *departamento*–year. Second, we account for spatial autocorrelation following Conley (1999). This assumes that the error terms in each municipality are correlated with those of other municipalities within a fixed radius, with linear distance decay. We report results using radii of 500 km and 800 km. Panels C to E of Appendix Tables O.10 and O.11 show that these alternative methods yield standard errors similar to those in our main estimation, and the statistical significance of our estimates is practically unchanged.

4.3 Mechanism: Weather Shocks and Consumption Smoothing

We now examine the effects of weather shocks using information from the three survey waves of the ELCA household survey. Our goal is to assess whether the household-level responses align with the aggregate patterns in land sales and farm size, and whether they support our interpretation of land sales as a consumption-smoothing device. Similar to Section 2, we estimate:

$$y_{h,t} = \alpha\, TempShocks_{v(h),t} + X'_{v(h),t}\tau + \iota_h + \kappa_t + \psi_{h,t}, \tag{7}$$

where $y_{h,t}$ denotes an outcome for household h in survey round $t \in \{2010, 2013, 2016\}$. The weather shock variable, $TempShocks_{v(h),t}$, is analogous to the municipality-level measure described in Section 2.1, but here we use the finer geographic information available in the survey to compute atypical temperature days at the *vereda* v level, a smaller administrative unit than the municipality.[29] Vector $X_{v(h),t}$ includes controls for log aggregate rainfall in the *vereda* at time t and its five lags. ι_h and κ_t denote household and year fixed effects, respectively. Finally, $\psi_{h,t}$ denotes the error term, which we cluster at the *vereda* level.

Table 3 reports the estimates.[30] We find that weather shocks reduce both the likelihood that households own any land and the average farm size, and increase the share of farmers reporting land sales (columns 2 to 4). The adverse effects of weather shocks, however, extend beyond land transactions. We observe higher sales of livestock (column 7) and a decline in the principal-component index of asset ownership (column 8), which aggregates household durables such as appliances, vehicles, and other assets. Appendix Table O.13 disaggregates these results and shows that the decline is driven by lower ownership probabilities across a broad range of assets. Column 1 further shows that affected households are more likely to migrate. These results are consistent with previous evidence of consumption smoothing mechanisms among rural households—see for example, Rosenzweig and Wolpin (1993); Aragón et al. (2021); Ibáñez et al. (2022). Furthermore, results reported in Appendix Table O.15 show that our findings on land fragmentation are not driven by households splitting: we find a positive effect on household splitting—with some members leaving the household—but there is no effect on households splitting while retaining land within the same municipality.

Despite these consumption smoothing strategies, we find that weather shocks still have a sizable impact on consumption—an 11% decline per 100 additional atypical-temperature days, as shown in column 5. This effect is about four times smaller than the 47% decline in income reported in column 6. The larger decline in income relative to consumption provides evidence of incomplete consumption smoothing, consistent with prior work for developing countries (Morduch, 1995; Dercon, 2002, 2004; Kazianga and Udry, 2006). Although our estimated effects on income and consumption are sizable, the underlying shock is also large, corresponding to a two-standard-deviation increase in atypical temperature days over two years. For comparison, Llerena Pinto et al. (2025) document a 53% income decline among

[29] Municipalities are the smallest official administrative division in Colombia. For some administrative purposes, rural areas within municipalities are further divided into *veredas*. *Veredas* operate under the executive power of municipalities' mayors but have their own democratically elected Community Action Boards (*Juntas de Acción Comunal).*

[30] To maintain a consistent estimation sample across outcomes, the results presented in Table 3 impute land-ownership values as zero for households that either migrate to urban areas or do not respond to the agricultural module of the survey. Results that restrict the sample to households that completed the agricultural module yield qualitatively similar but slightly stronger patterns and are reported in Appendix Table O.12.

Table 3: Weather Shocks and Household Decisions

	Household Migrated (1)	Household Has Land (2)	Household Sold Land (3)	Farm Size (Hectares) (4)	Consumption (logs) (5)	Income (logs) (6)	Sold Animals (7)	Asset Index (8)
TempShocks	0.089***	-0.055*	0.029***	-0.481*	-0.111***	-0.465***	0.046***	-0.112***
	(0.033)	(0.030)	(0.010)	(0.272)	(0.038)	(0.069)	(0.016)	(0.028)
Observations	13,442	13,442	13,442	13,442	13,442	12,773	12,634	12,596
R^2	0.532	0.564	0.412	0.605	0.707	0.647	0.932	0.585
Mean Dep. Var	0.14	0.83	0.03	2.36	2.80	2.06	0.67	-0.00

Notes: This table shows estimates of the effects of weather shocks on household decisions using a three-wave panel survey of rural households (2010, 2013, 2016). The dependent variables, from left to right, are: a dummy indicating whether the household migrated between survey waves; a dummy indicating whether the household owns any land; a dummy indicating whether the household sold any land; log farm size; log per-capita consumption; log per-capita income; a dummy indicating whether the household sold any livestock; and a principal-components asset index based on household assets and durable goods. Values for the land-ownership variables (columns 2–4) are imputed as zero for households that did not answer the survey's land module. All monetary values are expressed in 2016 Colombian pesos (in millions). All regressions control for log aggregate rainfall at t and its five lags and include household and time fixed effects. *Mean Dep. Var.* reports the sample mean of the untransformed dependent variable. Standard errors clustered at the *vereda* level are reported in parentheses. Data from ELCA survey. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

poor households in Ecuador following two consecutive rainfall shocks of two standard deviations below the mean; Hirvonen (2016) estimates imply a 11% reduction in per capita consumption from a one-standard-deviation increase in growing-season temperature in Tanzania; and Gao and Mills (2018) show that below-average rainfall reduces per adult equivalent consumption by about 18% among rural households in Ethiopia. These comparisons suggest that, despite their sizable magnitude, our estimates fall within the range of effects documented in related settings.

Together, the household-level evidence indicates that our measure of weather shock captures a negative income shock and that households use several coping mechanisms to smooth consumption after the shock, including selling land and other assets. These findings support interpreting the land sales observed in the administrative data as reflecting households' efforts to smooth consumption.

4.4 Alternative Mechanisms

Our results so far indicate that weather shocks increase the share of households selling their land to smooth consumption, and that this mechanism ultimately leads to a reduction in average farm size, by inducing newcomers to buy land. In the model developed in the next section, this entry of new farmers results from landless agents crowding out potential land purchases by incumbent farmers, whose incomes are also affected by the shock. Below, we discuss and discard five alternative mechanisms through which weather shocks might decrease average farm size.

Land-Contiguity and Demand Hold Back. We consider whether lack of contiguity between large and small farms might constrain land consolidation by limiting the expansion of larger farms following adverse shocks. To test this hypothesis, we construct a measure of contiguity between large and small farms in each municipality using 2017 land-registry maps. Within each municipality, we calculate the share of farms below the 10^{th} percentile of the size distribution that are contiguous to at least one farm above the 90^{th} percentile. Municipalities with a share above the national median are classified as having high contiguity. We estimate equation (4) on the number of owners and on mean and median farm size, adding an interaction term between the weather shock and an indicator for high contiguity. Panel A of Table 4 presents the results. The interaction coefficient is small and statistically insignificant for the number of owners and mean farm size, indicating similar fragmentation patterns across high- and low-contiguity municipalities. Column 3 provides some evidence of a smaller decline in median farm size in high-contiguity municipalities; however, the total effect of the shock remains negative in these municipalities. Moreover, Appendix Table O.14 shows that this smaller decline in median farm size is not robust when using an alternative contiguity measure based on GPS coordinates from the 2014 National Agricultural Census.

Alternative Land Uses. We explore whether farm size decreases due to the conversion of agricultural land into residential or recreational uses after the shock. Given that this conversion is more likely near large cities, we estimate the effect of weather shocks on the number of owners and on mean and median farm size, allowing for heterogeneity across municipalities with high and low distance to large cities. We classify municipalities as having high distance if they lie farther than the national median distance. Panel B of Table 4 shows that the interaction term is close to zero and statistically insignificant for mean and median farm size, in columns 2 and 3, and—contrary to what this mechanism would predict—we find a larger increase in the number of owners in municipalities farther from major cities.

Household Splits and Inheritances. A third alternative explanation is that the decline in average farm size reflects intra-household reallocation rather than land sales. This could occur if weather shocks induce household splits, with some members leaving and taking part of the family land, or if they trigger the formal registration of inheritances. We assess both possibilities. First, Appendix Table O.3 shows that inheritances are substantially less frequent than land sales, averaging about 2 transactions per municipality-year compared to 13 land sales. Additional estimations, not reported here, indicate that an increase of 100 atypical-temperature days raises inheritance registrations by 0.35 (s.e. 0.102). This implies that for each additional inheritance associated with atypical temperature days, there are roughly seven additional land sales. Second, Appendix Table O.15 shows that weather shocks increase household splits, but not splits in which a split-off household member remains in the same municipality and owns land there. Together, these results indicate that the reduction in average farm size is unlikely to be driven by inheritances or household divisions.

Land Ceilings Regulations. A potential alternative explanation for the lack of land consolidation after the shock is related to institutional constraints embedded in Colombia's land regulations. In particular, Law 160 of 1994 establishes municipality-specific land ceilings that limit the amount of government-allocated land any individual can own (Arteaga, 2025). In Appendix OB.4, we explore whether the decrease in farm size is driven by municipalities with larger shares of government-allocated land, where the ceiling is more likely to bind. The results show that this is not the case. Our coefficient estimates do not change when we control for the share of government-allocated land in each municipality and year, and there are no statistically significant differences in the effects of the shock across municipalities with high and low shares of government-allocated land. This suggests that the land ceiling regulation cannot explain the observed decrease in farm size after weather shocks.

Conflict. A final alternative mechanism relates to conflicts. Previous work finds that income shocks in developing countries might intensify violent conflict—e.g., Miguel et al. (2004)—which, in turn, might lead to land sales and land fragmentation to avoid victimization. Here, we check if our results are driven by conflict conditions in affected municipalities. Appendix Table O.17 includes controls for forced displacement and homicide rates to capture the influence of violent conflict. Our point estimates are largely unaffected by such controls.

4.5 Complementary Evidence from Survey Responses

The reduced-form results presented above show that land sales rise following weather shocks and suggest that many buyers are individuals who previously did not own land in the *departamento*. This section presents complementary evidence from an original qualitative survey that we collected to validate our interpretation and provide insight into the characteristics of land buyers. The survey was conducted by telephone in September 2025 with 151 rural community leaders representing more than 7,000 farmers across 30 municipalities in 18 *departamentos*. It gathers information on local land market dynamics, communities' exposure to different types of shocks, and how these shocks affect land sales.

The main results are reported in Appendix Table O.18. The following key findings emerge. First, land sales are a common mechanism for land allocation, as 68% of respondents report that such transactions occur in their communities. Second, land sales serve as an adjustment

Table 4: Weather Shocks and Farm Size - Heterogeneous Effects

	Number of Owners (log) (1)	Mean Farm Size (log) (2)	Median Farm Size (log) (3)
	Panel A: Land Registry Map - Contiguous Plots		
$TempShocks$	0.012**	-0.014**	-0.032**
	(0.005)	(0.006)	(0.014)
$TempShocks \times High$	-0.004	0.004	0.024*
	(0.005)	(0.006)	(0.013)
Observations	9,499	9,499	9,499
R^2	0.991	0.993	0.973
	Panel B: Distance to major city		
$TempShocks$	0.010*	-0.013**	-0.027*
	(0.006)	(0.006)	(0.015)
$TempShocks \times High$	0.011*	-0.006	0.007
	(0.006)	(0.006)	(0.012)
Observations	9,683	9,683	9,683
R^2	0.991	0.993	0.974

Notes: This table reports the estimated effect of weather shocks on the farm-size distribution, allowing for heterogeneous effects by plot contiguity and distance to a large city. Dependent variables are the log number of land owners in the municipality (column 1), log average farm size (column 2), log median farm size (column 3). The independent variable is the total number of atypical temperature days in the past two years divided by 100. *High* in Panel A denotes a dummy equal to one for municipalities with an above-median share of farms below the 10th percentile of the size distribution that are contiguous to at least one farm above the 90th percentile in the 2017 land-registry map. In Panel B, it denotes a dummy equal to one for municipalities whose driving distance to the nearest major city is above the national median. All regressions include municipality and year fixed effects, as well as cumulative rainfall and its five lags, the accumulated number of land allocations, the area covered by the land registry, and an indicator for cadastral registry updates. Standard errors clustered at the municipality level reported in parenthesis. Data from the National Land Registry (*Catastro Nacional*), maintained by the National Geographical Institute (IGAC). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

mechanisms after rural households are exposed to income shocks. Roughly one third of respondents report that farmers sell land after experiencing any type of income-reducing shock, and 35% of respondents report that farmers sell land after experiencing weather shocks in particular. Third, the majority of respondents report that land buyers are not from within their communities. Specifically, 49% report that land is not bought by farmers from their community in normal times; in distress times, this number rises to 65 percent, which is consistent with negative weather shocks attracting households who are not affected by the shock.

In line with the results from this qualitative survey, in the model developed below, households exposed to weather shocks must sell land to smooth consumption, drawing in outside buyers unaffected by the shock. We refer to these buyers as workers for simplicity, though we interpret them more broadly—they may, for instance, be farmers from nearby municipalities outside the shock's reach. Crucially, our results do not depend on the precise origin of entrants, only on the fact that they are less affected by the shock.

5 A Quantitative Model of Farm-Size Dynamics

This section develops a dynamic, quantitative model of the farm size distribution to address three questions. First, how much do the consumption-smoothing motives implied by our reduced-form results affect the efficiency of the farm size distribution? Second, under what economic conditions do weather shocks generate land fragmentation rather than consolidation? Third, what are the aggregate productivity consequences of a changing weather regime? In developing our model, we strive for parsimony: we incorporate the essential structure needed to speak to the empirical patterns presented in previous sections, while abstracting from others. We lay out the model in Sections 5.1 to 5.3. Section 5.4 then describes our calibration and Section 5.6 examines how much risk distorts the farm size distribution. Section 6 turns to our counterfactual analyses.

5.1 The Model

The economy represents a municipality that operates in discrete time, indexed by t. Each period corresponds to one year and includes a single harvest season. The economy is endowed with a fixed supply of land L and a mass of heterogeneous agents N. There are two sectors: agriculture (a) and non-agriculture (n). Aggregate productivity in agriculture is denoted by Z_t^a, which will later be subject to weather shocks in our simulations. Wages in the non-agricultural sector, w_t, are exogenously given.

Agents. Agents live forever and their preferences are described by their long-life utility $U(c) = \mathbb{E}\left[\sum_0^\infty \beta^t u(c_t)\right]$, where c_t is consumption and $u(c_t)$ is their utility function. They are heterogeneous in their land ownership, ℓ_t, their idiosyncratic agricultural productivity, s_t^a, and their idiosyncratic non-agriculture productivity, s_t^n. We characterize each agent by a triple $\phi_t = (s_t^a, s_t^n, \ell_t)$.[31] Agents with positive land holding ($\ell_t > 0$) are referred to as farmers and earn farm income at the end of each season, while those without land ($\ell_t = 0$) are workers whose income are derived from wages in the non-agricultural sector.

In each period, agents face an exogenous discontinuation shock: with probability δ, they are forced to sell all the land they own and transition to work in the non-agricultural sector in period $t+1$. If they are not hit by this shock, they make occupational and land-purchasing

[31] For parsimony, we abstract from asset holdings other than land (e.g., a risk-free asset), keeping the state space minimal. Despite this simplification, the model performs well in predicting the reduced-form relationship between farm size and weather shocks—see Figure 5.

decisions in two stages. In the first stage, agents choose whether to be a farmer or a worker in period $t+1$—a *discrete choice* between $\ell_{t+1} > 0$ and $\ell_{t+1} = 0$. This decision depends on the value of becoming a farmer, $V^a(\phi_t)$, or a worker, $V^n(\phi_t)$, in $t+1$ given ϕ_t, plus idiosyncratic preference shocks for each occupation, ν_t^a and ν_t^n. In the second stage, conditional on choosing to farm, agents decide how much land to acquire ahead of the harvest season in $t+1$—a *continuous choice* over $\ell_{t+1} > 0$. At this point, they choose consumption and land purchases to maximize current utility, $u(c_t)$, plus the expected value of owning ℓ_{t+1} in $t+1$, and entering the next period with state ϕ_{t+1}, $V(\phi_{t+1})$, subject to their budget constraints.

Production. Agents with land earn income from farming, while those without land earn income from working in the non-agricultural sector. The earnings function is given by:

$$\pi(\phi_t) = \begin{cases} Z_t^a \left(s_t^a\right)^{1-\gamma} \left(\ell_t\right)^{\gamma} & \text{if } \ell_t > 0 \\ w_t^n s_t^n & \text{if } \ell_t = 0 \end{cases} \tag{8}$$

where γ is the share of land in production.[32]

Distributional Assumptions. For agents who own land ($\ell_t > 0$), agricultural productivity s_t^a follows a AR(1) stochastic process, $\log s_t^a = \rho \log s_{t-1}^a + \sigma^a \epsilon_t^a$, where $\epsilon_t^a \sim \mathcal{N}(0,1)$ is an innovation shock, $\rho \in (0,1)$ is a persistence parameter, and $\sigma^a > 0$ is the volatility of ϵ_t^a. We denote by $g^a\left(s_{t+1}^a | s_t^a\right)$ the transition probability that results from this stochastic process. Agents without land ($\ell_t = 0$) draw their idiosyncratic agricultural productivity from the stationary distribution of s_t^a, denoted by $\tilde{g}^a(s_t^a)$. Every agent also draws a non-agriculture skill from the distribution $\tilde{g}^n(s_t^n)$. Finally, preference shocks for each sector, ν_t^a and ν_t^n, are drawn independently from an extreme value type I distribution (EVT1) with an inverse-dispersion parameter κ and a mean normalized to zero.

Timing. The sequence of events within each period t unfolds as follows. First, agents observe their productivities, s_t^a and s_t^n, their preference shocks, ν_t^a and ν_t^n, and a fraction δ of them are hit by the discontinuation shock. Second, conditional on these realizations, agents choose their occupation for the next period $t+1$. Third, agents earn income, consume, and buy or sell land. Finally, they transition to period $t+1$.

[32] Appendix OC.1 shows how variable labor and capital can be incorporated.

Optimal Choices. Agents' optimization problem can be written in recursive form.[33] At the start of each period, they choose their occupation based on the value function:

$$V(\phi_t) = \mathbb{E}\big[\,(1-\delta)\max\{V^a(\phi_t) + \nu_t^a, V^n(\phi_t) + \nu_t^n\} + \delta V^n(\phi_t)\,\big] \tag{9}$$

where the value of choosing to be a farmer is

$$\begin{aligned} V^a(\phi_t) &= \max_{c_t,\,\ell_{t+1}>0} \{u(c_t) + \beta\,\mathbb{E}\,[V(\phi_{t+1}) \mid \phi_t]\} \\ \text{s.t.} \quad c_t &= \pi(\phi_t) + p_t(\ell_t - \ell_{t+1}) \end{aligned} \tag{10}$$

and the value of becoming a worker is

$$\begin{aligned} V^n(\phi_t) &= u(c_t) + \beta\,V_0(\phi_t) \\ \text{s.t.} \quad c_t &= \pi(\phi_t) + p_t\,\ell_t \end{aligned} \tag{11}$$

We denote by $\ell^*(\phi_t)$ the optimal land purchase choice for ℓ_{t+1}. The term $V_0(\phi_t)$ is the exogenous value of becoming a worker, defined as $V_0(\phi_t) = V_0 + \tau^a$ if $\ell_t > 0$ and $V_0(\phi_t) = V_0 + \tau^n$ if $\ell_t = 0$. The outside option V^0 relates to the employment opportunities unaffected by local weather shocks—such as the local urban center or work in another municipality. Sectoral switching costs, denoted by $\{\tau^a, \tau^n\}$, capture in a reduced-form way agents' access to this option: for example, τ^a might capture search frictions that limit agents' access to jobs elsewhere, and τ^n might capture land market frictions that prevent workers from buying a plot of land. We take a broad interpretation of this outside option—it may also capture, for example, migration to the agricultural sector in another region. In what follows, we define $V_0^a = V_0 + \tau^a$ and $V_0^n = V_0 + \tau^n$.[34]

Law of Motion for the Type Distribution. Let $g(\phi_t)$ denote the density of agents of type ϕ_t at time t. We now describe how this density evolves into $g(\phi_{t+1})$. In period t, the mass of agents of type ϕ_t who choose to become farmers in period $t+1$ is

$$h^a(\phi_t) = \mu_t(\phi_t)\,g(\phi_t) \tag{12}$$

[33] As we focus on the stationary equilibrium, we suppress the dependence of the value function on the farm size distribution.

[34] We assume that the probability of returning to local agriculture after exiting is sufficiently low that the value of the outside option is unaffected by changes in the local value of farming.

where $\mu_t(\phi_t)$ is the probability that an agent of type ϕ_t becomes a farmer in $t+1$. Conversely, the mass of agents who become a worker in $t+1$ is

$$h^n(\phi_t) = [1 - \mu_t(\phi_t)]g(\phi_t). \tag{13}$$

Given the properties of the EVT1 distribution and the discontinuation probability δ, the probability that an agent of type ϕ_t becomes a farmer in period $t+1$ can be written as

$$\mu_t(\phi_t) = (1-\delta)\frac{\exp(\kappa V^a(\phi_t))}{\exp(\kappa V^a(\phi_t)) + \exp(\kappa V^n(\phi_t))} \tag{14}$$

Using equation (12), the density of agents who become farmers in period $t+1$ is

$$g(\phi_{t+1}|\ell_{t+1} > 0) = \int_{\phi_t} \Delta(\ell^*(\phi_t))\, g\left(s^a_{t+1}|s^a_t\right) \tilde{g}^n\left(s^n_{t+1}\right) h^a(\phi_t)\, d\phi_t, \tag{15}$$

where $\Delta(\ell^*(\phi_t))$ is the Dirac delta.[35] Similarly, using equation (13), the density of agents who become workers in period $t+1$ is

$$g(\phi_{t+1}|\ell_{t+1} = 0) = \int_{\phi_t} \tilde{g}^a\left(s^a_{t+1}\right) \tilde{g}^n\left(s^n_{t+1}\right) h^n(\phi_t)\, d\phi_t. \tag{16}$$

Resource Constraints and Market Clearing. To define the equilibrium, we specify the resource constraints. First, the land market must clear in every period t: aggregate land supply L must equal aggregate land demand,

$$L = \int \ell^*(\phi_t)\, h^a(\phi_t)\, d\phi_t. \tag{17}$$

Second, occupational choices must be consistent with the evolution of the population measure. The total mass of agents N is conserved across periods,

$$N = \int_{\phi_t} [h^a(\phi_t) + h^n(\phi_t)]\, d\phi_t, \tag{18}$$

which simply restates the conservation of agents implied by the law of motion for $g(\phi_t)$.

[35] In practice, in the computational implementation, which is a discretization approximation, this becomes equal to an indicator function that equals 1 when $\ell_{t+1} = \ell^*(\phi_t)$ and 0 otherwise.

5.2 Stationary Equilibrium

Given distributions $\{g^a, \tilde{g}^a, \tilde{g}^n\}$, a mass of agents $\{N\}$, a land endowment $\{L\}$, a production parameter $\{\gamma\}$, a discontinuation probability $\{\delta\}$, a dispersion parameter for preference shocks $\{\kappa\}$, aggregate agricultural productivity $\{Z^a\}$, a non-agricultural wage $\{w^n\}$, an outside option $\{V^0\}$, and sectoral switching costs $\{\tau^a, \tau^n\}$, a stationary competitive equilibrium consists of optimal land and consumption choices $\{\ell^*(\phi),\ c^*(\phi)\}$, a land price $\{p^*\}$, and an invariant distribution $\{g^*(\phi)\}$ such that: land and consumption choices are optimal and satisfy (9) to (11); optimal occupational choices and the discontinuation probability satisfy (14); the stationary distribution is consistent with (12), (13), (15) and (16); and the resource constraints in (17) and (18) hold.

5.3 Stochastic Economy

We now characterize the evolution of an economy subject to a sequence of weather shocks, following the approach of Boppart et al. (2018) and Auclert et al. (2021). To this end, we first parametrize how weather shocks ω_t influence aggregate agricultural productivity:

$$Z^a(\omega_t) = \bar{Z}\frac{2}{1+\exp(\xi\omega_t)}. \tag{19}$$

Here, ξ governs the sensitivity of agricultural productivity to weather shocks, and $\bar{Z}$ is a scale parameter. We assume that ω_t follows an autoregressive AR(1) stochastic process, $\omega_t = \rho^\omega \omega_{t-1} + \sigma^\omega e_t^\omega$, where e_t^ω is distributed normal $\mathcal{N}(0,1)$. This distributional assumption allows us to replicate the actual distribution of weather shocks in the data. We also note that our parametric form implies $Z^a(\omega_t = 0) = \bar{Z}$. For simplicity, we assume that the non-agriculture sector is not affected by weather shocks, so the wage w_t^n remains fixed. This assumption is consistent with evidence showing that agriculture is more sensitive to climate variation than non-agricultural sectors—see, for example, Nath (2025).

To simulate the evolution of an endogenous variable y_t in an economy exposed to a sequence of weather shocks, we compute

$$dy_t = \sum_s \widehat{dy}_s\, \omega_{t-s}, \tag{20}$$

where dy_t denotes the change in y_t, $\widehat{dy}_s$ is the model-implied response of y_t in period s to one-period perturbation of ω_0 that does not persist into subsequent periods (i.e. an MIT shock at $t = 0$), and ω_{t-s} is the realized weather shock in period $t - s$.[36] The change in y_t

[36] We compute the fully non-linear impulse-response to a one-standard deviation shock in ω_t using the

at time t is thus obtained by summing the effects of all past shocks, each weighted by the corresponding dynamic response of the economy estimated from the perturbation. In other words, dy_t reflects the accumulated effects of the entire history of shocks, with each shock contributing according to the strength of the economy's response s periods after it occurs. For example, the total change in the number of farms in period T combines the one-period response to the shock in $T-1$, the two-period response to the shock in $T-2$, and so forth.

5.4 Calibration

To simulate the model, we need to assign values to the following parameters: $\{\beta,\, \kappa,\, V_0^a,\, V_0^n,\, N,\, L, \gamma,\, \rho,\, \sigma,\, w^n, \mu^n,\, \sigma^n,\, \delta,\, \bar{Z}^a, \xi,\, \rho^\omega,\, \sigma^\omega\}$. To do so, we calibrate the stationary equilibrium of the model and its transition dynamics to different data moments. Table 5 summarizes the resulting parameter values.[37]

We start by setting the values of a few parameters before solving the model. For preferences, we choose a discount factor of $\beta = 0.96$, following Greenwood et al. (2019), who estimate this parameter in a developing-country context, and we assume log utility. For the production function, we set the share of land to $\gamma = 0.28$, consistent with the estimates of Avila and Evenson (2010) for Colombia. We estimate the persistence of idiosyncratic agricultural productivity, ρ, using farm-level longitudinal data from Colombia (the same data we use in Section 4.3), by computing the autocorrelation of model-implied measures of farm productivity over three years.[38] The weather-shock parameters ρ^ω and σ^ω—the persistence of weather shocks and the variance of their innovations—are estimated directly from our weather-shock data. We set $\delta = 0.02$, based on the exit rate of large farms with properties above the 90th percentile (approximately 8 hectares).[39] Finally, without loss of generality, we normalize $\bar{Z} = 1$, the land endowment to $L^S = 1$, and the wage to $w_t^n = 1$.

For the remaining parameters $\{\kappa,\, V_0^a,\, V_0^n,\, N,\, \sigma, \mu^n,\, \sigma^n\}$, we search for the values that make several model-generated statistics related to the farm-size distribution and the farmer's exit probabilities match their counterparts in the data. First, we adjust κ to match the exit probability of medium-sized farms (between 0.1 and 8 hectares) and V_0^a to match the exit

sequence-space Jacobian method developed in Auclert et al. (2021). The impulse response function relies on certainty equivalence: agents behave as if uncertain future aggregate variables were equal to their expected values and therefore do not internalize higher moments of the aggregate shock, such as its variance. See Reiter (2018) for a discussion of the advantages and limitations of this approach relative to alternative computational methods. Appendix OC provides further details on the implementation.

[37] Appendix OD provides details on the calibration and Appendix Section OD.1 discusses how we solve for the stationary equilibrim.

[38] Appendix OD.2 describes in more detail how we estimate the persistence of idiosyncratic productivity.

[39] See Appendix OD.3 for details on the estimation of exit rates.

Table 5: Summary of Calibration

Parameter	Description	Target/Source	Value
Panel (a): Preferences			
β	Discount rate	Greenwood et al. (2019)	0.96
κ	Dispersion of preference shocks	Exit of medium farms	8.5
V_0^a	Outside option for farmers	Exit of small farms	-97
V_0^n	Outside option for workers	Share of farmers	-12
Panel (b): Technology and Endowments			
N	Total mass of agents	Avg. farm size	0.54
L	Total land supply	Normalized	1
γ	Share of land / Span-of-control	Avila and Evenson (2010)	0.28
Panel (c): Idiosyncratic shocks			
ρ	AR(1) coefficient of s_t^a	Estimated from farm-level data	0.75
σ	Variance of shock to s_t^a	Variance of farm-size distribution	2.6
w_t	Wages	Normalization	1
μ^n	Average of draws of s_t^n	Average farm size of entrants	0.51
σ^n	Variance of draws of s_t^n	Income dispersion of workers	0.53
δ	Exogenous exit probability	Exit rate of large farms	0.02
Panel (d): Aggregate Productivity			
$\bar{Z}$	Agricultural TFP shifter	Normalized	1
ξ	Impact of ω_t on TFP	Reduced-form impact on consumption	0.61
ρ^ω	AR(1) coefficient of ω_t	Estimated from temperature shocks	0.41
σ^ω	Standard error of shock to ω_t	Estimated from temperature shocks	0.2

probability of very small farms (smaller than 0.1 hectares).[40] Second, we calibrate the total mass of agents, N, to match the average farm size, and we choose the variance of the innovation to idiosyncratic agricultural productivity, σ, to match the variance of farm size in the 2014 National Agricultural Census. We calibrate μ^n to match the average size of new farms, and set σ^n directly from the coefficient of variation of workers income, obtained from the longitudinal household survey. Finally, we calibrate V_0^n to match the average rural share of the population across *departamentos* in Colombia, which equals 0.37.

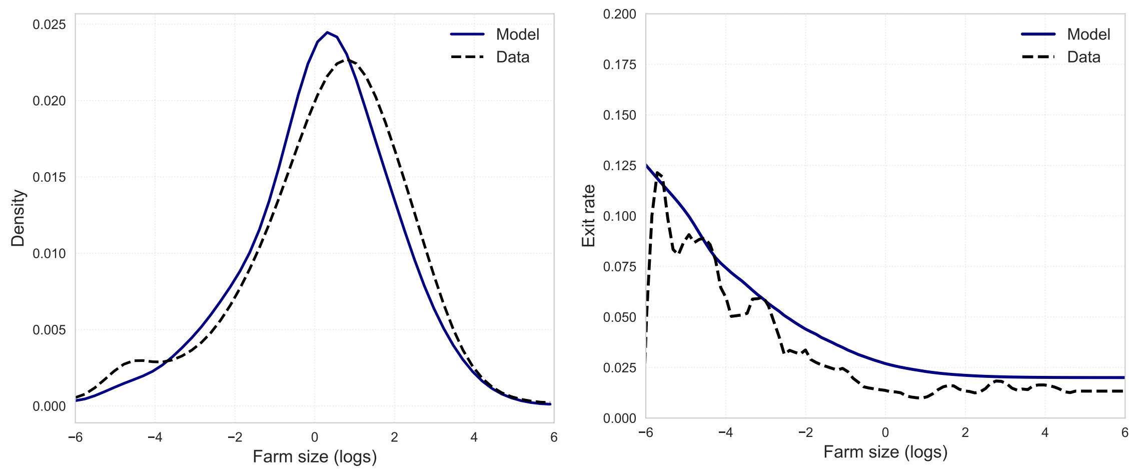
After calibrating the model in its stationary equilibrium, we choose the value of ξ that makes the simulated evolution of the stochastic economy reproduce the reduced-form regression results of log of consumption on weather shocks estimated in the data. Specifically, we target a value of -0.11 for small farmers in the model, reported in column (5) of Table 3.[41]

[40]The parameter κ is chosen to match the exit rate of medium-sized farms. Intuitively, it controls how sharply exit probabilities respond to changes in the relative value of farming versus exiting as farm size changes. If κ is too small, exit probabilities fall too quickly with farm size, and the model understates exit among medium-sized farms. See Panel (b) of Figure 4.

[41]We also test if results differ if we simulate several regions in our model and run the exact same specification as our reduced-form results, in which we add region and time fixed effects. Since the shock is already exogenous in the model, our calibration results are virtually the same. See Appendix Figure O.11

Figure 4: Farm Size Distribution and Exit Rate of Farmers (Model vs. Data)

(a) Farm Size Distribution (b) Exit Probability



Notes: This figure shows the farm-size distribution generated by the model and the distribution observed in the data. It also reports the exit rate by farm size in the model and in the data. Appendix OD explains how we estimate annualized exit rates across the farm-size distribution in the data. The source of the farm-size data is the 2014 Agricultural Census.

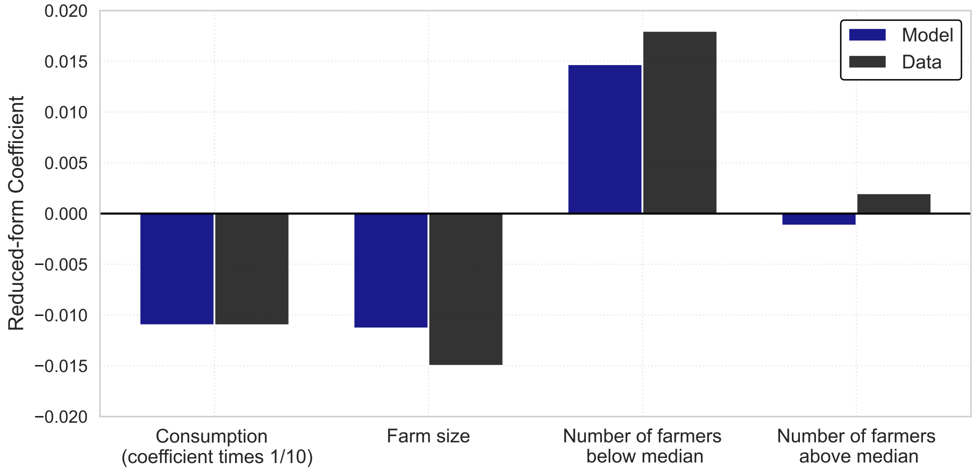
5.5 Fit of the Model

Focusing on the stationary equilibrium, Panel (a) of Figure 4 shows that the model generates a farm-size distribution that closely tracks the empirical one. Panel (b) shows that the model also matches farmer's exit rates across different parts of the farm-size distribution.

Turning to the stochastic economy, Figure 5 reports the reduced-form effects of weather shocks on key variables of interest. The first two bars show that, given our calibration of ξ, the model exactly replicates the observed decline in consumption in the data. Reassuringly, beyond this targeted moment, the model also reproduces several key patterns consistent with the mechanisms emphasized in our analysis. It broadly matches the observed changes in average farm-size and captures the fact that most of this adjustment is driven by smaller farmers (those below the median).

In brief, our model can quantitatively reproduce key features of the dynamics of the farm-size distribution observed in the data. First, it matches both the empirical farm-size distribution and the exit rates across farms of different sizes. Second, it captures how consumption and the farm-size distribution respond to weather shocks. Given the model's ability to replicate these patterns, we use it in Section 6 to simulate alternative shocks, better understand the underlying mechanisms, and assess the effects of changes in the distribution of weather shocks, ω_t. Before turning to those exercises, we use the model to ask what consumption-smoothing motives imply for the efficiency of the farm-size distribution.

Figure 5: Reduced-Form Impact of Weather Shocks (Model vs. Data)



Notes: This figure shows the impact of weather shocks in the Colombian data and the impact generated by simulating the stochastic economy in the model. We divide the consumption coefficient by 10 for ease of visualization. Our calibration sets ξ to match the reduced-form effect of weather shocks on consumption estimated from the Colombian longitudinal data. The remaining coefficients are not directly targeted in the calibration.

5.6 Risk and Agricultural Productivity

In an efficient allocation, the marginal product of land would be equalized across farmers. In our model, however, two forces prevent this equalization. First, farmers choose how much land to purchase before observing their realized farm productivity, s_t^a. This timing creates a mismatch between the period of land choice and the period in which productivity is realized—a constraint that a central planner allocating land prior to the realization of idiosyncratic shocks would also face. Second, because farmers rely on land sales to smooth consumption, exposure to risk distorts their land holdings. A central planner able to fully insure farmers would eliminate this channel and its associated distortion.

Specifically, the marginal product of land should be equalized across farmers in an efficient economy, so that farm size is proportional to farm productivity, s_t^a. The amount of land operated by a farmer with productivity s_t^a would be $\ell(s_t^a) = \dfrac{s_t^a}{\int s_t^a dG(\phi_t|\ell_t > 0)} L$. Panel (a) of Appendix Figure O.12 compares the optimal and the model's farm-size distributions, holding fixed the number of farmers. In our model, small farms are larger than under the efficient allocation because small farmers accumulate land for precautionary motives. Larger farmers, by contrast, operate less land than optimal because the higher precautionary demand for land among smaller farmers crowds them out. Quantitatively, the standard deviation of the farm-size distribution predicted by the model is about 20 percent lower than that of the optimal distribution. Panel (b) shows that more productive farmers have a higher marginal product of land in equilibrium. Thus, the efficient allocation would reallocate land away from less productive toward more productive farmers and increase the proportion of large farms. In our calibration, implementing the efficient allocation would increase aggregate output by 42 percent.

6 Counterfactual Analysis of Weather Shocks

With the model calibrated, we conduct two types of counterfactual exercises. First, we examine the economy's response to a single weather shock by evaluating the impulse-response functions. This exercise serves two purposes: to clarify the economic mechanisms that can rationalize the reduced-form results presented in Sections 2 and 4, and to study the dynamic implications of weather shocks and assess how persistent their effects are over time.

Second, we study the economy's response to different sequences of weather shocks, drawing on our definition of a stochastic economy in Section 5.3. We consider alternative shock distributions that feature a small increase in the probability of extreme temperatures. We examine how these small changes affect the farm-size distribution and aggregate agricultural productivity.

To solve for the economy's response to a sequence of shocks, we follow Auclert et al. (2021) and Boppart et al. (2018), which captures local household responses around the steady state.[42] This approach is well-suited in our case for three reasons. First, we evaluate the impact of a change in the number of shock realizations, not in the risk of the shock itself, which the method would not capture. Second, we restrict attention to small changes in the probability of extreme temperatures, consistent with the local linearization. Third, the method is most reliable when aggregate TFP fluctuations are small relative to cross-sectional dispersion in idiosyncratic productivity—so that precautionary saving is governed primarily by idiosyncratic risk (Reiter, 2018). Both conditions hold here: aggregate TFP fluctuations are roughly 60 times smaller than the cross-sectional dispersion in idiosyncratic productivity.[43] In the end of the section, we verify the robustness of our results by allowing farmers to fully anticipate the average TFP change, and by inflating idiosyncratic shocks to reflect the increase in aggregate variance of the shock.

[42] A global solution method would face a severe curse of dimensionality: under rational expectations, agents must forecast the entire cross-sectional distribution, which becomes a state variable in the Bellman equation. The Krusell and Smith (1998) approach mitigates this using low-dimensional moments, but remains costly here because land prices are determined by a non-trivial market-clearing condition (equation 17)—see Yang et al. (2025).

[43] The derivative of aggregate log-TFP with respect to ω_t is $\xi/2 = 0.305$. The stationary standard deviation of ω_t is $\sigma_\omega/\sqrt{1-\rho_\omega^2} \approx 0.22$, implying a standard deviation of aggregate log-TFP of $0.305 \times 0.22 \approx 0.067$, against an idiosyncratic standard deviation of $\sigma/\sqrt{1-\rho^2} \approx 3.93$.

6.1 Weather Shocks and Transition Dynamics

We simulate an economy that experiences a temporary, unexpected shock that increases the weather-shock variable ω_t by one standard-deviation, with an autocorrelation of 0.4—consistent with our estimates from the data. As shown in Panel (a) of Figure 6, this shock generates a drop in agricultural TFP of about 6 percent at $t = 0$.[44] The impact on TFP is short-lived, however: productivity returns to its average level in less than ten years.

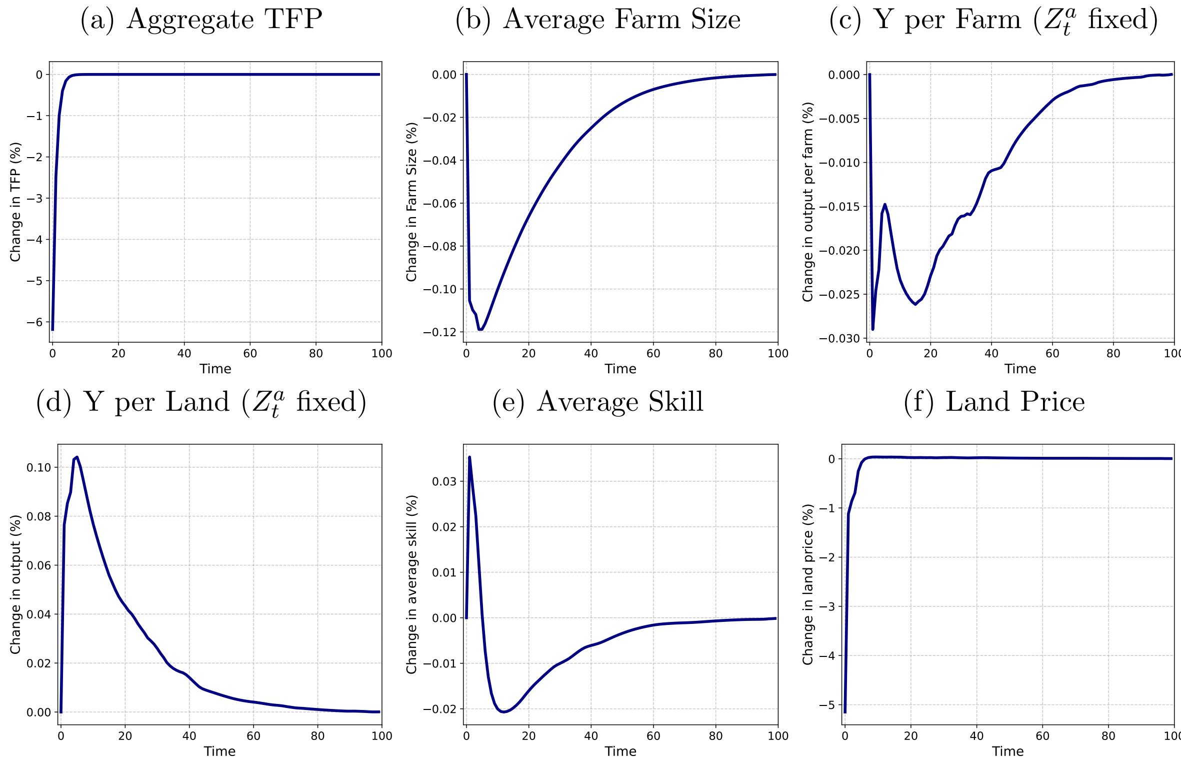
Effects on farm size and agricultural productivity. In the aftermath of the shock, the average farm size declines by about 0.12 percent, as shown in Panel (a) of Figure 6. This contraction is driven by the entry of new, smaller farmers into agriculture. Although aggregate TFP recovers within a few years, the adjustment in farm size is much slower: even twenty years after the shock, the average farm size remains roughly 0.06 percent below its pre-shock level. The increase in the number of farms leads to a decline in output per farm, above and beyond the direct negative effect of the weather shock on aggregate TFP. Panel (c) of Figure 6 illustrates this by showing the reduction in agricultural output per farm net of TFP changes—that is, isolating the economy's endogenous response. Output per farm falls by about 0.02 percent, and, once again, the recovery is slow.

Panel (d) shows that, despite the decline in output per farm, output per unit of land actually rises once we ignore the direct impact on the TFP. This pattern reflects the entry of additional farmers into agriculture, which raises the marginal product of land while lowering the marginal product of labor. The result is consistent with the common finding that regions with a high concentration of small farms exhibit higher land yields but lower output per worker. Panel (e) shows that, after the initial exit of small and low-productivity farms, the continued inflow of new low-skill entrants reduces average agricultural skill. This occurs because entrants draw their skills randomly from the stationary distribution, whereas incumbent farmers in the stationary equilibrium self-select into agriculture based on their comparative advantage. As a result, incumbents are positively selected on skill, which raises average productivity, while the entry of new farmers dilutes the effect of this selection on average skill.

Effects on land prices, consumption, and exit rate. Panel (f) of Figure 6 shows that land prices fall by about 5 percent at $t = 0$ when agents are hit by the shock. This decline occurs because the negative shock leads small farmers to sell land in order to smooth consumption—see Panel (a) of Figure 7. Larger farmers, by contrast, adjust their landholdings only min-

[44] To clarify what leads to this 6 percent, notice that, given our estimate of $\xi = 0.61$ and $\omega^{\xi} = 0.2$, we obtain $0.936 = 2/(1 + \exp(0.6 \times 0.2))$ from equation (19).

Figure 6: Transition Dynamics in response to a Weather Shock



Notes: This figure shows the impulse-response functions of the economy following a one-standard deviation increase in weather shocks. Panel (a) shows the impact on the aggregate TFP term, Z_t^a. Panel (b) shows the transition dynamics of average farm size, Panel (c) the agricultural output per farm, Panel (d) the agricultural output per unit of land, Panel (e) the average skill of farmers at the start of the period, and Panel (f) the land price.

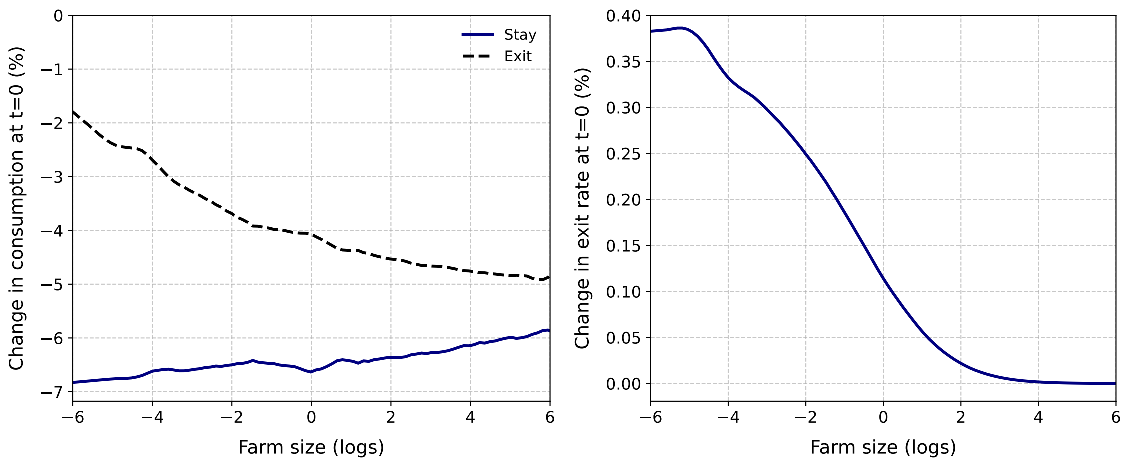
imally. At higher consumption levels, the marginal utility loss from a negative shock is smaller under concave utility, reducing their incentive to liquidate productive assets. In this sense, wealthier farmers are partially self-insured against shocks.

The change in consumption shown in Panel (a) of Figure 7 highlights the trade-off farmers face when confronted with the shock. Among smaller farms, those who remain in agriculture experience a sharp decline in consumption—between 6 and 7 percent. This reduction is particularly severe given their lower baseline utility, which implies a higher marginal utility of consumption. In contrast, small farmers who choose to sell all their assets and exit agriculture face a more modest decline in consumption of about 2—4 percent. The consumption response among larger farms is much smaller. Consequently, the exit probability—shown in Panel (b)—is substantially higher for smaller farms.

The role of productivity and land prices on exit and entry decisions. Appendix OE provides analytical comparative statics that help characterize the mechanisms shaping farmers' exit and workers' entry decisions. These results separate the effect of a decline in aggregate

Figure 7: Exit Rate and Consumption Responses by Farm Size at $t = 0$

(a) Consumption (b) Exit Probability



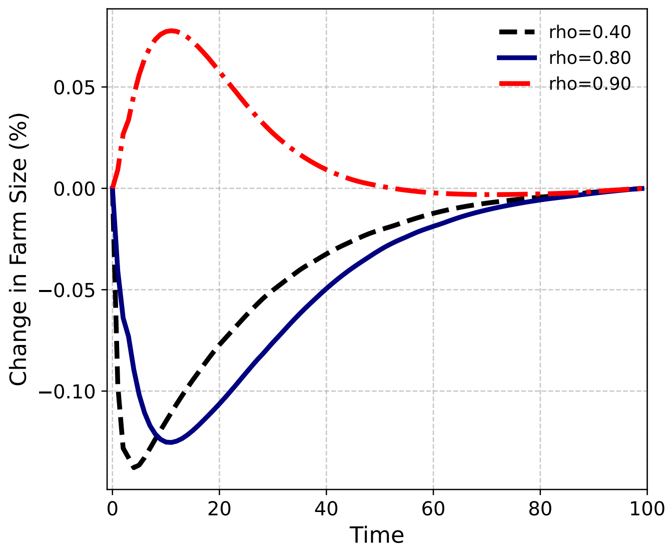
Notes: Panel (a) shows the consumption response to the shock at $t = 0$ by log farm size, separately for farmers who exit agriculture (dashed black line) and those who remain (blue line). Panel (b) shows the corresponding change in the exit rate.

productivity Z_t from the effect of a decline in the land price p_t. Consider first a transitory reduction in aggregate productivity Z_t, holding fixed the current land price p_t and the future value of farming relative to working. The effect on farmers' exit is unambiguous: lower current consumption, resulting from the decline in productivity, makes selling land more attractive by increasing the marginal utility of the additional consumption obtained from land sales. Intuitively, farmers move closer to subsistence levels of consumption, which makes them more willing to exit agriculture in exchange for consumption smoothing. Workers' entry decisions, in turn, are unaffected by the productivity decline, since they do not own land and the value of entering farming remains fixed.

Consider next a reduction in the land price p_t, holding fixed aggregate productivity Z_t and the future value of farming relative to working. Two opposing effects shape farmer' exit decisions. On the one hand, a wealth mechanism operates: farmers become poorer as the value of their land falls, which raises the marginal value of consumption and augments the to exit agriculture in order to smooth consumption. On the other hand, a cost mechanism operates: lower land prices make farming more attractive by reducing the cost of holding land. For workers, a reduction on land prices unambiguously increases the incentives to enter agriculture, because they do not suffer the asset-value loss associated with lower land prices, only the cost mechanism operates for them.

Putting these mechanisms together, the decline in agricultural productivity induces farmers to sell their land to smooth consumption, triggering a fall in land prices. This generates two opposing forces: a wealth effect that augments farmers' incentives to exit agriculture,

Figure 8: Farm Size response to Shocks with Different Persistence Structure



Notes: This figure shows the farm-size response to a one-time, one-standard deviation increase in the weather shock w_t at $t = 0$ under different persistence assumptions. We consider three shocks, with autocorrelations of 0.4, 0.8 and, 0.9, as indicated in the figure.

and a cost effect that attenuates them. Non-agricultural households are unaffected by the productivity decline itself, but the drop in land prices creates an incentive for them to enter.

The role of persistence. Our results indicate that when land prices fall, landless agents are more likely to enter agriculture. Whether this entry dominates exit by incumbent farmers depends critically on the persistence of the shock. Figure 8 illustrates this pattern by comparing how the same shock affects farm size under different autocorrelation assumptions. Under a transitory shock, we find a net inflow into agriculture and a reduction in average farm size; under a more persistent shock, the opposite occurs.

The comparative statics above help interpret this result. When productivity falls, farmers reduce their landholdings or exit altogether, putting downward pressure on land prices. If the shock is transitory, agricultural productivity returns quickly to its steady-state level, so the value of becoming a farmer for landless households is only mildly affected. The decline in land prices—without an equivalent decline in the present value of farming—therefore attracts landless agents into agriculture. In contrast, when the shock is highly persistent, the value of farming falls more substantially. The decline in land prices is then insufficient to attract workers, while incumbent farmers have even stronger incentives to exit. Appendix Figure O.13 shows these effects on workers' entry and farmers' exit for alternative values of shock persistence.

6.2 Climate Change and Agricultural Productivity

To close our analysis, we simulate a stochastic economy under alternative distributions of weather shocks. We begin by modeling the stochastic process that generates weather shocks in the data. We divide each year into 365 days and assume that the number atypical days in a year is the outcome of a sequence of Bernoulli trials, where each day is atypical with probability 0.4. To match the autocorrelation observed in the data, we allow these Bernoulli trials to be correlated over time, following a Markov process in which the economy transitions between typical and atypical days.[45] We calibrate the persistence of this stochastic process so that the standard deviation of the annual sum of atypical days matches the standard deviation of the weather-shock series in the data.[46] This gives us a sequence of weather shocks that we use as our baseline scenario.

To produce alternative scenarios consistent with climate change, we increase the probability of an atypical day in the underlying stochastic process from 0.4 to 0.42 in the "low-increase" scenario and to 0.44 in the "high-increase" scenario. The resulting distributions are shown in Panel (a) of Figure 9. Relative to the baseline, the mean of the weather-shock series increases by 0.2 standard-deviation units in the low-increase scenario and by 0.35 standard-deviation units in the high-increase scenario. Through the mapping from weather shocks to agricultural TFP in equation (19), these shifts imply average declines in agricultural TFP of 6 percent and 10 percent, respectively.

Panel (b) of Figure 9 presents the effects of alternative weather regimes on agricultural output and average farm size. Starting from the stationary equilibrium, we simulate the full sequence of weather shocks under each scenario and use equation (20) to compute, in each period, the deviation of each variable from the baseline scenario. We then report the average of these period-by-period deviations over the simulated path. In the low-increase scenario, agricultural output is about 3 percent lower on average.

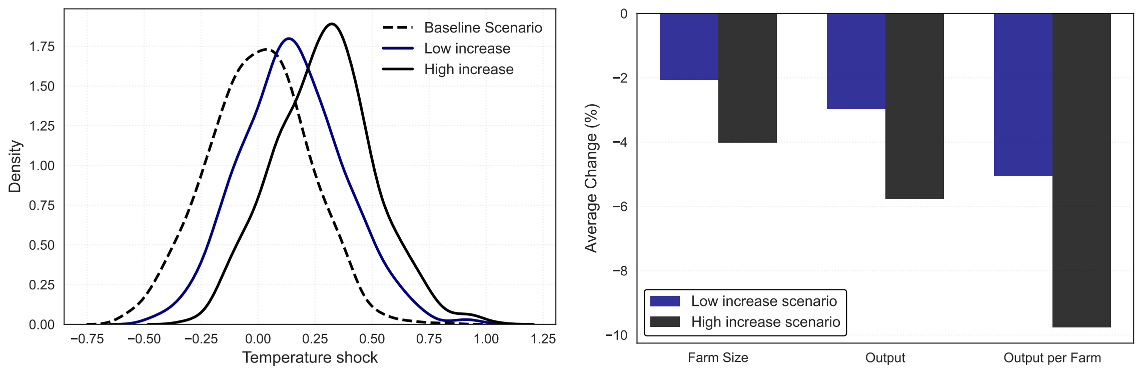
Because average farm size also falls—by roughly 2 percent—the reduction in output per farm is amplified to around 5.5 percent. These effects are quantitatively large. To put these numbers in perspective, the elasticity of output per worker with respect to the probability of an atypical weather day is approximately 1.1 percent ($5.5/5 \approx 1.1$). These results indicate that even modest shifts in the distribution of weather shocks can generate substantial aggregate productivity losses in agriculture through the mechanisms highlighted in this paper.

[45]See Appendix OD.4 for details on how we model this correlation.

[46]If atypical days were uncorrelated, we would have a sum of Bernoulli trials, which approximates a normal distribution with mean $n \times p$ and variance $n \times p \times (1 - p)$, where n is the number of trials and p is the probability of success. In the data, however, the empirical distribution of weather shocks exhibits a larger variance because atypical days are correlated over time.

Figure 9: Response to a Change in the Distribution of Weather Shocks

(a) Weather Shock Distribution Scenarios (b) Change Relative to Baseline Scenario



Notes: Panel (a) shows the distributions of weather-shock sequences generated under alternative probabilities of an atypical day within a year. The "Low increase" scenario raises this probability from 0.4 to 0.42, and the "High increase" scenario raises it from 0.4 to 0.44. Panel (b) shows the resulting effects on average farm size, agricultural output, and output per farm. Starting from the stationary equilibrium, we simulate the full sequence of weather shocks under each scenario, compute the period-by-period deviation of each variable from the baseline scenario using equation (20), and report the average deviation over the simulated path.

Finally, we consider the case in which households fully anticipate the average reduction in agricultural TFP implied by climate change. To do so, we compute a new steady state in which agricultural TFP is permanently lower by the average reduction induced by climate change. Because the change is now permanent, farmers exit agriculture and average farm size rises. Output per farmer nonetheless falls, since farmers' exit is not strong enough to fully offset the lower TFP. Starting from this anticipated steady state, we then trace the economy's response to the same sequence of weather shocks as above. We find that the reduction in average farm size induced by the increase in weather shocks still amplifies the drop in output per farmer significantly, by approximately 40 percent—see Appendix Figure O.14.[47]

7 Conclusion

Development economics has long examined both the implications of the farm-size distribution and the effects of uninsured income shocks, but largely in isolation. Yet because land is both a productive asset and a consumption-smoothing device for rural households, how households cope with uninsured shocks shapes who farms, at what scale, and— by influencing the

[47] In a complementary exercise, we also allow farmers to react to the increase in aggregate risk by inflating the variance of the idiosyncratic productivity shock by the corresponding amount, which captures a precautionary savings motive. The results are largely unchanged.

distribution of farm sizes—the aggregate productivity of agriculture. This paper brings these two strands of the literature together and provides evidence on an important, underexplored link between the two.

Using household-level data from a broad range of developing countries, we document substantial churning in land operated over time and heterogeneous exit probabilities across the farm-size distribution. Adverse weather shocks reduce farmers' survival probabilities but, strikingly, sharply increase the share of landless households that begin operating a positive amount of land.

Drawing on rich administrative data from Colombia, we find that weather shocks intensify land-market activity and increase the number of farmers in a region, compressing average farm size and generating a pronounced rise in smallholdings. This result is not obvious ex ante, as adverse shocks could in principle drive land consolidation if larger, relatively less affected farmers absorbed smaller farms in distress. The fact that we find the opposite to be true in the data establishes a new connection between income shocks and the farm-size distribution, which we trace to households' use of land as a consumption-smoothing device in the absence of insurance.

To understand the mechanisms at work and assess the aggregate implications of individuals' choice to enter or exit agriculture, we develop a heterogeneous-agent dynamic model in which households choose both their occupation and the scale of land operation. We calibrate the model to match a series of empirical moments of the Colombian data, including our reduced-form impact of weather shocks. The model endogenously produces a farm size distribution and allows us to study transition dynamics. Because shocks are short-lived, small farmers liquidate assets to smooth consumption while the value of future farming remains largely unchanged. This makes entry into farming more attractive for non-farm households, who crowd out potential farm expansion by incumbent large farmers, whose incomes are also affected by the shock, generating the documented decrease in farm size. Counterfactual simulations reveal sizable productivity losses: a 5 percent increase in the frequency of atypical temperature days reduces aggregate agricultural output by roughly 3 percent and output per worker by approximately 5 percent.

More broadly, our results identify a new mechanism contributing to low agricultural productivity and to the prevalence of smallholder farming across developing countries: uninsured income shocks and the use of land as a consumption-smoothing device. This mechanism is particularly relevant given evidence that farm sizes are shrinking in developing countries while expanding in developed ones (Lowder et al., 2016). The farm size dynamics we document are consistent with key empirical regularities in the literature—notably, low output per worker but high output per unit of land in smallholder-dominated regions. Work such as Foster and Rosenzweig (2022) points to significant productivity gains from reducing the prevalence of small farms; our findings suggest that uninsured shocks are an important driver of that prevalence. A natural direction for future research is to examine how changes in the risk environment itself—through, for example, the provision of agricultural insurance—would reshape the farm size distribution and its consequences for productivity. A second direction could aim at better understanding how farmers' adaptation to changing weather regimes would influence the link between uninsured income shocks and the farm-size distribution.

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Online Appendices for "Farm Size Distribution, Weather Shocks, and Agricultural Productivity"

(Not for Publication)

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OA Data

This section describes in detail all the data sources we use. Appendix Table O.3 provides summary statistics for our main datasets.

OA.1 Household Surveys Data

Household survey data for Colombia comes from ELCA, the *Encuesta Longitudinal Colombiana de la Universidad de los Andes*, a longitudinal household survey with three waves (2010, 2013, and 2016), collected by Universidad de Los Andes. The survey is nationally representative for urban areas and representative at the rural level for four agroecological regions. For the estimations in Section 2, we use the 2010 and 2013 rounds, for comparability with other countries. For the rest of the analysis, we include all survey rounds. More details of ELCA are provided in Appendix OA.6.

Data for Mexico comes from ENNVIH, the *Encuesta Nacional sobre Niveles de Vida de los Hogares* (also known as MxFLS), a nationally representative longitudinal survey conducted in three waves: 2002, 2005, and 2010. We use the 2005 and 2010 rounds in our analyisis. For Peru, we use ENAHO, the *Encuesta Nacional de Hogares sobre Condiciones de Vida y Pobreza*, a continuous, nationally representative household survey conducted annually since 2007. Its rotating panel structure allows households to be tracked over multiple years. While ENAHO interviews panel households every year, for comparability with other surveys we select households that were interviewed three years apart (i.e., at least four consecutive years) between 2007 and 2023.

Finally, the EHCVM (*Enquête Harmonisée sur le Conditions de Vie des Ménages*) are nationally representative household surveys with two rounds of data (2017-2018 and 2021-2022) for Benin, Burkina Faso, Côte D'Ivoire, Guinea Bissau, Mali, Niger, Senegal, and Togo.

OA.2 Land Transaction Data

Recipients of land in Colombia must register their property with the local public notary, and all formal land transactions involving the plot—including mortgages—must be recorded in the national land registry maintained by the *Superintendencia de Notariado y Registro* (SNR, National Superintendence of Notaries)—the government agency that regulates and oversees Colombia's notarial and public-registry system. Our land-transaction dataset contains the complete transaction history of all land plots granted by the government to private individuals between 1900 and 2010. We match the location of the property in the SNR records to the official list of Colombian municipalities provided by DANE, Colombia's National Statistical Agency.[48] Using this information, we construct a balanced annual panel at the municipality level with data on the number of full and

[48]Municipalities are the smallest official administrative division, of which there are 1,123 in total.

partial land sales, mortgages, and government land allocations.

The Colombian government has carried out the free allocation of public idle lands (*baldíos*) to private individuals continuously since the beginning of the twentieth century. These allocations have become the largest and most consequential land reform policy instrument employed by the national government (Albertus, 2015). Formally, a *baldío* allocation is an administrative resolution issued by the national government to transfer state-owned vacant land to a private party. In practice, this allocation process consisted largely two types of programs: frontier-settlement schemes, through which unused public lands are granted to poor smallholders, and initiatives focused on titling state-owned lands that had previously been occupied informally (Ibáñez and Muñoz, 2010).

The bulk of government-owned land allocations began during the period of the U.S. Alliance for Progress with the enactment of the Social Agrarian Reform Act (Law 135) in 1961, which created the land reform agency (INCORA, later renamed as INCODER, and currently the National Land Agency, ANT). During the second half of the twentieth century, land-allocation laws were amended on three occasions (Law 01 of 1968, Law 30 of 1988, and Law 160 of 1994) but the explicit objective of the policy consistently remained the reduction of land inequality and the provision of land to landless farmers (CNMH, 2016). Figure O.3 shows the evolution of *baldíos* allocations since 1901, the vast majority of which occurred between 1960 and 1990. In terms of both the number of beneficiaries and the amount of land allocated, the scale of the policy has been substantial: more than 550,000 land properties have been granted to private individuals in 1,034 of the 1,123 existing municipalities. These properties account for 23 million hectares—more than half of the privately held land in the country (Sánchez and Villaveces, 2016; Arteaga et al., 2017).

Land petitioners undergo an administrative process with the national land agency to determine whether they meet the legal requirements to become beneficiaries. Although these requirements have changed over time, the most important conditions involve owning no other land and having an income below a specified threshold. Under the current legislation, the process formally consists of nine steps, which include placing an announcement of the intended allocation in a local newspaper and conducting a physical inspection of the land to be granted. Although the procedure is officially expected to take 60 days, in practice allocation processes are considerably longer, and some can take years (Gutiérrez Sanín, 2019).

Appendix Figure O.4 shows the evolution of the average and median size of allocated properties since 1960. The overwhelming majority of allocations made during the 1961-2014 period consisted of relatively small land parcels, with a median allocation size across municipalities of 6.6 hectares. Importantly for this paper, Law 160 of 1994 established a ceiling on the amount of government-allocated land that a single individual can own. This limit, defined by the municipality-specific Agricultural Family Unit (UAF), restricts the ability of relatively larger farmers to purchase land that was originally government-owned. In Appendix section OB.4, we show that these land ceilings do not drive our results.

The universe of land allocations made by the government during the 1901–2011 period is registered in the System of Information for Rural Development (SIDER) dataset currently maintained by the ANT. After receiving the property, beneficiaries must register it with the local public notary, and all formal land transactions carried out on the property (including mortgages) are subsequently registered and stored in a dataset maintained by the National Superintendence of Notaries (SNR), the government agency that supervises regional notaries and keeps a record of all real estate market transactions conducted among private parties.[49]

Our main source of data is the transaction history in the SNR for all *baldío* allocations whose beneficiaries registered their property with a notary, thereby completing the administrative process required to obtain a formal property right.[50] We focus primarily on land-purchase transactions, which include either the transfer of an entire property from one individual to another or the subdivision and sale of only a fraction of the original property. We refer to these as *full sales* or *partial sales* respectively. We also study mortgages, as they may serve as an important adjustment margin when households face negative productivity shocks. For each transaction between two parties, we observe the property's location, the transaction date, and the type of transaction.

OA.3 Farm Size Distribution

For more than 50 years, the National Geographical Institute of Colombia (IGAC) has collected and managed the national cadastre. This system records information on the location, size, ownership, and economic purpose of all real estate properties in every Colombian municipality with the exception of the *departamento* of Antioquia, which operates its own independent cadastral system (Ibáñez et al., 2012).

By law, IGAC must conduct comprehensive cadastral updates in every municipality every five years. Between these broad updates, the dataset is updated annually based on new property registrations submitted by landowners, which allows us to measure the annual variation in the farm-size distribution. In practice, however, information is not always updated regularly, and the interval between cadastral updates varies across municipalities.[51] Martinez (2019) shows that the

[49]The history of the transactions carried out over a property, known as the Certificate of Liberty and Tradition (*Certificado de Libertad y Tradición*) is public information and can be consulted —upon payment of a small fee—for any property with a real estate registration number on SNR's website.

[50]Although registration was not automatic and a non-negligible number of beneficiaries did not complete this final step (Faguet et al., 2020), Appendix Figure O.5 shows that allocations and real estate registrations track each other closely over time, suggesting that the vast majority of allocated properties were ultimately registered.

[51]There are currently 80 municipalities in which IGAC has not yet implemented census-level broad updates of the cadastral information system. These municipalities rely instead on a self-reported system (*Catastros Fiscales*), under which landowners register their properties at regional IGAC offices.

timing of IGAC updates is not driven by changes in local economic conditions (e.g., property booms).

In our study, we use municipal-level aggregate information from all farms in IGAC's cadastre that are (i) privately owned, and (ii) classified as having an agricultural economic purpose. This amounts to roughly 40 million hectares of land. We construct a yearly municipal panel with the number of farms, the number of owners, median farm size, and average farm size within size ranges as defined by (Ibáñez et al., 2012). Because data from the land registry is only available for the period 2000-2011, we restrict our analysis to these years. We exclude from our final sample of municipalities (for both for the transaction-level data and the land registry data) large metropolitan areas and municipalities with very few properties registered (i.e. below the 1^{st} percentile). Our final sample consists of 864 municipalities, which together account for 85.3% of the country's rural population.

OA.4 Land Transaction Data - Landless Buyers

The observed increase in the number of landowners per municipality, shown in Figure O.9 and Table 2 indicates that large landowners within the municipality are not expanding their operations by purchasing the small plots sold after the shocks. We cannot rule out, however, the possibility that these plots are being acquired by large landowners from nearby municipalities.

To investigate this possibility, we compile yearly lists of landowners at the *departamento* level using the history of land transactions. We define an individual as a landowner in year t if they appear at any point before year t in the land-registry data on the receiving end of a transaction—whether a sale, an inheritance, or a government allocation—and construct a list of current owners for each year in our sample. We then match the names of individuals buying land on a given year with the list of current landowners to determine whether those buyers owned any land in any other municipality of the same *departamento*.

Since we do not have information on the ID numbers of buyers and owners, we match individuals in both lists by first and last name, assuming that all registries under the same name within a *departamento* correspond to the same individual. Because names may contain misspellings or variation in the ordering of first and last names, we implement this matching using several string-matching algorithms that vary in how conservative they are respect to type-I errors, including exact matching, bigram similarity, and Jaro-Winkler distance. We then compare the results across these approaches.

OA.5 Weather Shocks

We define weather shocks at the level of each administrative unit to account for the variation in climatic conditions across areas. The shocks are based on the unit's own distribution of weather realizations, which we compute using long-run daily weather measurements (similar, for example, to Kaur (2019)). This differs from definitions based on a fixed temperature threshold—more suitable for analyses focused on a specific region or crop (see, for example, Ibáñez et al. (2022)).

We construct measures of weather shocks using the ERA5 data set, provided by the Copernicus Climate Change Service (C3S) of the European Centre for Medium-Range Weather Forecasts (ECMWF). This dataset contains global reanalysis temperature data with a horizontal resolution of 0.25×0.25 degrees (approximately $28km \times 28km$ depending, on the longitude) at an hourly frequency.[52] We use the temperature of the atmosphere two meters above the surface (in degrees Kelvin) from 1979 to 2016 for all ERA5 pixels. For each pixel, we compute the average temperature for each day d, and obtain the average daily temperature of each municipality-day pair (i, d) by taking a weighted average of the pixels within the municipality, using as weights each pixel's area relative to the municipality's total area.[53]

We compute the historical quarterly distribution of daily temperatures by considering all temperature measurements for municipality-day pairs (i, d) in calendar-quarter q throughout the period 1979-2016. For each municipality, this results in four distributions—one for each quarter of the year. We compute the 20^{th} and 80^{th} percentiles of each distribution and define the average temperature of a given municipality-day as atypically high if it is above the 80^{th} percentile of the corresponding distribution for m in quarter q. Similarly, we classify a day as having atypically low temperatures if it falls below the 20^{th} percentile that distribution.

Finally, for each year t, we sum the number of atypically high or low temperature days. In our baseline specifications, we estimate effects on outcomes measured at the municipality-year (i, t) level and use as our preferred measure of weather shocks the total number of days with atypical temperatures over the previous two years (i.e., $t-1$ and $t-2$). Appendix Figures O.6 and O.7 show the spatial and temporal variation of the these temperature-shock measures across municipalities. This definition of weather shocks has two advantages. First, it accounts for seasonality at the calendar-quarter level since the underlying distribution is quarter-specific. For example, because some quarters of the year are typically hotter, we classify a day as atypically hot only if its temperature is high relative to the historical distribution for that same quarter. Second, the

[52] Reanalysis weather information from the ERA5 results from the combination of climate models with observational data from satellites and ground sensors.

[53] In the survey data, we use "municipality" to refer to the administrative unit identified by *Admin 2* codes. In Colombia, these units correspond to municipalities, but in other countries they can have different names, such as provinces in Burkina Faso or cercles in Mali. The only exception is Peru, where we use *Admin 3* units.

measure is municipality-specific, recognizing that a given absolute temperature may be atypically high—and harmful—in one location but not in another.

In our estimations we also control for total rainfall. To construct this measure, we use the ERA5 monthly precipitation reanalysis data with resolution 0.1×0.1 degrees (approximately $9\ km \times 9\ km$ depending on the longitude) and apply the conversion factor provided by the Copernicus Climate Change Service (C3S) to obtain a measure of total monthly precipitations in cubic milliliters for each pixel. We then compute a weighted average across the pixels within each municipality to obtain monthly rainfall, using as weights the pixel's area relative to the municipality's total area. For each year, we sum precipitation across months to construct a measure of total rainfall for the municipality-year pair (i, t).

OA.6 Colombian Household Survey, ELCA

To study how farmers' decisions respond to weather shocks, we use the Colombian Longitudinal Survey conducted by the Universidad de los Andes (ELCA). The ELCA consists of a sample of 4,800 rural households interviewed over three survey rounds—a baseline in 2010 and follow-ups in 2013 and 2016. The rural sample is representative of small agricultural producers in four micro-regions: Atlantic, Central, Coffee-Growing, and South. Within each region, municipalities and *veredas* were randomly chosen. The baseline sample includes 17 municipalities.

In the follow-up rounds, enumerators attempted to reinterview all households and, when a household had split or migrated, tracked the household head, spouse, and children under nine in 2010. The attrition rate after three waves (by 2016) was 13.5%. The household questionnaire collected detailed information on land ownership and migration histories. We focus on how migration, farm size, land ownership, assets, and household consumption respond to weather shocks. Panel B of Appendix Table O.3 contains descriptive statistics for the ELCA panel. On average, 12% of households migrated, 88% owned some land, and the average size was 2.5 hectares, with 75% of farms smaller than 3 hectares.

OB Additional Reduced-Form Results

OB.1 Alternative Methods for Identifying Previous Owners

We show that the results in Table 1—wich distinguish purchases made by landless individuals from those made by buyers who already own some land elsewhere—are robust to the choice of matching algorithm and to the specific sampling properties of the land-transaction data.

First, because the matching of buyers and owners is based on first and last names—which are subject to misspellings—we report results obtained using different string-matching algorithms (exact match, bigram, and Jaro-Winkler). These algorithms vary in how conservative they are with respect to the risk of incorrectly matching two distinct individuals with similar names (type II error) or failing to match the same individual across lists due to spelling variation (type I error). Panels A, B, and C of Table O.4 show that our results do not depend on the choice of matching algorithm.

Second, because the land-transaction data are available only for plots which were at some point allocated by the government to private individuals, it is possible that some landowners who hold other types of land do not appear in our owner lists are therefore incorrectly classified as landless. This omission could bias the coefficient on sales to landless buyers (column 4 in Table 1) upwards. To gauge the severity of this bias, we re-estimate the regressions using only municipalities in which the share of private farmland originiating from government allocations is above the median. In these municipalities, the likelihood of incorrectly classifying landowners as landless should be lower, since a larger share of land transactions is observed. Columns 4, 5, and 6 in Table O.4 report the coefficients from this restricted sample. The estimated share of weather-driven land purchases made by landless buyers is nearly identical to that in the full sample, suggesting that any bias arising from this type of misclassification is small.

OB.2 Flexible Lag Specification

We estimate a flexible-lag specification that exploits the temporal variation in the occurrence of extreme temperature events, allowing us to estimate separate coefficients for the impact of weather shocks occurring in each year between t and $t-7$. This specification accommodates the possibility that households may exhaust alternative coping mechanisms after consecutive years of adverse weather conditions, and that the effects may take time to materialize in land transactions or in the land registry.

Specifically, we estimate a regression of the form:

$$s_{i,t} = \sum_{j=0}^{7} \beta_j AnnualAtypicalDay_{i,t-j} + X'_{i,t}\delta + \eta_i + \kappa_t + \varepsilon_{i,t}, \tag{O.1}$$

where $s_{i,t}$ denotes the number of land transactions (sales or mortgages), the log number of farm owners, or the log average and log median farm size in municipality i and year t. Our main variable of interest, $AnnualAtypicalDay_{i,t-j} = \sum_{d=1}^{365} AtypicalDay_{i,d,t-j}$, measures the number of days with atypical temperatures in the municipality i during year $t-j$. We include the contemporaneous value and seven lags of this variable .

The vector $X_{i,t}$ includes time-varying municipality characteristics. In particular, it contains rainfall at t and seven lags; the cumulative number of farms allocated in the municipality up to t, which controls for the stock of land on which we observe transactions; a dummy variable indicating whether a cadastral update occured; and the total municipal land area recorded in the registry, which accounts for changes in registry coverage. The model also includes municipality fixed effects η_i to control for time-invariant heterogeneity across municipalities, and year fixed effects κ_t to absorb time-specific shocks to land markets and farm size that affect all municipalities. The coefficients β_j capture the reduced form effects of contemporaneous and lagged days of extreme temperature on land transactions and farm size. Although distress land sales are likely to occur soon after a negative income shocks, notarizing a transaction and updating the property information in the land registry are the final steps in the land-transfer process and can take several months or even years.[54]

Appendix Figure O.8 presents the coefficient estimates from equation O.1 using the total number of land sales and land mortgages per municipality as dependent variables. The figure shows that increases in the frequency of extreme temperature events lead to a rise in land sales for up to two years after the shock. Extreme temperature events also increase the number of land mortgages, with statistically significant effects persisting up to five years. Appendix Figure O.9 shows the coefficient estimates for atypical temperature days and their lags, with the log number of landowners (panel a), and log mean and log median farm size (panel b) as dependent variables in equation O.1. The figure indicates that atypical temperature days increase the number of landowners and reduce farm size, with statistically significant effects emerging two years after the shock and persisting up to seven years. The longer lag in the effects observed in the registry data is consistent with the fact that land transactions are reported to the land registry only at the final stage of the purchase process. Moreover, because the land registry captures the stock of properties, these shocks are likely to have lasting effects, unless offset by subsequent land transactions.

OB.3 Expected Weather Shock

This section describes the procedure used to construct an expected weather shock variable. The purpose of this variable is to control for the possibility that municipalities follow different temperature trends that are mechanically correlated with the probability of experiencing atypical-temperature days. We follow the counterfactual-control approach developed by Jones et al. (2026). Specifically, we construct a variable $\widehat{TempShock}_{i,t}$ that captures the expected number of atypical-temperature days implied by municipality-specific temperature trends, and include it as a control

[54] In this process, buyers and sellers first sign a transaction agreement that specifies a sequence of payment installments. The signing of the public deed at a notary's office typically occurs upon the final payment. The signed deed is then submitted to the local land registry office to formalize the transfer of ownership. We observe land transactions on the date when the public deed is signed, and we observe changes in average farm size and in the number of owners for the year in which the deeds are submitted to the registry office.

in our main specifications.

Let i index municipalities, and let ℓ index a daily temperature observation for municipality i. For example, ℓ could correspond to the observed daily temperature in municipality i on March 3, 2005. Each observation ℓ has a year, a month, and a quarter. We denote these by $r(\ell)$, $m(\ell) \in \{1, \ldots, 12\}$, and $q(\ell) \in \{1, \ldots, 4\}$, respectively. Thus, if ℓ corresponds to March 3, 2005, then $r(\ell) = 2005$, $m(\ell) = 3$, and $q(\ell) = 1$.

Recall the construction of our realized weather shock. Using the original daily temperature data, we compute the 20th and 80th percentiles of the daily temperature distribution within each municipality and quarter of the year, denoted by $\mu^{20}_{i,q}$ and $\mu^{80}_{i,q}$. These thresholds are fixed for each municipality and quarter of the year, and are computed using the 1979–2016 temperature data. We then define an atypical-temperature day as a day whose temperature falls below $\mu^{20}_{i,q(\ell)}$ or above $\mu^{80}_{i,q(\ell)}$. Our realized weather shock, $TempShock_{i,t}$, is the number of atypical-temperature days in municipality i over years $t-1$ and $t-2$.

To construct the expected weather shock, we proceed in four steps. First, we collapse the daily temperature data to the municipality-month-year level and estimate municipality-month-specific temperature trends:

$$\overline{Temp}_{i,m,r} = \alpha_{i,m} + \beta_{i,m} r + \varepsilon_{i,m,r} \tag{O.2}$$

where $\overline{Temp}_{i,m,r}$ is the average temperature in municipality i, month of the year m, and year r. The parameter $\alpha_{i,m}$ is a municipality-month intercept, $\beta_{i,m}$ is the municipality-month-specific linear temperature trend, and $\varepsilon_{i,m,r}$ is the residual term. We estimate this equation separately for each municipality and month of the year using the 1979–2016 temperature data, and save the estimated slope $\widehat{\beta}_{i,m}$.

Second, we merge the estimated trends back into the original daily temperature data and remove the municipality-month-specific trend from each daily temperature observation:

$$Temp^D_{i,\ell} = Temp_{i,\ell} - \widehat{\beta}_{i,m(\ell)} r(\ell) \tag{O.3}$$

where $Temp^D_{i,\ell}$ is the de-trended temperature for daily observation ℓ in municipality i.

Third, for each year $t = 1998, \ldots, 2011$, we use the de-trended daily temperatures to construct the temperature distribution that would be expected in municipality i in year t given its municipality-month-specific temperature trend:

$$Temp^{CF}_{i,\ell}(t) = Temp^D_{i,\ell} + \widehat{\beta}_{i,m(\ell)} t \tag{O.4}$$

This step takes each de-trended daily temperature observation and shifts it according to the municipality-month temperature trend evaluated at year t. In the implementation, we construct this distribution using only daily temperature observations from before the regression sample, 1979–1997. Let $\mathcal{P}_i$ denote the set of pre-sample daily observations for municipality i. Thus, for each municipality i and year t, the projected distribution consists of the values $Temp^{CF}_{i,\ell}(t)$ for all $\ell \in \mathcal{P}_i$. This distribution has one observation for each available pre-sample daily temperature observation in municipality i.

Fourth, for each municipality and year t, we compute the share of projected daily temperatures $Temp^{CF}_{i,\ell}(t)$ that fall below $\mu^{20}_{i,q(\ell)}$ or above $\mu^{80}_{i,q(\ell)}$, using the same fixed municipality-quarter thresholds used to construct our realized weather shock. These thresholds are not recomputed using the projected temperatures, and they do not vary by year. We define the expected number of atypical-temperature days in municipality i and year t as:

$$ExpAtypicalDay_{i,t} = 365 \times \frac{1}{|\mathcal{P}_i|} \sum_{\ell \in \mathcal{P}_i} \mathbb{1}\left[Temp^{CF}_{i,\ell}(t) < \mu^{20}_{i,q(\ell)} \text{ or } Temp^{CF}_{i,\ell}(t) > \mu^{80}_{i,q(\ell)}\right] \tag{O.5}$$

Multiplying by 365 expresses this share as an annual number of atypical-temperature days.

Finally, to match the timing of our main weather shock variable, we define the expected weather shock as the sum of the expected number of atypical-temperature days in the previous two years:

$$\widehat{TempShock}_{i,t} = ExpAtypicalDay_{i,t-1} + ExpAtypicalDay_{i,t-2} \tag{O.6}$$

For ease of exposition, we include this variable divided by 100 in the regressions.

OB.4 The Land Ceiling Regulation

We investigate whether our findings on the absence of land consolidation are driven by institutional factors stemming from Colombia's land regulation policies. As discussed in Section 4.4, Law 160 of 1994 established municipality-specific land ceilings that cap the amount of land originally granted by the government that any private individual can accumulate. This restriction could help explain the lack of land consolidation at the upper end of the farm-size distribution, since it limits the ability of large landholders to acquire additional parcels whose provenance was a government allocation.[55]

To test if these restrictions are driving our results, we re-estimate the model in (4), adding an interaction term between the shock variable and a dummy indicating whether the municipality is above the median in the share of its area that was at some point part of a government allocation. The intuition behind this test is that land ceilings apply only to allocated land, not to other types of farmland. Therefore, if these restrictions were responsible for the land-fragmentation patterns shown in Table 2, we would expect the effects to be concentrated in municipalities where a large

[55]The explicit purpose of these land ceilings, as stated in the law, was precisely to prevent land concentration by large landholders.

share of agricultural land originates from government allocations.

As columns 4-6 in Appendix Table O.16 show, adding the interaction term has no effect on the estimated impact of weather shocks on average farm size or on the total number of owners, although we do find a marginally significant interaction with median farm size. Moreover, including the continuous measure of the share of government-allocated land as a control (columns 1-3) has virtually no influence on the magnitude or precision of the original estimates. We interpret these results as evidence that our main findings of are not driven by the specific institutional features of Colombia's land-regulation.

OC Model Details and Extensions

OC.1 Extension with capital and labor employment

We consider an alternative production function in which farmers choose both capital and labor. The production function is given by:

$$y_t = Z_t^a (s_t^a)^{1-\gamma} (\ell_t^\alpha n_t^\lambda k_t^\rho)^\gamma \tag{O.7}$$

where Z_t^a is aggregate agricultural productivity, s_t^a is idiosyncratic agricultural productivity, ℓ_t is land, n_t is labor employed in production, k_t is capital, and r_t is the exogenous rate at which farmers rent capital. The maximization problem associated with the production function above yields the following expression for the farmer's profits:

$$\pi_t = \left[Z_t^a (s_t^a)^{1-\gamma} (\ell_t^\alpha)^\gamma \left(\frac{\lambda r_t}{\rho w_t} \right)^{\rho\gamma} \right]^{\frac{1}{1-\gamma(\rho+\lambda)}} \left[\left(\frac{1}{(\rho+\lambda)\gamma} \right)^{\frac{\gamma(\rho+\lambda)}{1-\gamma(\rho+\lambda)}} - \left(\frac{1}{(\rho+\lambda)\gamma} \right)^{\frac{1}{1-\gamma(\rho+\lambda)}} \right] [r_t]^{\frac{\gamma(\rho+\lambda)}{1-\gamma(\rho+\lambda)}} \tag{O.8}$$

The equation above can be rewritten as

$$\pi_t = \tilde{Z}_t^a (s_t^a)^{\tilde{\gamma}} (\ell_t)^{\tilde{\alpha}} \tag{O.9}$$

where we defined

$$\tilde{Z}_t^a \equiv \left[Z_t^a \left(\frac{\lambda r_t}{\rho w_t} \right)^{\lambda\gamma} \right]^{\frac{1}{1-\gamma(\rho+\lambda)}} \left[\left(\frac{1}{(\rho+\lambda)\gamma} \right)^{\frac{\gamma(\rho+\lambda)}{1-\gamma(\rho+\lambda)}} - \left(\frac{1}{(\rho+\lambda)\gamma} \right)^{\frac{1}{1-\gamma(\rho+\lambda)}} \right] [r_t]^{\frac{\gamma(\rho+\lambda)}{1-\gamma(\rho+\lambda)}},$$

$\tilde{\gamma} \equiv \frac{1-\gamma}{1-\gamma(\rho+\lambda)}$, and $\tilde{\alpha} \equiv \frac{\gamma\alpha}{1-\gamma(\rho+\lambda)}$. This is equivalent to equation (8) in the main text.

OC.2 Solving for the transition dynamics using the SSJ matrix

In Sections 5.3 and 6 of the main text, we simulate the transition dynamics of the economy when it is hit by an unexpected weather shock—that is, an MIT shock. In general, computing the full sequence of endogenous prices that clears markets in every period can be computalionally intensive. We therefore follow the approach developed in Auclert et al. (2021), which relies on the sequence-space Jacobian (SSJ) to solve for this sequence of prices more efficiently. We construct the sequence-space Jacobian matrix and use it to update the land price within our algorithm. In our setting, the SSJ is the matrix that maps a change in the price of land in period s into the corresponding change in land demands in period t.

The algorithm used to solve for the path of land prices proceeds as follows. We beging with an initial guess for the sequence of prices $\mathbf{p}^0$, of length T, where T is the number of periods that we consider in the transition dynamics. Given this price sequence and a vector describing the path of aggregate TFP, $\mathbf{Z}$, over that same horizon, we compute the vector of excess land demand in each period t. We denote the resulting vector of excess demand under this initial guess of prices and aggregate TFP as Δ^g. We then update the sequence of land prices using the sequence-space Jacobian as follows:

$$\mathbf{p}^{g+1} = \mathbf{p}^g - \mathbf{J}^{-1} \times \Delta^g$$

where g indexes the iteration and $\mathbf{J}$ is the sequence-space Jacobian matrix. The matrix $\mathbf{J}$ gives the derivative of excess demand in period t (row) with respect to a change in land price in period s (column). In principle, one could compute the Jacobian numerically by evaluating each derivative element-by-element. However, Auclert et al. (2021) show how to construct this matrix much more efficiently by exploiting the structure of the model's backward- and forward-induction steps. Following their approach, we construct

$$F = \begin{bmatrix} Y_0 & Y_1 & \dots & Y_{T-1} \\ E_0^{'} D_0 & E_0^{'} D_1 & \dots & E_0^{'} D_{T-1} \\ \vdots & \vdots & & \vdots \\ E_{T-2}^{'} D_0 & E_{T-2}^{'} D_1 & \dots & E_{T-2}^{'} D_{T-1} \end{bmatrix}$$

where Y_s is the excess demand for land in period $t = 0$ resulting from a shock to land price in period s; E_t is a matrix representing the expected land demand in period t for each agent type present at $t = 0$, assuming all agents follow the stationary-equilibrium land and occupational choices; and D_s is the change in the density of each agent type from period $t = 0$ to period $t = 1$ following a shock to land prices in period s. The matrix F is referred to as the *fake news* matrix because it captures how excess demand responds when agents first learn, at $t = 0$, that land prices will rise in period s, but then discover at $t = 1$ that this anticipated shock will not occur. Once F has been constructed, we can obtain the sequence-space Jacobian by computing, for each period t and each price change in period s, $J_{t,s} = J_{t-1,s-1} + F_{t,s}$.

Using the sequence-space Jacobian to update prices allows us to find the solution quickly. Under our main calibration, the model converges in fewer than 10 iterations, with the sum of excess demand across periods falling below 10^{-4}—recall that we normalize the land supply one.

OC.3 Simulating the Stochastic Economy

Following Boppart et al. (2018), we compute the evolution of the stochastic economy using equation (20). To do so, we first obtain the impulse response of the economy's endogenous variables to an MIT shock at $t = 0$. Let $\widehat{dy_s}$ denote the corresponding response in period s. We recover $\widehat{dy_s}$ using the sequence-space Jacobian, embedded in an algorithm that solves for the full nonlinear equilibrium path of prices and endogenous variables. This approach allows us to compute impulse responses without relying on the fully linearized general-equilibrium solution of Auclert et al. (2021).[56]

OD Calibration Details

OD.1 Solving for the Stationary Equilibrium

We solve for the stationary equilibrium using a grid of 300 points for land ownership and 15 points for the idiosyncratic agricultural productivity shock. This gives a 300×15 matrix of agricultural income. Similarly, we set a grid of 300 points for the idiosyncratic non-agricultural productivity and 15 points for the idiosyncratic non-agricultural productivity shock.[57] This gives us a matrix of non-agriculture income for landless agents. Having the two matrices with the same dimensions simplifies the solution algorithm. We space the grid of land and wages according to a log-normal distribution and construct the non-agriculture productivity grid using Rouwenhorst's approach, implemented in the companion package distributed with Auclert et al. (2021).

The algorithm follows a standard solution method. Because we have a discrete-choice problem, the agents' problem presents non-convexities that prevent us from applying the endogenous-grid method. We therefore use a simple value function iteration.

[56] Auclert et al. (2021) develop a complete linearized general-equilibrium solution method.

[57] Because farmers do not earn a wage and workers do not earn farm revenues within each period, we do not need to consider the full matrix of all combinations of non-agriculture skill, agricultural skill, and land ownership.

OD.2 Estimating ρ

To estimate the persistence in the farm productivity shock, we use our longitudinal household survey data, which contain information on total agricultural output per farm and landholdings. Using this information, we construct the following measure of $s^a_{i,t}$:

$$\widehat{s}^a_{i,t} = \left[\frac{y_{i,t}}{Z^a_t \ell^\gamma_{i,t}}\right]^{\frac{1}{1-\gamma}}$$

where $y_{i,t}$ and $\ell_{i,t}$ are our measured values of output and land, respectively. In what follows, we assume that Z^a_t is not systematically correlated with $\ell_{i,t}$. We then assume that the farm productivity of farmer i in period t is given by

$$\log \widehat{s}^a_{i,t} = \rho \log \widehat{s}^a_{i,t-1} + \sigma \epsilon_{i,t}$$

where $\epsilon_{i,t}$ is the productivity shock. Since the gap between survey waves in the data is three years, what we effectively observe is

$$\log \widehat{s}^a_{i,t} = \rho^3 \left(\log \widehat{s}^a_{i,t-3}\right) + \sigma \left(\rho^2 \epsilon_{i,t-2} + \rho \epsilon_{i,t-1} + \epsilon_{i,t}\right)$$

We estimate the regression above while adding a control for the probability of exiting during these three years, which we estimate using a logit model with a polynomial in initial farm size as the explanatory variable. Our final estimation is

$$\log \widehat{s}^a_{i,t} = \rho^3 \left(\log \widehat{s}^a_{i,t-3}\right) + \beta \widehat{P}^{exit}_{i,t-3} + \sigma \left(\rho^2 \epsilon_{i,t-2} + \rho \epsilon_{i,t-1} + \epsilon_{i,t}\right),$$

where $\widehat{P}^{exit}_{i,t-3}$ is the predicted probability that the farmer exited over the three-year period.[58] This procedure gives us the value of ρ.

OD.3 Estimating Annual Exit Rates from the Data

The longitudinal household survey data from Colombia spans a three-year window. To estimate the annualized exit rate used to calibrate the model, we proceed as follows. First, we groups farms into size bins and compute the average exit rate within each bin. We then compute the annualized rate using the following formula:

$$\widehat{p}_{survive}(b) = (1 - p_{exit}(b))^3 \tag{O.10}$$

[58]Accounting for exit lowers the estimated persistence of farm productivity shocks, ρ, but the resulting change in point estimates is small.

where $\widehat{p}_{survive}(b)$ is the share of farmers in farm-size bin b who survive over a three years period in the ELCA dataset. Using the formula above, we compute $p_{exit}(b) = 1 - [\widehat{p}_{survive}(b)]^{1/3}$, which is the statistic we use in our calibration.

OD.4 Alternative Weather Shock distribution

In Section 6.2, we simulate alternative sequences of weather shocks by increasing the probability of extreme events. To do so, we model the data-generating process for our weather-shock variable. Specifically, we treat each day as a Bernoulli draw indicating whether it is atypical, and define the weather-shock variable as the sum of 712 such daily draws (two years). Each day has a baseline probability of 0.4 of being atypical. We introduce serial correlation by assuming that these draws follow a two-state Markov chain. If day t is atypical, the probability that day $t+1$ is also atypical is $1-0.6(1-\zeta)$. If day t is typical, the probability that day $t+1$ is also typical is $1-0.4(1-\zeta)$. A higher value of ζ implies stronger persistence in weather conditions. The Markov chain is therefore given by:

$$\begin{pmatrix} 1-0.6(1-\zeta) & 0.6(1-\zeta) \\ 0.4(1-\zeta) & 1-0.4(1-\zeta) \end{pmatrix}. \tag{O.11}$$

OE Analytical Results

To better understand how the change in price shapes agents' decision to become a farmer, we study analytically the impact of a small change in land price and the aggregate productivity. We do so using equation (14), ignoring general equilibrium effects related to changes in the farm-size distribution and applying the envelope conditions.

OE.1 TFP shock and the probability of becoming a farmer

For a given agent, $\phi_t = (s_t^a, s_t^n, \ell_t)$, the impact of a small change in Z_t^a on its probability of becoming a farmer, $\mu_t(\phi_t)$ is

$$\frac{\partial \mu_t(\phi_t)}{\partial Z_t} = \kappa(1-\delta)\mu_t(\phi_t)\left(1-\mu_t(\phi_t)\right)\underbrace{\left[u'\left(c_t^F(\phi_t)\right) - u'\left(c_t^W(\phi_t)\right)\right]}_{>0}\frac{\partial \pi_t(\phi_t)}{\partial Z_t^a} > 0 \tag{O.12}$$

where $c_t^F(\phi_t)$ is consumption if the agent becomes a farmer, $c_t^W(\phi_t)$ is consumption if the agent becomes a worker, and $u'()$ is the marginal utility of consumption. Notice that, because $\frac{\partial \pi_t(\phi_t)}{\partial Z_t^a} = 0$ for workers, their decision to become farmers is unaffected by the change in aggregate agricultural TFP. Note that $u'\left(c^F(\phi_t)\right) > u'\left(c^W(\phi_t)\right)$, since consumption is always lower when agents buy land. Therefore, a reduction in Z_t^a always reduces the probability that agents become farmers.

This equation also shows that farm productivity s_t^a and total land ℓ_t have an ambiguous effect on the magnitude of the impact of Z_t^a on the measure of farmers $\mu_t(\phi_t)$. On the one hand, more productive or skilled farmers—those with higher s_t^a and ℓ_t—are more exposed to aggregate shocks because their profits are more sensitive to changes in productivity, implying a larger derivative $\frac{\partial \pi_t(\phi_t)}{\partial Z_t^a} > 0$. On the other hand, these same farmers tend to have higher consumption levels, which reduces the marginal-utility gap between farming and working; that is, $u'\left(c^F(\phi_t)\right) - u'\left(c^W(\phi_t)\right)$ is smaller. Because of the curvature of the utility function, this dampens the response of their occupational choice to aggregate shocks.

OE.2 Price shock and the probability of becoming a farmer.

The impact of a small shock to land prices on the probability of becoming a farmer, $\mu_t(\phi_t)$, is given by:

$$\frac{\partial \mu_t(\phi_t)}{\partial p_t} = \kappa(1-\delta)\mu_t(\phi_t)\left(1-\mu_t(\phi_t)\right)\left[\underbrace{\left[u'\left(c_t^F(\phi_t)\right) - u'\left(c_t^W(\phi_t)\right)\right]\ell_t}_{\text{income effect (+)}} - \underbrace{u'\left(c^F(\phi_t)\right)\ell_t^*(\phi_t)}_{\text{land price effect (-)}}\right] \lessgtr 0. \tag{O.13}$$

In this case, two mechanisms operate in opposite directions. First, there is an income effect: when land prices rise, agents who own more land become richer. This mechanism moves in the same direction of prices, and its influence is stronger when farmers are poorer, since the marginal utility difference $u'\left(c^F(\phi_t)\right) - u'\left(c^W(\phi_t)\right)$ is larger. Notice that agents with no land are not affected by the income effect.

Second, there is a land price effect, which moves in the opposite direction of the income effect. When land prices fall, agents are more likely to enter or remain in farming because they can purchase more land. For agents outside of agriculture, this is the only mechanism that affects their decision, since they do not own land and therefore are not affected by the income effect.

OF Tables and Figures

Table O.1: Characteristics of Households by Farming Status

	Entrants	Incumbents	Exiters	Landless	All
Household head literate	0.499	0.418	0.477	0.627	0.493
Household head's occupation is agriculture	0.262	0.816	0.656	0.112	0.555
Household located in rural area	0.715	0.901	0.743	0.384	0.714
Above country median household size	0.320	0.493	0.440	0.346	0.431
Above country median household head age	0.421	0.504	0.528	0.458	0.485
Above country median land size	0.355	0.516	0.340	-	0.476
Above country median household expenditure	0.394	0.441	0.487	0.632	0.500
Number of households	4,121	29,146	4,718	16,844	54,829

Notes: West Africa–EHCVM (2018, 2021); Mexico–ENNVIH (2002, 2005); Peru–ENAHO (2007–2023). All variables are measured at baseline, except land size for entrants, which is measured at $t = 2$. Country-specific median land size is computed excluding landless households.

Table O.2: Effect of Weather Shocks on Households' Entry and Exit from Agriculture

	LatAm (1)	LatAm (2)	W. Africa (3)	W. Africa (4)	All (5)	All (6)
TempShock	-0.035	0.006	-0.082***	-0.095***	-0.047**	-0.035
	(0.026)	(0.023)	(0.029)	(0.031)	(0.022)	(0.023)
TempShock × No Farm	0.266***	0.072*	0.189***	0.132***	0.203***	0.125***
	(0.060)	(0.041)	(0.045)	(0.030)	(0.052)	(0.037)
TempShock × Small Farm	-0.035	0.006	0.001	-0.001	-0.009	-0.000
	(0.023)	(0.028)	(0.016)	(0.014)	(0.017)	(0.017)
N	24074	24074	85584	85584	109658	109658
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Year x Region FE	Yes	Yes	Yes	Yes	Yes	Yes
Year x Group FE	No	Yes	No	Yes	No	Yes

Notes: This table shows results from estimating equation (3) using data from household surveys in West Africa (EHCVM 2018–2021), and Latin America (ENNVIH 2002–2005; ENAHO 2007–2023).

Table O.3: Descriptive Statistics

	Mean	Std. Dev.	Min	Max
	(1)	(2)	(3)	(4)
	Panel A: Municipality (N= 864)			
SNR				
Number of total land sales	12.62	24.75	0	292
Number of full land sales	10.83	21.63	0	281
Number of partial land sales	1.79	6.05	0	133
Number of mortages	2.62	7.54	0	172
Number of inheritances	2.00	4.14	0	48
Land Registry				
Number of owners	2,501.13	2,160.43	21	18,768
Mean of farm size	32.06	106.71	1	1,693
Median of farm size	16.15	86.73	0	1,438
Controls				
Number of total allocations	444.68	679.23	1	6,550
=1 if land registry update	0.07	0.25	0	1
Registered area (1000 ha.)	41,255.76	87,792.98	340	1,475,761
Accumulated precipitation	3,516.60	2,647.67	372	21,144
Days of atypical high temperature	120.47	75.50	5	483
Days of atypical low temperature	154.92	72.23	1	401
Days of atypical temperature	275.39	43.56	157	485
	Panel B: ELCA - Household (N= 4,280)			
=1 if HH has land	0.92	0.27	0	1
=1 if farm size $\leq$ 3 ha	0.75	0.43	0	1
=1 if HH migrated	0.10	0.30	0	1
=1 if HH sold animals	0.69	0.46	0	1
Asset index	0.00	0.39	-1	3
Farm size (ha.)	2.65	5.62	0	118
Consumption per capita	2.79	2.52	0	104
Accumulated precipitation	3,617.71	2,457.81	729	22,934
Days of atypical high temperature	264.14	118.34	64	528
Days of atypical low temperature	50.80	43.60	0	176
Days of atypical temperature	314.94	84.67	174	529

Notes: This table presents summary statistics for each estimation sample. Panel A describes the variables used for municipality-level estimations. Total number of sales includes full and partial sales during the year. Full sales correspond to transfers of the entire property to another owner, while partial sales transfer only a fraction of the property. The number of total allocations corresponds to the cumulative sum of government-allocated farms in the municipality from 1901 until the year of observation. Panel B summarizes the data used for estimations at the household-year level. This data comes from three rounds (2010, 2013 and 2016) of ELCA, a panel of rural households collected by Universidad de los Andes. Climate data used to compute the number of atypical temperature days and the accumulated precipitation come from the Copernicus Climate Change Service (*C3S*). Days with atypical temperature refer to the total number of days across the two prior years with abnormally high or low temperatures. Accumulated precipitation is the volume of rainfall in milliliters for year t.

Table O.4: Effect of Temperature Shock on Land sales: Landless vs Already Owners - Alternative Matching Algorithms

	Full Sample			Above median land share from govt. allocations		
	Total Sales	Landless Buyers	Already Owners	Total Sales	Landless Buyers	Already Owners
	(1)	(2)	(3)	(4)	(5)	(6)
	Panel A: Fuzzy Matching - Bigram					
$TempShock_{i,t}$	2.537***	1.866***	0.671***	3.524***	2.679***	0.845***
	(0.536)	(0.408)	(0.192)	(0.855)	(0.654)	(0.300)
Observations	10,021	10,021	10,021	5,097	5,097	5,097
R^2	0.905	0.889	0.838	0.899	0.877	0.839
Mean Dep. Var.	12.62	9.37	3.25	20.91	15.47	5.44
	Panel B: Fuzzy Matching - Jaro Winkler					
$TempShock_{i,t}$	2.537***	1.402***	1.135***	3.524***	2.080***	1.445***
	(0.536)	(0.325)	(0.291)	(0.855)	(0.526)	(0.466)
Observations	10,021	10,021	10,021	5,097	5,097	5,097
R^2	0.905	0.857	0.889	0.899	0.845	0.885
Mean Dep. Var.	12.62	6.10	6.52	20.91	10.04	10.87
	Panel C: Exact Matching					
$TempShock_{i,t}$	2.537***	1.780***	0.757***	3.524***	2.546***	0.978***
	(0.536)	(0.412)	(0.190)	(0.855)	(0.663)	(0.297)
Observations	10,021	10,021	10,021	5,097	5,097	5,097
R^2	0.905	0.889	0.837	0.899	0.877	0.839
Mean Dep. Var.	12.62	9.62	3.00	20.91	15.90	5.01

Notes: This table presents coefficient estimates from equation (4) in a balanced panel at the municipality-year level using as outcome land sales. Columns (1)–(3) report estimates for the full sample, while columns (4)–(6) restrict the sample to municipalities with an above-median share of land allocated by the government. Within each set of columns, we show results for total sales, sales to landless buyers, and sales to existing owners, respectively. Buyer type is identified by checking whether the buyer's name appears in earlier SNR records (sales, inheritances, or government allocations) within the same *departamento*. Panels present results across alternative matching procedures used to classify buyers (Fuzzy Matching–Bigram, Fuzzy Matching–Jaro Winkler, and Exact Matching). The independent variable is the total number of atypical temperature days in the past two years divided by 100. All regressions include municipality and year fixed effects, as well as cumulative rainfall and its five lags, the accumulated number of land allocations, the area covered by the land registry, and an indicator for cadastral registry updates. See the text for details. Standard errors clustered at the municipality level reported in parenthesis. Data from the National Superintendency of Notaries (SNR). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table O.5: Weather Shocks and Number of owners, by Initial Size Decile

	Number of owners by initial distribution quantiles (q_m^j)									
	q_m^1 (1)	q_m^2 (2)	q_m^3 (3)	q_m^4 (4)	q_m^5 (5)	q_m^6 (6)	q_m^7 (7)	q_m^8 (8)	q_m^9 (9)	q_m^{10} (10)
$TempShocks$	0.032**	0.017*	0.026***	0.016*	0.022***	0.008	0.006	-0.001	0.001	-0.000
	(0.016)	(0.010)	(0.008)	(0.008)	(0.006)	(0.007)	(0.006)	(0.006)	(0.006)	(0.004)
Observations	10,004	9,967	9,942	9,893	9,996	9,958	10,017	9,981	9,996	10,017
R^2	0.940	0.971	0.981	0.982	0.986	0.985	0.985	0.990	0.990	0.993

Notes: This table reports the estimated effect of weather shocks on the log number of land owners within fixed segments of the initial farm-size distribution. The dependent variables are the log number of owners in municipality–year whose total landholdings fall into each of the ten farm-size bins defined by the municipality's baseline (year 2000) land-distribution deciles. The independent variable is the total number of atypical-temperature days in the past two years divided by 100. Each column corresponds to a different regression. All regressions include municipality and year fixed effects, as well as cumulative rainfall and its five lags, the accumulated number of land allocations, the area covered by the land registry, and an indicator for cadastral registry updates. Standard errors clustered at the municipality level reported in parenthesis. Data from the National Land Registry (*Catastro Nacional*), maintained by the National Geographical Institute (IGAC). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table O.6: Weather Shocks and Land Sales - Alternative Shock Definitions

	Total (1)	Full (2)	Partial (3)	Mortg. (4)
Panel A: Aggregates over t to $t-2$				
TempShocks	3.036***	2.521***	0.515**	1.256***
	(0.527)	(0.504)	(0.221)	(0.252)
Observations	10,021	10,021	10,021	10,021
R^2	0.906	0.903	0.636	0.758
Mean Dep. Var	12.62	10.83	1.79	2.62
Panel B: Aggregates over $t-1$ to $t-3$				
TempShocks	1.870***	1.455***	0.416	0.956***
	(0.466)	(0.432)	(0.265)	(0.207)
Observations	10,021	10,021	10,021	10,021
R^2	0.905	0.902	0.636	0.757
Mean Dep. Var	12.62	10.83	1.79	2.62
Panel C: Threshold 5, 95				
TempShocks	4.290***	3.150***	1.140***	1.922***
	(0.867)	(0.804)	(0.370)	(0.368)
Observations	10,021	10,021	10,021	10,021
R^2	0.906	0.902	0.636	0.757
Mean Dep. Var	12.62	10.83	1.79	2.62
Panel D: Threshold 1.5SD				
TempShocks	3.689***	2.765***	0.925***	1.762***
	(0.752)	(0.695)	(0.317)	(0.314)
Observations	10,021	10,021	10,021	10,021
R^2	0.906	0.902	0.636	0.758
Mean Dep. Var	12.62	10.83	1.79	2.62
Panel E: Municipality Daily Temperature Distribution				
TempShocks	0.941**	0.804**	0.137	0.353
	(0.372)	(0.315)	(0.208)	(0.217)
Observations	10,021	10,021	10,021	10,021
R^2	0.905	0.902	0.635	0.756
Mean Dep. Var	12.62	10.83	1.79	2.62
Panel F: Daily Temperature Distribution 1990 – 2011				
TempShocks	2.248***	1.887***	0.361	1.004***
	(0.546)	(0.504)	(0.238)	(0.231)
Observations	10,021	10,021	10,021	10,021
R^2	0.905	0.902	0.635	0.757
Mean Dep. Var	12.62	10.83	1.79	2.62

Notes: This table presents coefficient estimates from equation (4) in a balanced panel at the municipality-year level, using various measures of land-market activity as outcomes. The dependent variable in column 1 is the total of land sales (full + partial), in column 2 the number of full sales, in column 3 the number of partial sales, and in column 4 the number of land mortgages. The independent variable is the total number of atypical-temperature days in the previous years, divided by 100. Each panel reports results using alternative definitions of atypical days and lag structures. Panels A and B define atypical days with the 20^{th} and 80^{th} percentiles of the municipality-quarter distribution of daily temperature, as in the main specification, but aggregate them over $t-2$ to t and $t-3$ to $t-1$, respectively. Panels C to F aggregate over $t-2$ to $t-1$ as in the main specification, but use alternative thresholds to define atypical-temperature days. In Panel C, atypical days are those falling below the 5^{th} or above the 95^{th} percentile of the temperature distribution. In Panel D, the thresholds are defined as *mean* $\pm$ *1.5 standard deviations*. Panels E and F use the 20^{th} and 80^{th} percentiles as in the main specification but use alternative constructions of the temperature distribution. Panel E does not condition on the quarter of the year. Panel F restricts the data to 1990–2011 to compute the distribution. All regressions include municipality and year fixed effects, as well as cumulative rainfall and its five lags, the accumulated number of land allocations, the area covered by the land registry, and an indicator for cadastral registry updates. See the text for details. Standard errors clustered at the municipality level reported in parenthesis. Data from the National Superintendency of Notaries (SNR). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table O.7: Weather Shocks and Average Farm Size - Alternative Shock Definitions

	Number of Owners (logs)	Mean Farm Size (logs)	Median Farm Size (logs)
	(1)	(2)	(3)
Panel A: Aggregates over t to $t-2$			
TempShocks	0.011**	-0.013**	-0.024*
	(0.005)	(0.005)	(0.013)
Observations	10,021	10,021	10,021
R^2	0.992	0.993	0.974
Panel B: Aggregates over $t-1$ to $t-3$			
TempShocks	0.022***	-0.021***	-0.028**
	(0.005)	(0.006)	(0.012)
Observations	10,021	10,021	10,021
R^2	0.992	0.993	0.975
Panel C: Threshold 5, 95			
TempShocks	0.024***	-0.028***	-0.036**
	(0.008)	(0.008)	(0.015)
Observations	10,021	10,021	10,021
R^2	0.992	0.993	0.974
Panel D: Threshold 1.5SD			
TempShocks	0.016**	-0.018***	-0.027**
	(0.006)	(0.007)	(0.014)
Observations	10,021	10,021	10,021
R^2	0.992	0.993	0.974
Panel E: Municipality Daily Temperature Distribution			
TempShocks	0.011**	-0.012**	-0.015
	(0.004)	(0.005)	(0.010)
Observations	10,021	10,021	10,021
R^2	0.992	0.993	0.974
Panel F: Daily Temperature Distribution 1990 – 2011			
TempShocks	0.013***	-0.013**	-0.021*
	(0.005)	(0.005)	(0.012)
Observations	10,021	10,021	10,021
R^2	0.992	0.993	0.974

Notes: This table reports the estimated effect of weather shocks on the farm-size distribution. Dependent variables are the log number of land owners in the municipality (column 1), log average farm size (column 2), log median farm size (column 3). The independent variable is the total number of atypical-temperature days in the previous years, divided by 100. Each panel reports results using alternative definitions of atypical days and lag structures. Panels A and B define atypical days with the 20^{th} and 80^{th} percentiles of the municipality-quarter distribution of daily temperature, as in the main specification, but aggregate them over $t-2$ to t and $t-3$ to $t-1$, respectively. Panels C to F aggregate over $t-2$ to $t-1$ as in the main specification, but use alternative thresholds to define atypical-temperature days. In Panel C, atypical days are those falling below the 5^{th} or above the 95^{th} percentile of the temperature distribution. In Panel D, the thresholds are defined as *mean ± 1.5 standard deviations*. Panels E and F use the 20^{th} and 80^{th} percentiles as in the main specification but use alternative constructions of the temperature distribution. Panel E does not condition on the quarter of the year. Panel F restricts the data to 1990–2011 to compute the distribution. All regressions include municipality and year fixed effects, as well as cumulative rainfall and its five lags, the accumulated number of land allocations, the area covered by the land registry, and an indicator for cadastral registry updates. See the text for details. Standard errors clustered at the municipality level reported in parenthesis. Data from the National Land Registry (*Catastro Nacional*), maintained by the National Geographical Institute (IGAC). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table O.8: Weather Shocks, Land Sales, and Average Farm Size - Hot and Cold Days

	Land sales				Owners & Farm size (logs)		
	Total	Full	Partial	Morgt.	Number of Owners	Mean Farm Size	Median Farm Size
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Cold TempShocks	3.507***	2.534***	0.972***	1.300***	0.017**	-0.019**	-0.015
	(0.634)	(0.593)	(0.276)	(0.341)	(0.007)	(0.008)	(0.014)
Hot TempShocks	2.394***	1.937***	0.457*	1.009***	0.014***	-0.014***	-0.023*
	(0.552)	(0.516)	(0.236)	(0.233)	(0.005)	(0.005)	(0.012)
Observations	10,021	10,021	10,021	10,021	10,021	10,021	10,021
R^2	0.906	0.902	0.636	0.757	0.992	0.993	0.974
Mean Dep. Var	12.62	10.83	1.79	2.62	2,501.13	32.06	16.15

Notes: This table presents coefficient estimates from equation (4) in a balanced panel at the municipality-year level, using various measures of land-market activity and farm size as outcomes. The dependent variable in column 1 is the total of land sales (full + partial), in column 2 the number of full sales, in column 3 the number of partial sales, in column 4 the number of land mortgages, in column 5 the log number of land owners in the municipality, in column 6 log average farm size, and in column 7 the log median farm size. The independent variables are the number of days with average temperature below the 20^{th} percentile of the municipality–quarter daily temperature distribution (*Cold Temp Shock*) and above the 80^{th} percentile (*Hot Temp Shock*) over the previous two years. All regressions include municipality and year fixed effects, as well as cumulative rainfall and its five lags, the accumulated number of land allocations, the area covered by the land registry, and an indicator for cadastral registry updates. See the text for details. Standard errors clustered at the municipality level reported in parenthesis. Outcomes data from the National Superintendency of Notaries (SNR) and National Land Registry (*Catastro Nacional*), maintained by the National Geographical Institute (IGAC). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table O.9: Weather Shocks, Land Sales, and Average Farm Size - Evapotranspiration Index (SPEI)

	Land sales				Owners & Farm size		
	Total	Full	Partial	Morgt.	Number of Owners (logs)	Mean Farm Size (logs)	Median Farm Size (logs)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: SPEI continuous							
$SPEI(-)$	0.859**	0.441	0.418**	-0.349*	0.013**	-0.017**	-0.042**
	(0.349)	(0.324)	(0.171)	(0.183)	(0.006)	(0.007)	(0.019)
Observations	10,021	10,021	10,021	10,021	10,021	10,021	10,021
R^2	0.905	0.902	0.636	0.756	0.992	0.993	0.975
Mean Dep. Var.	12.62	10.83	1.79	2.62	2,501.13	32.06	16.15
Panel B: SPEI discrete							
Months ($SPEI < -2$)	0.846**	0.647**	0.199	0.389***	0.006*	-0.006	-0.012
	(0.348)	(0.322)	(0.160)	(0.121)	(0.003)	(0.004)	(0.009)
Observations	10,021	10,021	10,021	10,021	10,021	10,021	10,021
R^2	0.905	0.902	0.635	0.756	0.992	0.993	0.974
Mean Dep. Var.	12.62	10.83	1.79	2.62	2,501.13	32.06	16.15

Notes: This table presents coefficient estimates from equation (4) in a balanced panel at the municipality-year level, using various measures of land-market activity and farm size as outcomes, but using a different data source to construct weather shocks. The dependent variable in column 1 is the total of land sales (full + partial), in column 2 the number of full sales, in column 3 the number of partial sales, in column 4 the number of land mortgages, in column 5 the log number of land owners in the municipality, in column 6 log average farm size, and in column 7 the log median farm size. Panel A measures the shock as the average negative annual Evapotranspiration Index (SPEI) over the previous two years; Panel B as the number of months with SPEI below -2 in either of the previous two years. See text for more details. All regressions include municipality and year fixed effects, as well as cumulative rainfall and its five lags, the accumulated number of land allocations, the area covered by the land registry, and an indicator for cadastral registry updates. Standard errors clustered at the municipality level reported in parenthesis. Outcomes data from the National Superintendency of Notaries (SNR) and National Land Registry (*Catastro Nacional*), maintained by the National Geographical Institute (IGAC). SPEI data from the SPEIbase v.2.9 (Beguería et al., 2023). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table O.10: Weather Shocks and Land Sales - Alternative Specifications and Clusters

	Total (1)	Full (2)	Partial (3)	Mortg. (4)
Panel A: Controls for *departamento* specific linear trends				
TempShocks	2.532***	2.018***	0.514**	1.059***
	(0.535)	(0.502)	(0.224)	(0.241)
Observations	10,021	10,021	10,021	10,021
R^2	0.905	0.902	0.636	0.757
Mean Dep. Var	12.62	10.83	1.79	2.62
Panel B: Controls for expected shock				
TempShocks	2.551***	2.029***	0.522**	1.054***
	(0.534)	(0.503)	(0.230)	(0.237)
Observations	10,021	10,021	10,021	10,021
R^2	0.906	0.903	0.636	0.758
Mean Dep. Var	12.62	10.83	1.79	2.62
Panel C: Municipality and *departamento* × year clusters				
TempShocks	2.537***	2.013***	0.523*	1.046***
	(0.786)	(0.732)	(0.284)	(0.388)
Observations	10,021	10,021	10,021	10,021
R^2	0.905	0.902	0.636	0.757
Mean Dep. Var	12.62	10.83	1.79	2.62
Panel D: Spatial autocorrelation (500 Km radius)				
TempShocks	2.537***	2.013***	0.523***	1.046***
	(0.717)	(0.629)	(0.192)	(0.405)
Observations	10,021	10,021	10,021	10,021
R^2	0.038	0.032	0.012	0.037
Mean Dep. Var	12.62	10.83	1.79	2.62
Panel E: Spatial autocorrelation (800 Km radius)				
TempShocks	2.537***	2.013***	0.523**	1.046**
	(0.778)	(0.684)	(0.205)	(0.427)
Observations	10,021	10,021	10,021	10,021
R^2	0.038	0.032	0.012	0.037
Mean Dep. Var	12.62	10.83	1.79	2.62

Notes: This table presents coefficient estimates from equation (4) in a balanced panel at the municipality-year level, using various measures of land-market activity as outcomes and different econometric specifications. The dependent variable in column 1 is the total of land sales (full + partial), in column 2 the number of full sales, in column 3 the number of partial sales, and in column 4 the number of land mortgages. The independent variable is the total number of atypical-temperature days in the past two years, divided by 100. All regressions include municipality and year fixed effects, as well as cumulative rainfall and its five lags, the accumulated number of land allocations, the area covered by the land registry, and an indicator for cadastral registry updates. Panel A adds an interaction between the administrative division *departamento* and a linear trend. Panels B follows Jones et al. (2026) by controlling for expected number of atypical temperature days in each municipality, given municipality- specific temperature trends. Panels C and D present the baseline results under alternative inference procedures. In Panel C, standard errors are clustered by municipality and by *departamento*–year. Panels D and E account for spatial autocorrelation across municipalities within a 500 km radius (Panel D) and a 800 km radius (Panel E) (Conley, 1999). In all other panels, standard errors are clustered at the municipality level. Standard errors are reported in parentheses throughout. Data from the National Superintendency of Notaries (SNR). For reference, the average distance between municipalities within a *departamento* is 89 km, while 500 km corresponds approximately to the maximum distance between municipalities in the same *departamento*. We use the `acreg` Stata routine from Colella et al. (2019) to implement this method.

Table O.11: Weather Shocks and Average Farm Size - Alternative Specifications and Clusters

	Number of Owners (logs)	Mean Farm Size (logs)	Median Farm Size (logs)
Panel A: Controls for *departamento* specific linear trends			
TempShocks	0.015***	-0.015***	-0.022*
	(0.005)	(0.005)	(0.012)
Observations	10,021	10,021	10,021
R^2	0.992	0.993	0.975
Mean Dep. Var	2,501.13	32.06	16.15
Panel B: Controls for expected shock			
TempShocks	0.015***	-0.015***	-0.021*
	(0.005)	(0.005)	(0.012)
Observations	10,021	10,021	10,021
R^2	0.992	0.993	0.974
Mean Dep. Var	2,501.13	32.06	16.15
Panel C: Municipality and *departamento* × year clusters			
TempShocks	0.015**	-0.015**	-0.021
	(0.006)	(0.007)	(0.013)
Observations	10,021	10,021	10,021
R^2	0.992	0.993	0.974
Mean Dep. Var	2,501.13	32.06	16.15
Panel D: Spatial autocorrelation (500 Km radius)			
TempShocks	0.015**	-0.015***	-0.021***
	(0.007)	(0.006)	(0.007)
Observations	10,021	10,021	10,021
R^2	0.673	0.072	0.011
Mean Dep. Var	2,501.13	32.06	16.15
Panel E: Spatial autocorrelation (800 Km radius)			
TempShocks	0.015**	-0.015***	-0.021***
	(0.007)	(0.006)	(0.006)
Observations	10,021	10,021	10,021
R^2	0.673	0.072	0.011
Mean Dep. Var	2,501.13	32.06	16.15

Notes: This table reports the estimated effect of weather shocks on the farm-size distribution using different econometric specifications. Dependent variables are the log number of land owners in the municipality (column 1), log average farm size (column 2), log median farm size (column 3). The independent variable is the total number of atypical-temperature days in the past two years, divided by 100. All regressions include municipality and year fixed effects, as well as cumulative rainfall and its five lags, the accumulated number of land allocations, the area covered by the land registry, and an indicator for cadastral registry updates. See the text for details. Standard errors clustered at the municipality level reported in parenthesis. Panel A adds an interaction between the administrative division *departamento* and a linear trend. Panels B follows Jones et al. (2026) by controlling for expected number of atypical temperature days in each municipality, given municipality- specific temperature trends. Panels C and D present the baseline results under alternative inference procedures. In Panel C, standard errors are clustered by municipality and by *departamento*–year. Panels D and E account for spatial autocorrelation across municipalities within a 500 km radius (Panel D) and a 800 km radius (Panel E) (Conley, 1999). In all other panels, standard errors are clustered at the municipality level. Standard errors are reported in parentheses throughout. Data from the National Land Registry (*Catastro Nacional*), maintained by the National Geographical Institute (IGAC). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. For reference, the average distance between municipalities within a *departamento* is 89 km, while 500 km corresponds approximately to the maximum distance between municipalities in the same *departamento*. We use the `acreg` Stata routine from Colella et al. (2019) to implement this method.

Table O.12: Weather Shocks and Household Decisions, Excluding Migrant Households

	Household Migrated (1)	Household Has Land (2)	Household Sold Land (3)	Farm Size (Hectares) (4)	Consumption (logs) (5)	Income (logs) (6)	Sold Animals (7)	Asset Index (8)
TempShocks	0.089***	-0.101***	0.038***	-0.443	-0.111***	-0.465***	0.046***	-0.112***
	(0.033)	(0.023)	(0.012)	(0.295)	(0.038)	(0.069)	(0.016)	(0.028)
Observations	13,442	11,778	11,778	11,778	13,442	12,773	12,634	12,596
R^2	0.532	0.557	0.428	0.627	0.707	0.647	0.932	0.585
Mean Dep. Var	0.14	0.92	0.03	2.65	2.80	2.06	0.67	-0.00

Notes: This table shows estimates of the effects of weather shocks on household decisions using a three-wave panel survey of rural households (2010, 2013, 2016). The sample is further restricted to households that responded to the survey's agricultural module (thus excluding households that migrated to non-rural areas), with the exception of the first column, in which we include all households. The dependent variables, from left to right, are: a dummy indicating whether the household migrated between survey waves; a dummy indicating whether the household owns any land; a dummy indicating whether the household sold any land; log farm size; log per-capita consumption; log per-capita income; a dummy indicating whether the household sold any livestock; and a principal-components asset index based on household assets and durable goods. All monetary values are expressed in 2016 Colombian pesos (in millions). All regressions control for log aggregate rainfall at t and its five lags and include household and time fixed effects. *Mean Dep. Var.* reports the sample mean of the untransformed dependent variable. Standard errors clustered at the *vereda* level are reported in parentheses. Data from ELCA survey. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table O.13: Weather Shocks and Household Assets

	Refrigerators	Washing Machines	Blenders	Ovens	Microwave	Heaters	Showers	Air-condi.	Televis.	Radios
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
TempShocks	-0.058**	0.058**	-0.039	-0.015	-0.017	-0.019*	-0.063***	-0.099***	-0.049**	0.008
	(0.027)	(0.026)	(0.027)	(0.013)	(0.013)	(0.010)	(0.017)	(0.031)	(0.022)	(0.031)
Observations	12,596	12,596	12,596	12,596	12,596	12,596	12,596	12,596	12,596	12,596
R^2	0.663	0.629	0.550	0.438	0.438	0.402	0.550	0.435	0.524	0.531
Mean Dep. Var	0.63	0.24	0.70	0.04	0.03	0.02	0.05	0.06	0.84	0.53
	Sound Equipment	Video Equipment	Computers	Bikes	Motorcy.	Automob.	Houses	Office	Lots	Other Goods
TempShocks	-0.051**	0.006	-0.043**	-0.007	0.051**	-0.036***	-0.006	0.001	-0.005	-0.160***
	(0.020)	(0.023)	(0.018)	(0.028)	(0.025)	(0.011)	(0.012)	(0.003)	(0.006)	(0.038)
Observations	12,596	12,596	12,596	12,596	12,596	12,596	12,596	12,596	12,596	12,596
R^2	0.574	0.523	0.511	0.594	0.669	0.694	0.512	0.455	0.412	0.478
Mean Dep. Var	0.32	0.25	0.08	0.40	0.31	0.05	0.05	0.00	0.01	0.06

Notes: This table shows estimates of the effects of weather shocks on household assets using a three-wave panel survey of rural households (2010, 2013, 2016). The dependent variables are dummies corresponding to the options presented to households when asked: *"Does this household own any of the following assets?"* All regressions control for log aggregate rainfall at t and its five lags and include household and time fixed effects. *Mean Dep. Var.* reports the sample mean of the untransformed dependent variable. Standard errors clustered at the *vereda* level are reported in parentheses. Data from ELCA survey. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table O.14: Weather Shocks and Farm Size - Heterogeneous Effects by Contiguous Plots - Contiguity Measure from 2014 National Agricultural Census

	Number of Owners (logs) (1)	Mean Farm Size (logs) (2)	Median Farm Size (logs) (3)
$TempShocks$	0.019***	-0.019***	-0.022
	(0.006)	(0.007)	(0.015)
$TempShocks \times High$	-0.003	0.005	0.010
	(0.006)	(0.006)	(0.013)
Observations	8,575	8,575	8,575
R^2	0.991	0.993	0.973

Notes: This table reports the estimated effect of weather shocks on the farm-size distribution using an alternative measure of contiguity between small and large farms. The dependent variables are the log number of land owners in the municipality (column 1), the log average farm size (column 2), and the log median farm size (column 3). The independent variable is the total number of atypical-temperature days in the past two years divided by 100. The variable *High* indicates municipalities with above-median contiguity as determined from GPS coordinates in the 2014 National Agricultural Census; the interaction term tests whether weather shocks have differential effects in these municipalities. All regressions include municipality and year fixed effects, as well as cumulative rainfall and its five lags, the accumulated number of land allocations, the area covered by the land registry, and an indicator for cadastral-registry updates. Standard errors clustered at the municipality level are reported in parentheses. Data from the National Land Registry (*Catastro Nacional*), maintained by the National Geographical Institute (IGAC). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table O.15: Effect of Weather Shocks on Household Split

	HH split (1)	HH split in municipality (2)	HH split with land (3)
TempShocks	0.041**	0.035**	0.001
	(0.018)	(0.017)	(0.013)
Observations	13,442	13,442	13,442
R^2	0.387	0.372	0.359
Mean Dep. Var	0.06	0.05	0.04

Notes: This table shows estimates of the effects of weather shocks on household splits using a three-wave panel survey of rural households (2010, 2013, 2016). The dependent variable in column 1 equals one if the household experienced a split between survey waves. The dependent variable in column 2 equals one if at least one split-off household member remained in the same municipality. The dependent variable in column 3 equals one if at least one split-off household member remained in the same municipality and owned land there. The independent variable is the total number of atypical-temperature days in the past two years divided by 100. All regressions control for log aggregate rainfall at t and its five lags and include household and year fixed effects. Standard errors clustered at the vereda level are reported in parentheses. Data from ELCA. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table O.16: Weather Shocks, Farm Size, and Share of Government-Allocated Area

	Control: Share allocates			H_i: Share allocated		
	Number of Owners (logs) (1)	Mean Farm Size (logs) (2)	Median Farm Size (logs) (3)	Number of Owners (logs) (4)	Mean Farm Size (logs) (5)	Median Farm Size (logs) (6)
$TempShocks$	0.014***	-0.015***	-0.022*	0.014***	-0.016***	-0.031***
	(0.005)	(0.005)	(0.012)	(0.005)	(0.005)	(0.011)
$TempShocks \times H_i$				0.000	0.004	0.027*
				(0.008)	(0.009)	(0.015)
Observations	10,021	10,021	10,021	10,021	10,021	10,021
R^2	0.992	0.993	0.975	0.992	0.993	0.975

Notes: This table reports the estimated effect of weather shocks on the farm-size distribution taking into account the share of government allocated land in the municipality. The dependent variables are the log number of land owners in the municipality (columns 1 and 4), the log average farm size (columns 2 and 5), and the log median farm size (columns 3 and 6). The independent variable is the total number of atypical-temperature days in the past two years divided by 100. Columns 1–3 include the continuous measure of the share of municipal land that originated from government allocations as a control. Columns 4–6 add an interaction between the temperature-shock variable and an indicator for whether the municipality is above the median in the share of its area that was ever part of a government allocation. All regressions include municipality and year fixed effects, as well as cumulative rainfall and its five lags, the accumulated number of land allocations, the area covered by the land registry, and an indicator for cadastral-registry updates. Standard errors clustered at the municipality level are reported in parentheses. Data from the National Land Registry (*Catastro Nacional*), maintained by the National Geographical Institute (IGAC). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table O.17: Weather Shocks, Land Sales, and Average Farm Size - Conflict Controls

	Land sales				Owners & Farm size (logs)		
	Total	Full	Partial	Morgt.	Number of Owners (logs)	Mean Farm Size (logs)	Median Farm Size (logs)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Controls for displacement							
TempShocks	2.524***	2.000***	0.525**	1.049***	0.015***	-0.015***	-0.022*
	(0.537)	(0.505)	(0.229)	(0.237)	(0.005)	(0.005)	(0.012)
Observations	10,014	10,014	10,014	10,014	10,014	10,014	10,014
R^2	0.906	0.902	0.636	0.757	0.992	0.993	0.974
Mean Dep. Var	12.63	10.84	1.79	2.62	2,501.38	32.06	16.15
Panel B: Controls for homicide							
TempShocks	2.457***	1.938***	0.519**	1.016***	0.015***	-0.015***	-0.022*
	(0.540)	(0.509)	(0.226)	(0.237)	(0.005)	(0.005)	(0.012)
Observations	10,014	10,014	10,014	10,014	10,014	10,014	10,014
R^2	0.906	0.903	0.636	0.758	0.992	0.993	0.974
Mean Dep. Var	12.63	10.84	1.79	2.62	2,501.38	32.06	16.15

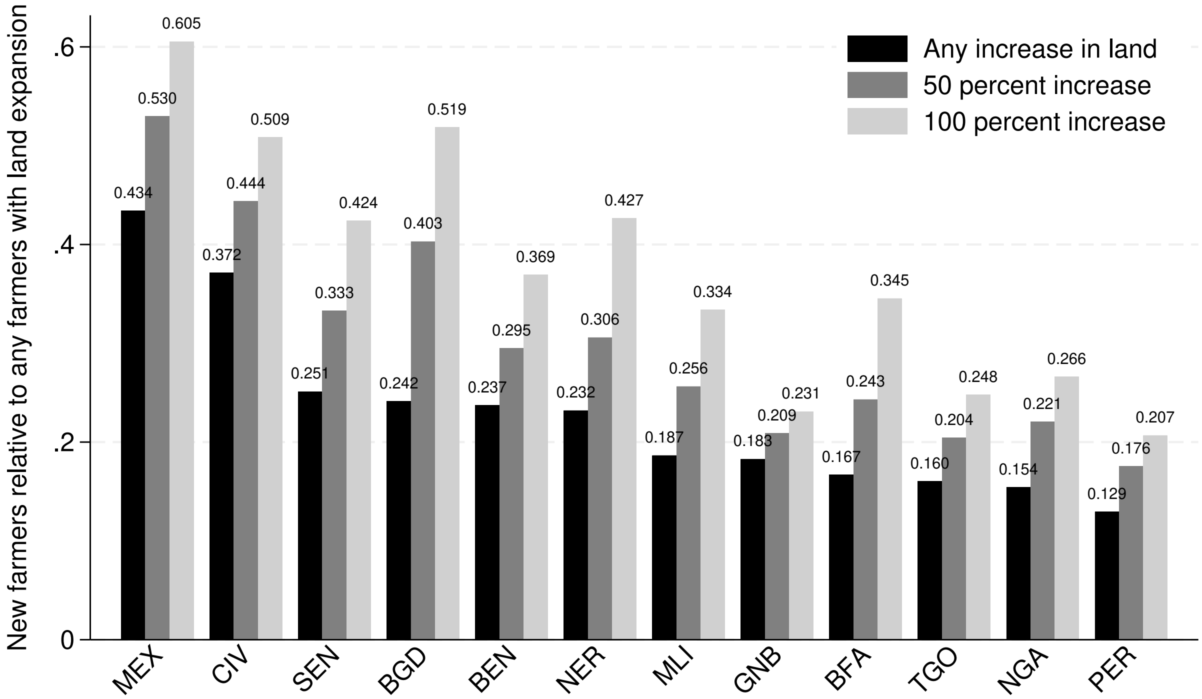
Notes: This table presents coefficient estimates from equation (4) in a balanced panel at the municipality-year level, using various measures of land-market activity and farm size as outcomes. The dependent variable in column 1 is the total of land sales (full + partial), in column 2 the number of full sales, in column 3 the number of partial sales, in column 4 the number of land mortgages, in column 5 the log number of land owners in the municipality, in column 6 log average farm size, and in column 7 the log median farm size. The independent variable is the total number of atypical-temperature days in the past two years, divided by 100. All regressions include municipality and year fixed effects, as well as cumulative rainfall and its five lags, the accumulated number of land allocations, the area covered by the land registry, and an indicator for cadastral registry updates. Panel A includes as an additional control the rate of forced displacement while Panel B the homicide rate. Both are expressed as a share of the municipality's 2005 population. Standard errors clustered at the municipality level are reported in parentheses. Data from the National Superintendency of Notaries (SNR).

Table O.18: Survey Results on Land Sales and Income Shocks

Panel A: Land Sales and Shocks	**Frequency**
(1) Land sales occur in the community	68.20%
(2) Weather shocks impacted farmers' income in recent years	53.60%
Among respondents reporting ocurrence of land sales:	
(3) Shocks of any type lead farmers to sell land	33.10%
Among respondents reporting ocurrence of land sales and weather shocks:	
(4) Weather shocks lead farmers to sell land	35.80%
Panel B: Types of Land Buyers	
In normal times land is bought by:	
(5) Landholders from the municipality	41.38%
(6) Landholders from outside the municipality	29.66%
(7) Non-landholders from the municipality	12.41%
(8) Non-landholders from outside the municipality	16.55%
During distress sales land is bought by:	
(9) Landholders from the municipality	36.25%
(10) Landholders from outside the municipality	35.00%
(11) Non-landholders from the municipality	10.00%
(12) Non-landholders from outside the municipality	18.75%

Notes: This table presents results from the survey "Funcionamiento de los Mercados de Tierras Rurales en Colombia" (Functioning of Rural Land Markets in Colombia). Frequencies in Panel A correspond to the share of respondents answering "yes" to each question. Questions 1 and 2 were asked of all respondents. Question 3 was asked only of respondents who reported land sales in the community in previous years. Question 4 was asked only of respondents who reported both land sales and weather shocks affecting farmers' income. Frequencies in Panel B correspond to the distribution of selected response categories, as respondents could choose more than one option. The questions under *"In normal times"* (5 to 8) were asked of respondents who reported land sales in the community, whereas the questions under *"During distress sales"* (9 to 12) were asked only of respondents who reported that shocks (of any type) lead farmers to sell land.

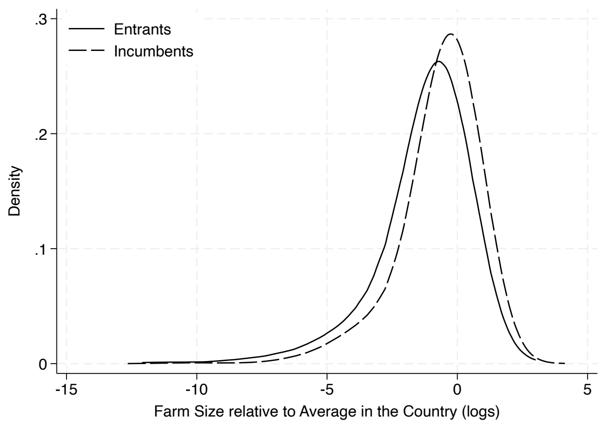
Figure O.1: Share of Entrants relative to All Farmers Increasing their Land Operations



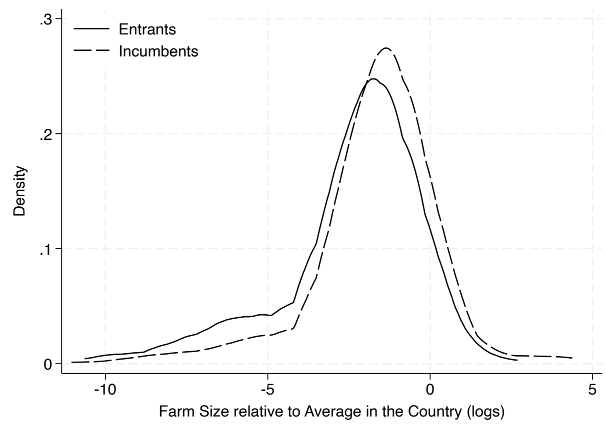
Notes: This figure shows the share of entrants among all farmers reporting an increase in their land operations across countries. Because there might be measurement error driving the changes in farm size distribution, we report the share based on different cutoffs of increase in land operations.

Figure O.2: Farm Size Distribution - Entrants vs. Incumbents

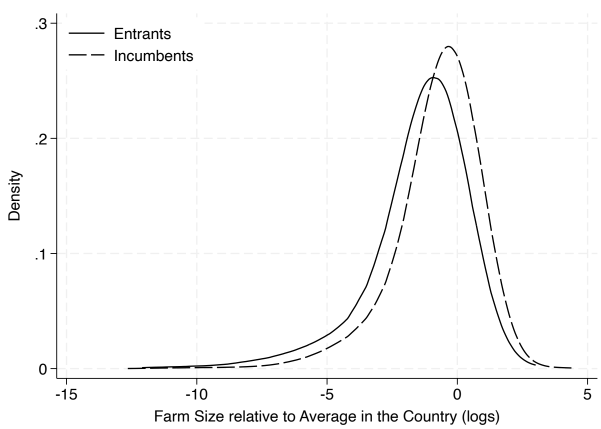
(a) West Africa



(b) Latin America

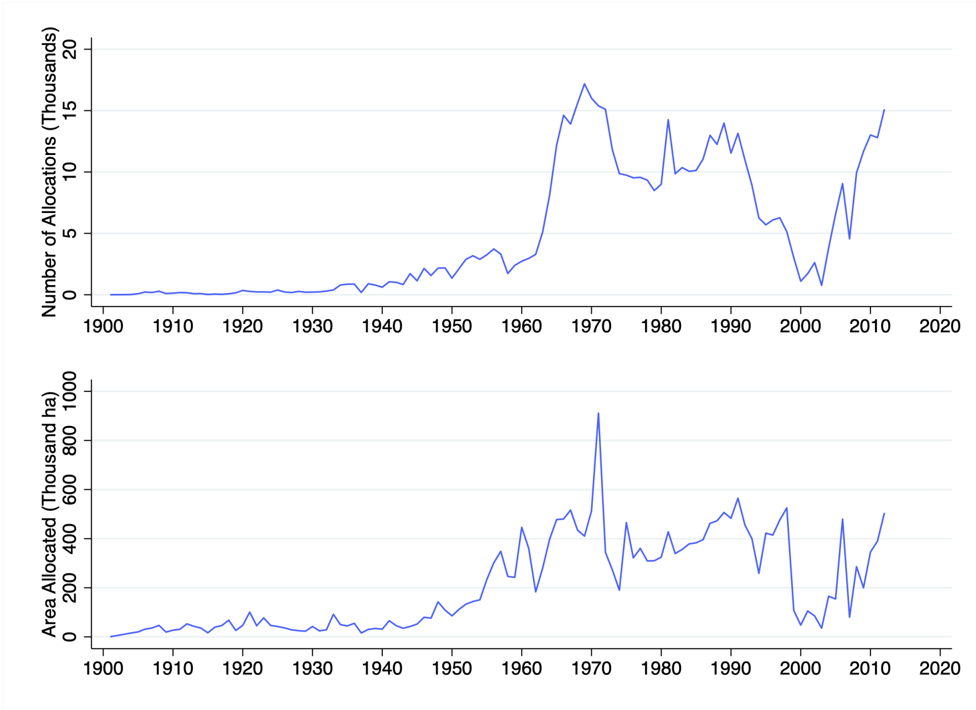


(c) All Countries



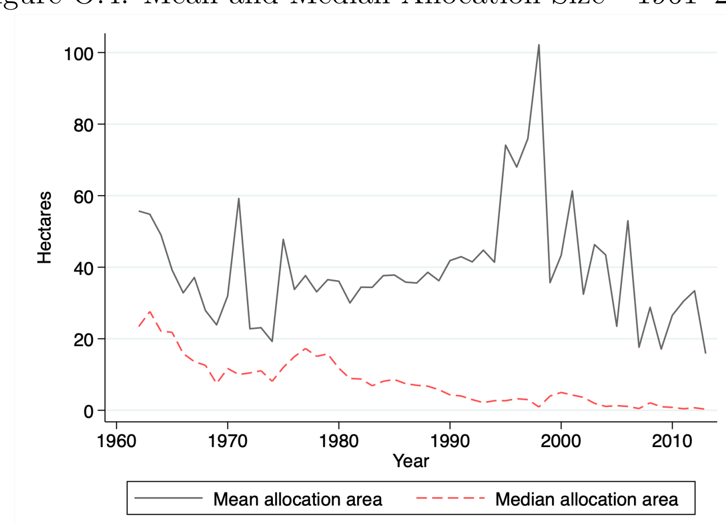
Notes: This figure shows the farm-size distribution for incumbents and new entrants. Observations are weighted by the inverse of the number of observations in each country, so that each country contributes equally. Given the three-year gap between survey waves, entrants operate farms that are, on average, 40 percent smaller than those of incumbents in West Africa and 70 percent smaller in Latin America.

Figure O.3: Land Allocations in Colombia - 1901–2012



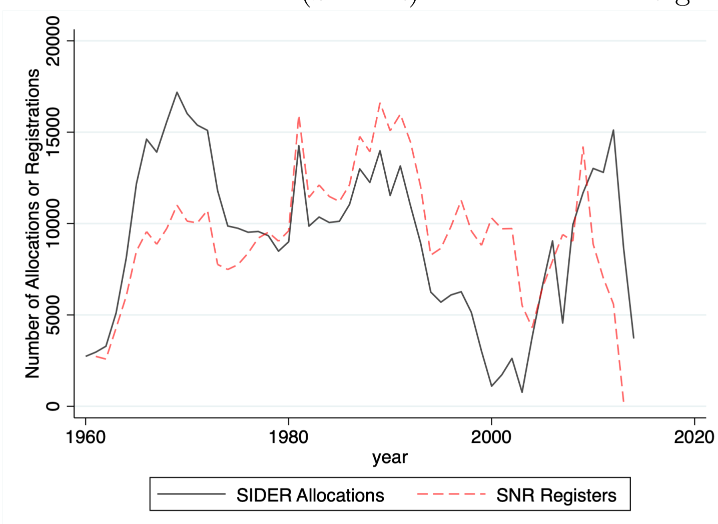
Notes: Data from the System of Information for Rural Development (SIDER)

Figure O.4: Mean and Median Allocation Size - 1961–2012



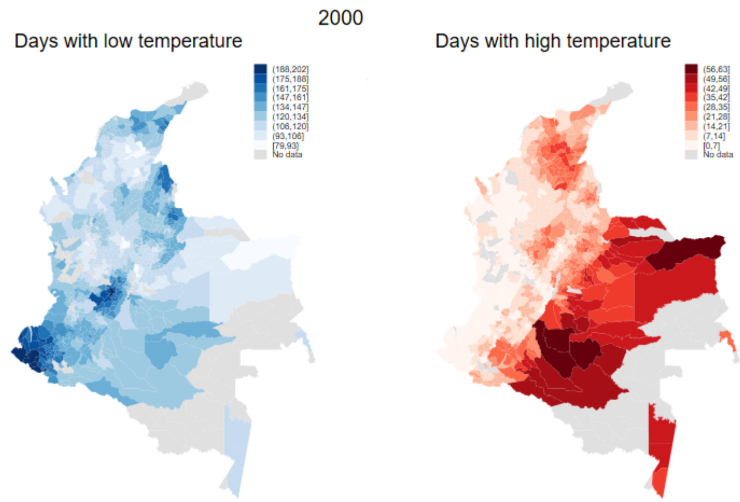
Notes: This figure shows the national yearly mean and median areas of land properties granted by the government through the public land–allocation program. Data from the System of Information for Rural Development (SIDER).

Figure O.5: Number of Allocations (SIDER) vs. Number of Registrations (SNR)

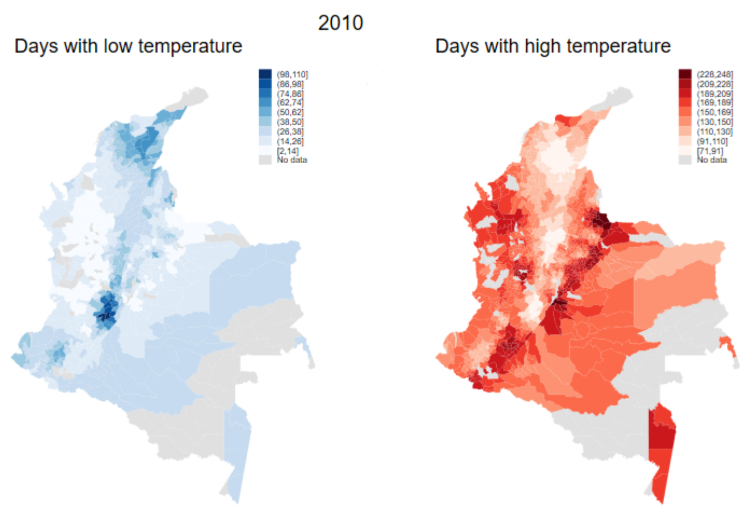


Notes: The figure compares the number of land properties allocated by the government through the public land–allocation program with the number of properties registered at local public notary offices as having been received from the government. Property registration is the final step in the allocation process and establishes the beneficiary's formal property rights over the granted plot. Data from the System of Information for Rural Development (SIDER) and the National Superintendency of Notaries (SNR).

Figure O.6: Weather Shocks Across Space



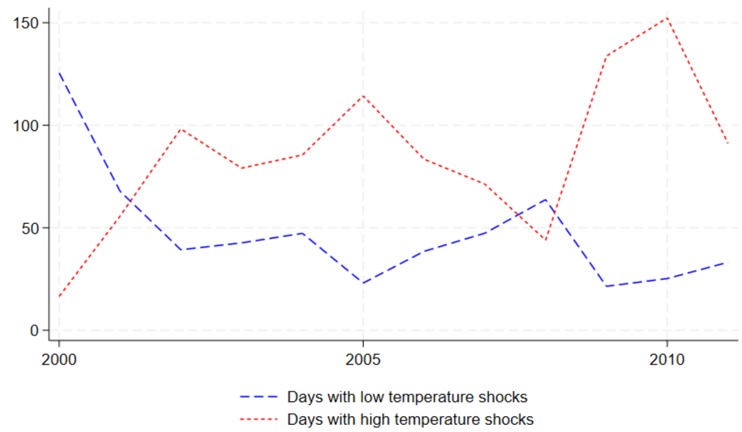
(a) Shocks in 2000



(b) Shocks in 2010

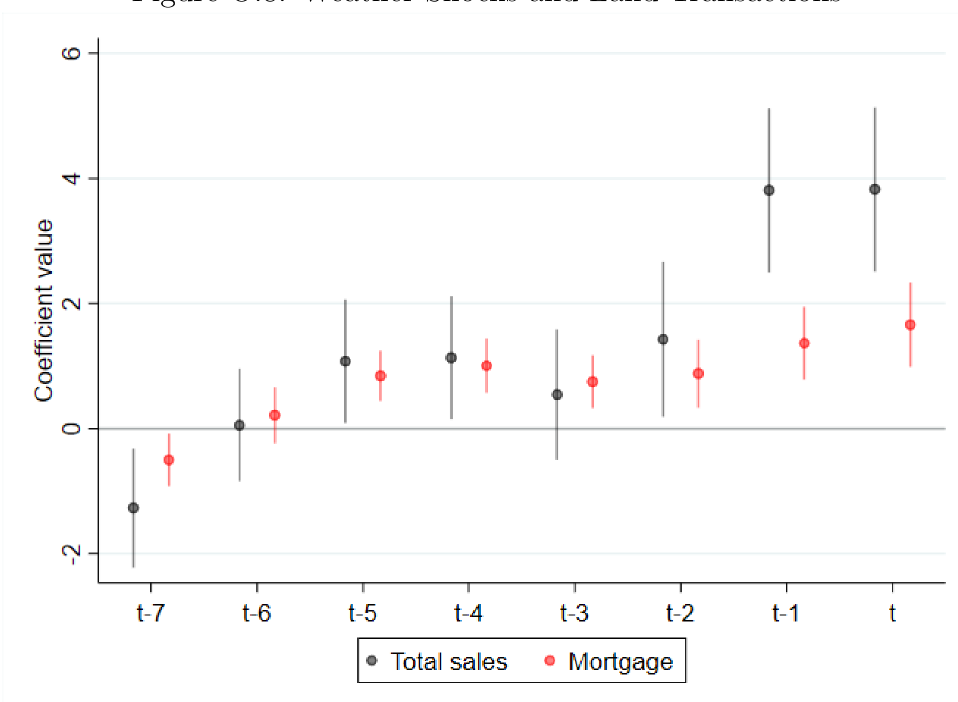
Notes: The figure shows the average number of days with extreme heat (red) and extreme cold (blue) across municipalities in our sample in 2000 and 2010. Data from the Copernicus Climate Change Service (C3S).

Figure O.7: Weather Shocks Across Time



Notes: The figure shows the average across municipalities of the number of days with atypical heat (red) and atypical cold (blue) in our sample for the 1979–2016 period. Data from the Copernicus Climate Change Service (C3S).

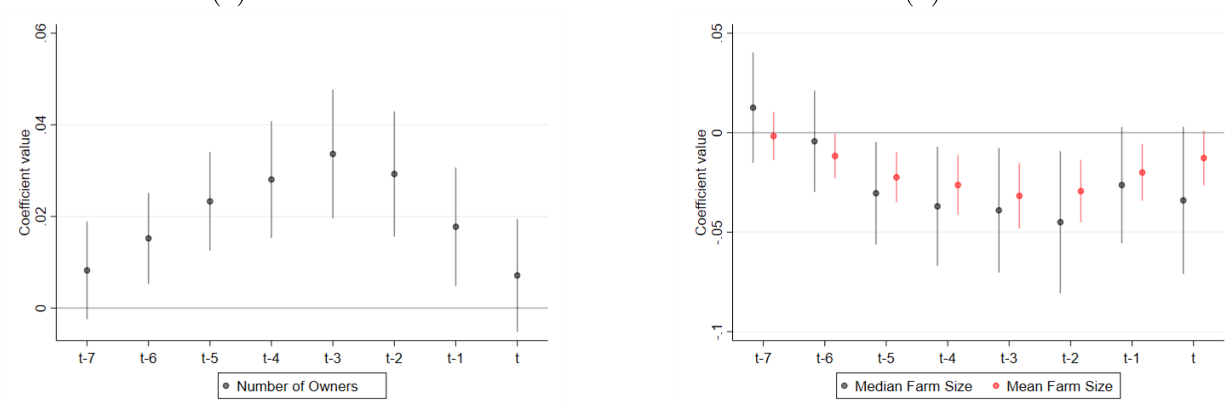
Figure O.8: Weather Shocks and Land Transactions



Notes: The figure presents the coefficient estimates for the number of days with atypical temperature (divided by 100) and its lags from equation (O.1). The outcomes are the total number of land sales (black) and the total number of land mortages (red). All regressions include municipality and year fixed effects, as well as cumulative rainfall and its five lags, the accumulated number of land allocations, the area covered by the land registry, and an indicator for cadastral registry updates. Standard errors clustered at the municipality level. Error bars indicate 95% confidence intervals. Data from the National Superintendency of Notaries (SNR).

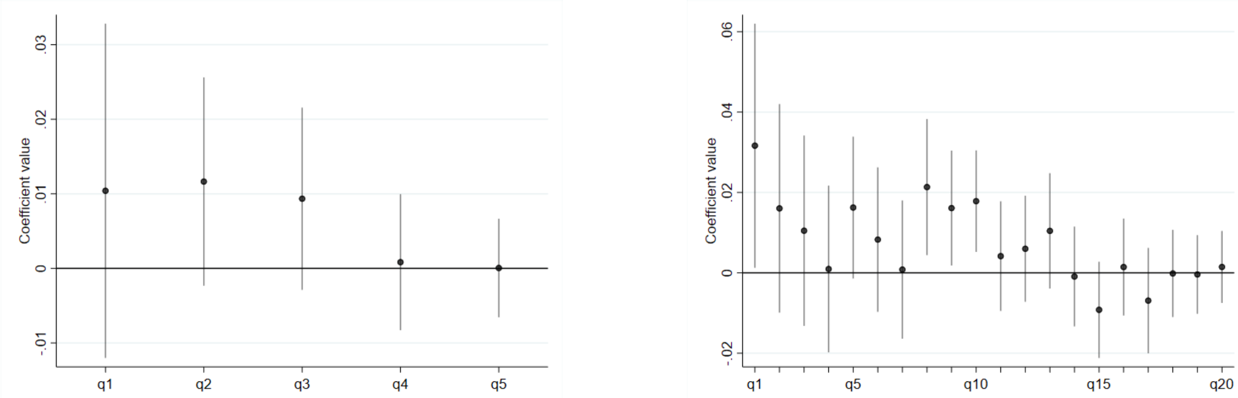
Figure O.9: Weather Shocks and Farm Size

(a) Owners (b) Area



Notes: The figure presents coefficient estimates for the number of days with atypical temperature (divided by 100) and its lags from equation (O.1). In Panel (a), the outcome is the number of owners, and in Panel (b), the outcomes are mean and median farm size. All regressions include municipality and year fixed effects, as well as cumulative rainfall and its five lags, the accumulated number of land allocations, the area covered by the land registry, and an indicator for cadastral-registry updates. Standard errors are clustered at the municipality level. Error bars indicate 95% confidence intervals. Data from the National Land Registry (*Catastro Nacional*), maintained by the National Geographical Institute (IGAC).

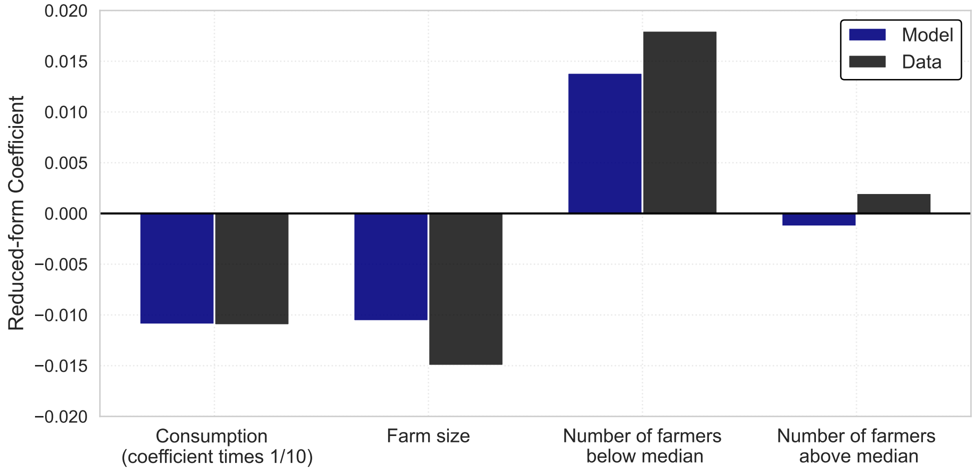
Figure O.10: Weather Shocks and Number of Owners by Initial Distribution Quantiles - Alternative Partitions



(a) Quintiles (b) Vigintiles

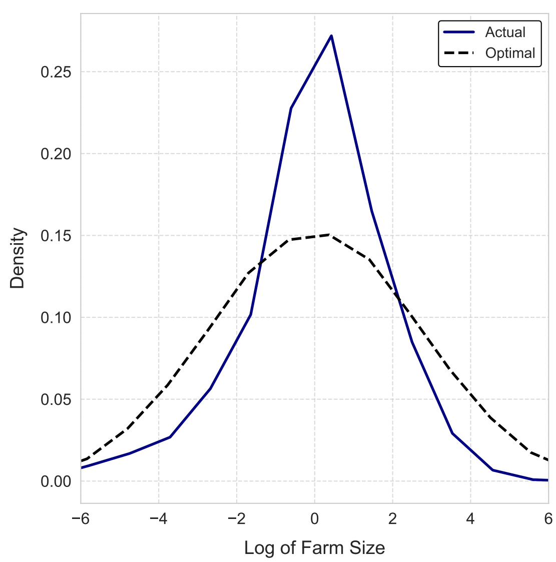
Notes: This figures reports estimates of γ^j for $j = 1, \ldots, 5$ (Panel a) and of f γ^j for $j = 1, \ldots, 20$ (Panel b) from equation (6). A separate regression is run for each of the quantiles of the initial municipality-level distribution of farm sizes. The outcome is the log number of owners whose farm sizes fall within the corresponding quantiles. The independent variable is the number of atypical temperature days in the past two years, divided by 100. All regressions include municipality and year fixed effects, as well as cumulative rainfall and its five lags, the accumulated number of land allocations, the area covered by the land registry, and an indicator for cadastral registry updates. Error bars indicate 95% confidence intervals. Standard errors are clustered at the municipality level.

Figure O.11: Reduced-form Impact of Weather Shocks with Multiple Regions (Model vs. Data)

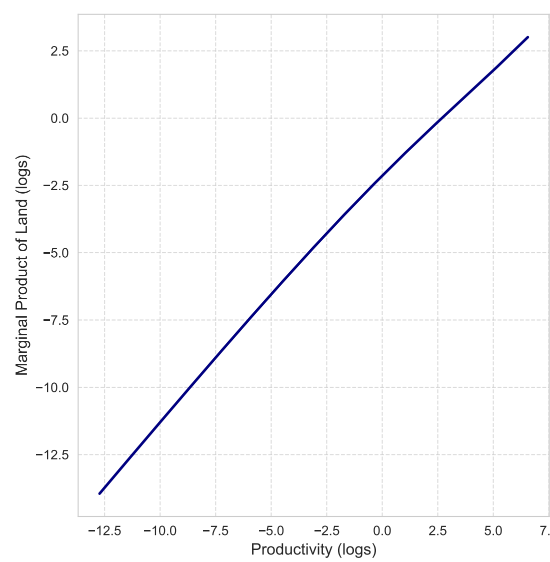


Notes: This figure shows the calibration of the model in which we simulate the evolution of many regions and estimate a fixed effect specification with region and time fixed effects. Since the shock was already exogenous, our calibration results are virtually the same.

Figure O.12: Optimal Farm Size Distribution



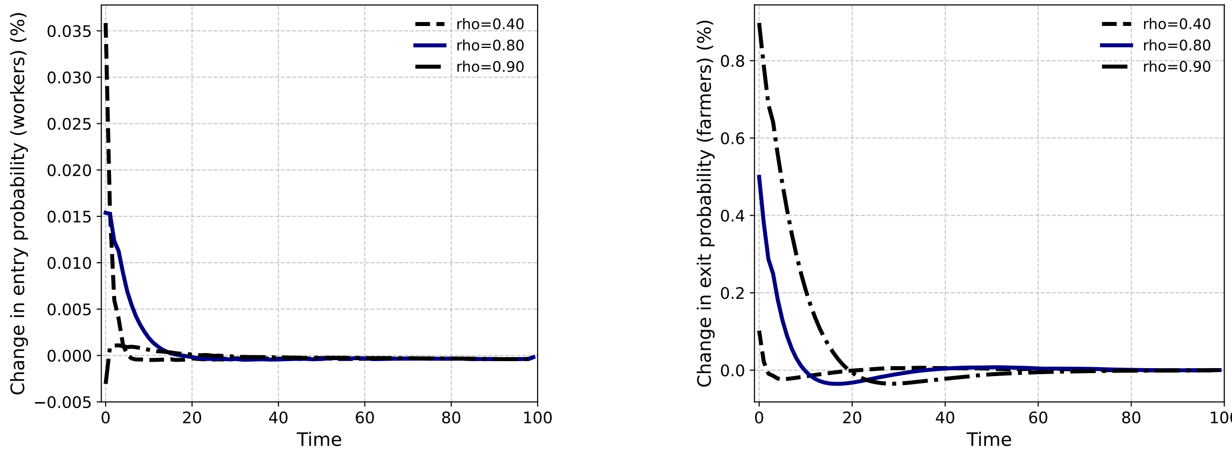
(a) Optimal vs. Actual Farm Size



(b) Marginal Product of Land

Notes: Notes: This figure compares the farm-size distribution generated by the model with a counterfactual allocation in which the marginal product of land is equalized across farmers, as in an economy with fully functioning rental markets. Panel (a) compares the actual and counterfactual farm-size distributions. Panel (b) shows the marginal product of land across farmers by productivity in the model allocation. The counterfactual farm size of a farmer with productivity s_t^a is computed as $\ell(s_t^a) = \frac{s_t^a}{\int s_t^a dG(\phi_t)} L$, where L is total land and $G(\phi_t)$ is the stationary distribution of farmer types.

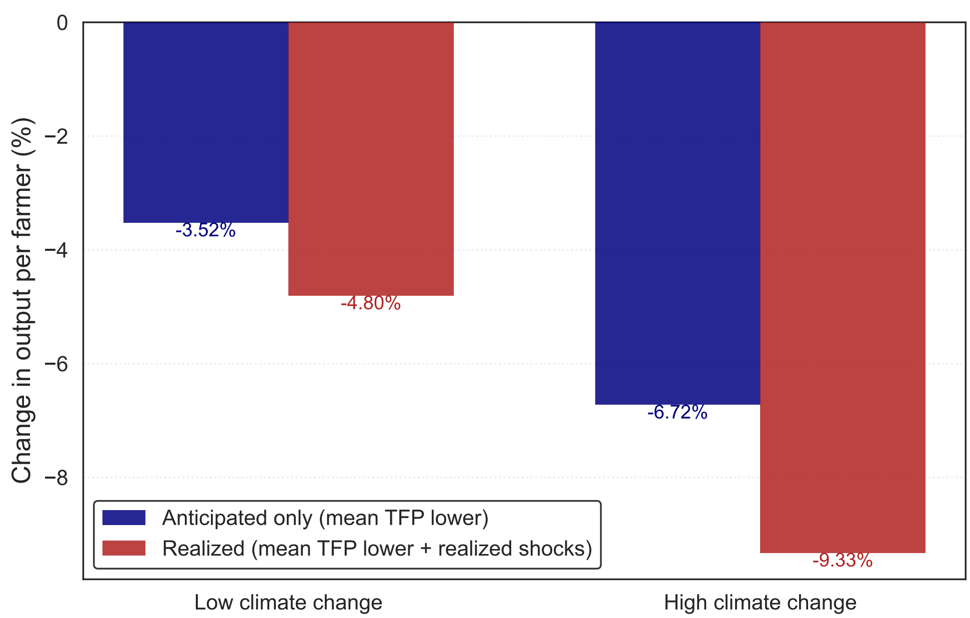
Figure O.13: Workers' Entry and Farmers' Exit with Different Shock Persistence



(a) Workers' Entry (b) Farmers' Exit

Notes: This figure shows the response of workers' entry and farmers' exit to a one-time, one-standard-deviation increase in the weather shock ω_t at $t = 0$ under different persistence assumptions. We consider three shocks, with autocorrelations of 0.4, 0.8, and 0.9, as indicated in the figure.

Figure O.14: Impact of Climate Change with Anticipation of Changes in Average Productivity



Notes: This figure shows the effect of the climate-change scenarios on output per farmer when households anticipate the average reduction in agricultural TFP. The "anticipated only" bars report the change in output per farmer in a new steady state with permanently lower agricultural TFP. The "Realized" bars add the effects of the corresponding sequence of weather shocks, starting from that new steady state. Changes are reported relative to the baseline steady state.